

**Graduation Project Report
(Academic Year 2025 – 2026)**

**A Website for Tourism & Travel
“Your Only Chance (YOC)”**

Submitted in Partial Fulfillment of the Requirements for the
Bachelor of Science in Management Information Systems (MIS)

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list of Abbreviations

- ☐ **YOC:** Your Only Chance (The Project Name)
- ☐ **AI:** Artificial Intelligence
- ☐ **MIS:** Management Information Systems
- ☐ **ERU:** Egyptian Russian University
- ☐ **DFD:** Data Flow Diagram
- ☐ **EERD:** Enhanced Entity Relationship Diagram
- ☐ **UML:** Unified Modeling Language
- ☐ **CMS:** Content Management System
- ☐ **GTM:** Go-To-Market Strategy
- ☐ **SWOT:** Strengths, Weaknesses, Opportunities, and Threats
- ☐ **UI:** User Interface
- ☐ **UX:** User Experience
- ☐ **HTML:** Hyper Text Markup Language
- ☐ **CSS:** Cascading Style Sheets
- ☐ **PHP:** Hypertext Preprocessor
- ☐ **SQL:** Structured Query Language (used in MySQL)
- ☐ **API:** Application Programming Interface
- ☐ **GDP:** Gross Domestic Product
- ☐ **SDGs:** Sustainable Development Goals
- ☐ **HTTPS:** Hypertext Transfer Protocol Secure
- ☐ **XAMPP:** Cross-Platform (X), Apache (A), MariaDB (M), PHP (P), and Perl (P)

chapter 1

Introduction

1.1 Introduction

Tourism is one of the most influential global industries, playing a major role in economic development and job creation worldwide. Countries like Egypt rely heavily on tourism due to their rich historical heritage, cultural diversity, and unique natural attractions (UN Tourism, 2024) [1].

Traditionally, travellers depended on travel agencies, printed brochures, and face-to-face consultations to plan their trips. Although traditional tourism companies provided structured services, they often lacked flexibility, real-time information, personalization, and transparency, which made the planning process slow and inconvenient (Werthner & Klein, 1999; Buhalis & Law, 2008) [2].

Our project aims to address these challenges by developing an integrated tourism website designed specifically for Egypt. The platform combines AI-powered personalized recommendations, virtual destination exploration (photos and videos), and a unified booking system for hotels, camps, and guided trips. It also includes a community space where travellers can share experiences and reviews. By integrating modern technologies, personalization tools, and reliable booking features, the project seeks to enhance the travel experience, simplify trip planning, and support Egypt's ongoing digital transformation in the tourism sector (Gretzel et al., 2015; Ricci et al., 2011; Ministry of Planning and Economic Development, 2016) [3], [5].

With the rise of digital transformation, the tourism industry witnessed a major shift. Online platforms enabled travellers to explore destinations, compare prices, read reviews, and book services more easily. These digital tools significantly improved accessibility, convenience, and user experience, giving travellers more control over their decisions (Buhalis & Law, 2008; Statista, 2024) [5], [7]. Recent academic studies also highlight the increasing role of artificial intelligence and machine learning in enhancing personalization and traveller satisfaction (Gretzel et al., 2015; Ricci et al., 2011) [3], [4].

1.2 Alignment with Egypt vision 2030

The YOC project fully aligns with the goals of Egypt Vision 2030, particularly under the pillars of Economic Development, Innovation, and Environmental Sustainability.

Vision 2030 emphasizes transforming Egypt into a leading global tourism destination by leveraging digital transformation, sustainable practices, and improved service quality all of which are core objectives of our project.

How Our Project Supports Egypt Vision 2030

1. Digital transformation in tourism

- Vision 2030 highlights the need for smart, technology-driven systems in tourism management.
- Our website applies AI-based recommendations, first look, and online booking to digitalize the Egyptian tourism experience directly supporting this transformation goal.

2. Sustainable and inclusive tourism

- The Vision calls for sustainable use of Egypt's cultural and natural heritage while ensuring accessibility to all travelers.
- Our platform promotes domestic and eco-friendly travel, encourages local guides and small tourism providers, and ensures equal access for both Egyptian and international tourists.

3. Economic growth and job creation

- Vision 2030 aims to increase tourism's contribution to GDP and expand employment opportunities.
- By connecting hotels, camps, and guides directly with travelers, our system reduces dependency on competitors, increases local business income, and creates new digital tourism jobs.

4. Promoting Egypt's heritage and image

- A major Vision 2030 goal is to enhance Egypt's global image through cultural promotion.
- The website supports this by offering virtual exploration of landmarks, interactive cultural content, and photos and videos experiences that showcase Egypt's heritage to the world.

5. Data driven decision making

- Vision 2030 encourages innovation through data and analytics.
- Our system collects and analyzes tourism data to help stakeholders understand visitor behavior, optimize offers, and support evidence based planning aligning with the Vision's innovation targets.

1.3 Literature review

1.3.1 Related work

Several online tourism and booking platforms have been developed to simplify travel planning and reservations. Among the most popular are Booking.com, TripAdvisor, and Egypt.travel, each focusing on specific aspects of the tourism experience.

- Booking.com mainly provides hotel and accommodation booking services worldwide. It offers a wide range of hotels, but it focuses more on the global market rather than specific local destinations.
- TripAdvisor allows users to review hotels, restaurants, and tourist attractions. It helps travelers plan their trips based on community reviews, yet it lacks real-time personalization and local cultural integration.
- Egypt.travel, the official tourism portal of Egypt, provides information about attractions and historical sites, but it does not allow users to make direct bookings or interact with local guides.

- Wego.com, mainly acts as a metasearch engine for flights and hotels, focusing on price comparison rather than personalized travel planning. It lacks AI-driven recommendations, virtual destination previews, community interaction, and guide-booking features.
- Misr Travel, is a long-established national company, but its digital presence remains outdated with limited mobile usability. The platform lacks personalization tools, interactive destination previews, and integrated online booking features.
- DeCastro Tours , provides local services, yet its website offers limited digital capabilities. It lacks advanced search tools, personalized trip suggestions, community interaction, and structured photo/video content for destinations.
- CWT specializes in corporate travel, but its system is not optimized for leisure tourism needs. It does not provide personalized trip tools, unified booking for different services, community features, or strong visual previews.
- EgyptAir Tourism – Karnak , offers official travel services, but its online platform lacks modern automation and user-friendly booking flow. It does not include AI-based recommendations, rich destination previews, or interactive customer engagement features.
- Travel Choice has a valid license but maintains a minimal digital footprint. It lacks an automated booking platform, personalized recommendations, community interaction, and detailed multimedia destination content.

- Blue Sky Travel is well-established, yet its website design is outdated and limited in functionality. It lacks advanced filtering, personalized suggestions, community features, and comprehensive multimedia destination previews.
- Flash Tours serves international markets, but its digital tools remain basic. The platform lacks modern UX, AI-based recommendation features, user-generated content, and strong destination media.
- Travco is a major tourism group, though its online booking experience is not fully modernized. It lacks automated filtering tools, AI-driven personalization, community engagement features, and rich visual exploration content.
- ETL Travel provides corporate and group services, but its website lacks clear structure and modern usability. It does not offer AI-powered suggestions, user-generated content, or consistent visual destination materials.
- Abercrombie & Kent – Egypt , offers luxury travel experiences, yet its digital platform provides limited personalization and local content. It lacks community interaction, rich destination previews, and flexibility for diverse traveler types.
- Memphis Tours has strong market presence, but some parts of the website lack transparency and smooth mobile performance. The platform does not support AI-based recommendations and offers limited multimedia previews.
- Look at Egypt Tours offers private trips but needs better navigation and UX. It lacks an instant assistance chatbot, multilingual depth, and integrated community features for user-shared experiences.

Other recent studies and university projects have explored similar concepts such as integrating AI-driven travel recommendation systems, first look, and personalized user experiences. However, most of them either focus on one feature or lack a complete integration between booking, personalization, and user interaction.

The proposed system aims to bridge these gaps by combining all these features into one platform. It provides an AI-based trip customization system, user-generated trip suggestions, first look, and a community, all focused specifically on Egyptian tourism.

1.3.2 Competitor analysis

	Booking	Trip advisor	Visit egypt	look at Egypt tours	Egypt travel	wego	Misir travel	Decastro tarvel	CWT	Egypt air travel	Travel choice	blue sky	Flash tour	travco	ETL travel	Memphis tours	YOC
Multi Language Support	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AI Customizer	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Hotel Booking	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Tour Booking	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Camp Booking	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Local Verified Guides	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Secure Payment	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Payment Options (Visa, Wallet, etc.)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Reviews & Ratings	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Community & Social Interaction	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Chatbot / 24/7 Support	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
User Interface	8.5/10	8/10	7/10	8/10	7/10	8/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10	8.5/10

1. Booking.com

Identified Gaps:

Booking.com lacks several destination-specific features, including verified local guides, camp booking services, and community-driven interaction. Moreover, its tour booking options are limited and not tailored to regional cultural experiences.

YOC's Advantage:

YOC addresses these limitations by integrating a verified local guide system, offering

comprehensive camp and adventure booking services, and incorporating a dedicated social community for travelers. This positions YOC as a more localized and culturally adaptive platform.

2. TripAdvisor

Identified Gaps:

TripAdvisor primarily functions as a review aggregator and does not provide direct booking or secure payment processes. Additionally, it lacks camp booking services and offers no formal mechanism for guide verification.

YOC's Advantage:

YOC combines direct booking with a secure digital payment system, offers camp and safari booking options, and employs a structured model for verifying local guides, thereby offering a more integrated user experience.

3. Visit Egypt

Identified Gaps:

Visit Egypt provides limited service coverage, lacking hotel booking, camp booking, and secure online payment features. Furthermore, its user interface quality is comparatively low, limiting usability.

YOC's Advantage:

YOC compensates for these deficiencies by providing an all-in-one tourism platform supported by a modern, user-centered design and a fully secure payment system.

4. Look at Egypt Tours

Identified Gaps:

The platform does not offer hotel or camp bookings and lacks social or community-focused features. It also operates mainly through predefined tour packages, limiting flexibility.

YOC's Advantage:

YOC offers hotels, camps, and customizable tours within a unified system, supported by

community interaction and user-generated reviews, thus enhancing user autonomy and engagement.

5. Egypt Travel

Identified Gaps:

The platform lacks camp booking capabilities, community features, a modern interface, and secure payment functionalities.

YOC's Advantage:

YOC introduces camp booking, a contemporary UI/UX, advanced security for payments, and an interactive community environment, delivering a more comprehensive service model.

6. Wego

Identified Gaps:

Wego provides limited tourism-oriented services, with no tours, camps, guides, or community engagement tools.

YOC's Advantage:

YOC fills these gaps by incorporating tour booking, camp experiences, local guide verification, and community-based interaction within the same ecosystem.

7. Misr Travel

Identified Gaps:

The platform does not offer modern secure payment infrastructure, has no community or camp booking features, and maintains a traditional, less intuitive interface.

YOC's Advantage:

YOC enhances user experience through a modern interface, secure payments, verified guides, and expanded offerings including camp booking.

8. Decastro Travel

Identified Gaps:

Decastro Travel exhibits limited digital functionality, lacking camp booking, community features, and a system for verifying local guides.

YOC's Advantage:

YOC integrates these missing features, offering a digital-first platform with robust guide verification and community interaction.

9. CWT

Identified Gaps:

As a corporate travel service, CWT does not provide tourism-related experiences such as tours, camps, or local guides, nor does it support community engagement.

YOC's Advantage:

YOC is purpose-built for tourism, offering a full spectrum of cultural, adventure, and leisure services that CWT does not cover.

10. Egypt Air Travel

Identified Gaps:

The service focuses primarily on flights and lacks hotel bookings, tours, camps, guides, and community engagement.

YOC's Advantage:

YOC functions as a holistic tourism platform, providing accommodations, tours, activities, guides, and camps within a single ecosystem.

11. Travel Choice

Identified Gaps:

Travel Choice does not offer camp experiences, guide verification, community features, or advanced payment systems.

YOC's Advantage:

YOC addresses all these gaps through integrated camp booking, secure payments, community tools, and a verified guide framework.

12. Blue Sky

Identified Gaps:

The platform lacks secure payment solutions, verified guides, community features, and camp booking services.

YOC's Advantage:

YOC introduces secure online payment capabilities, a structured verification process for guides, community interaction, and camps.

13. Flash Tour

Identified Gaps:

Flash Tour has a limited digital presence and does not include community engagement, camp booking, or guide verification functionalities.

YOC's Advantage:

YOC adopts a digital-first approach, offering modern platform features, camp booking, and verified guides.

14. Travco

Identified Gaps:

The company mainly provides packaged tours and does not support individual hotel bookings, camps, or community interaction.

YOC's Advantage:

YOC offers individualized bookings for hotels, camps, and tours, supported by a social community and verified local guides.

15. ETL Travel

Identified Gaps:

The platform lacks secure online payment systems, guide services, community functions, and camp booking options.

YOC's Advantage:

YOC overcomes these limitations by integrating secure payments, verified guides, community features, and adventure/camp booking.

16. Memphis Tours

Identified Gaps:

Memphis Tours does not provide camp booking, community interaction, or a verified guide framework. Its interface design is also relatively outdated.

YOC's Advantage:

YOC improves upon these weaknesses through a modern interface, camp booking services, community integration, and formal guide verification.

1.4 Problem statement

Despite the global rise in digital tourism, many travellers still face difficulties when planning trips within Egypt. Existing platforms often provide scattered and unorganized information, limited personalization, and no exploration tools. In addition, most systems fail to offer a unified platform that combines hotel, camp, and trip booking in one place. As a result, tourists struggle with confusion, poor decision-making, and time-consuming planning processes (El-Gohary, 2012) [3].

Studies further confirm that travellers expect centralized, reliable, and interactive digital solutions supported by modern technologies (Statista, 2024; Werthner & Klein, 1999) [7], [8].

1.5 Objectives

The main objective of this project is to develop an integrated tourism platform that enhances the digital travel experience in Egypt. The system aims to provide users with a seamless and unified platform where they can explore destinations, view first look, book hotels, camps, and guided trips in one place.

Additionally, the platform seeks to offer AI-powered personalized recommendations based on user preferences, interests, and budget to support informed decision-making (Ricci et al., 2011; Gretzel et al., 2015) [3].

Another key objective is to improve traveller engagement by providing a community space for sharing experiences, reviews, and travel tips. The platform also aims to increase reliability and trust through verified service providers, organized travel information, and user-friendly booking features.

Overall, the project's objective is to support Egypt's ongoing digital transformation in tourism by delivering a modern, efficient, and technologically advanced solution that addresses the limitations found in current tourism platforms (Ministry of Planning and Economic Development, 2016; Statista, 2024; El-Gohary, 2012) [3], [5], [8].

1.6 Gap Analysis Summary (Based on References websites and papers):

Academic and industry studies reveal several gaps in current tourism platforms. Research shows that many existing websites lack advanced personalization models and AI-driven recommendation systems capable of understanding traveller preferences (Ricci et al., 2011; Gretzel et al., 2015) [1], [3], [8].

Other studies highlight that digital tourism tools in Egypt still suffer from fragmented information, making it difficult for users to find complete and reliable travel details in one place (El-Gohary, 2012; Werthner & Klein, 1999) [2], [6], [8].

Additionally, global reports confirm that modern travellers expect more interactive features such as exploration, real-time support, and community-driven content (Statista, 2024; UN Tourism, 2024) [7], [8].

Our project solves these gaps by providing:

- A unified platform combining hotels, camps, and trip booking.
- AI-based personalized recommendations tailored to users.
- First look destination exploration before booking.
- A community space where travellers can share reviews, experiences, and travel tips to increase trust and engagement.
- A smart chatbot that provides instant assistance, trip guidance, and answers to common travel questions.

1.7 significance:

1.7.1 social significance

- Promotes Egypt's culture, heritage, and hospitality globally.
- Encourages sustainable and responsible tourism practices.
- Provides digital employment opportunities (guides, developers).
- Enhances the image of Egypt as a modern, tourist-friendly destination.

1.7.2 Economic significance

- Boosts tourism revenue through digital exposure and bookings.
- Supports small and medium-sized tourism enterprises.
- Attracts foreign currency inflow.
- Contributes to Egypt's GDP through the tourism and technology sectors.

1.8 System request

1.8.1 Business need

The Egyptian tourism industry lacks a centralized and personalized online platform that works to both international and local travelers. Current travel websites in Egypt offer limited customization, weak user interaction, and no AI-support.

This project aims to solve these issues by developing an intelligent tourism website that helps users easily plan, book, and customize their trips across Egypt.

1. To enhance Egypt's global visibility as a travel destination using digital technology.
2. To improve customer retention through personalized and AI-driven recommendations.
3. To increase booking efficiency by search, filter, and booking processes.
4. To improve customer support quality and reduce costs using an AI chatbot.
5. To gain competitive advantage over existing Egyptian platforms like Visit Guide and Halla Travel.
6. To expand market share among both local and international travelers.
7. To collect data insights about customer preferences for strategic marketing.
8. To increase user engagement through community and content-sharing features.
9. To build brand trust by featuring verified guides and ratings.
10. To enhance conversion rates by offering first look that encourage bookings.
11. To attract partnerships with hotels, guides, and local agencies through a professional digital presence.
12. To enable cross-selling and upselling of travel-related services (e.g., tours, activities, transportation).

13. To support scalable business growth, with a platform expandable to a mobile app (future phase).
14. To improve decision-making via analytics dashboards and customer behavior tracking.
15. To reduce operational costs associated with traditional tourism marketing and booking methods.
16. To strengthen Egypt's tourism brand identity through a modern, user-friendly digital platform.

1.8.2 Business requirements

The proposed system should:

1. Allow booking of hotels, camps, and guided trips.
2. Provide first look with images and videos for destinations.
3. Use AI to suggest trips based on interests and budget.
4. Include a chatbot assistant for real-time support and inquiries.
5. Enable users to share reviews, photos, and experiences in a community feed.
6. Offer filters and search tools to refine destinations by price, location, and type.

1.8.3 Business value

Revenue Growth:

The system enables direct bookings between tourists and service provider and, increasing overall profit margins.

Operational Efficiency:

booking system, AI customer support, and recommendation processes using AI reduces manual work and operational expenses.

Market Expansion:

By targeting both local and international travelers, the platform increases market reach and helps attract new customer segments.

Brand Competitiveness:

Offering advanced features like AI Planner , first look, and verified local guides positions the platform as a modern and trusted tourism solution improving competitive advantage in Egypt's tourism market.

Data Analytics & Strategic Decision-Making:

The system gathers and analyze user data, helping management make data-driven marketing and pricing decisions that improve profitability.

Customer Retention & Lifetime Value:

Personalized experiences, transparent rating, and continuous engagement encourage repeat bookings and long-term customer loyalty.

Partnership Opportunities:

Establishing collaborations with hotels, guides, and agencies generates multiple revenue streams through advertising, package deals, and cross-selling opportunities.

Scalability & Future Growth:

The platform's architecture allows easy expansion into mobile applications and regional markets, supporting long-term business sustainability.

1.8.4 Special issues or constraints

1. The system should be scalable to mobile applications in the future.
2. Development should be completed within the project timeline.
3. Requires integration with secure payment gateways and reliable hosting servers.

1.9 System scope

1.9.1 In-Scope

1. User Registration and Authentication

- Users can sign up, log in, and manage profiles securely using email or social accounts.

2. Tourism Services Booking

- Booking system for hotels, camps, and guided trips within Egypt only.
- Includes search and filter.

3. AI-Planner System

- The platform includes an intelligent trip planner powered by an external AI API using a Retrieval-Augmented Generation (RAG) framework. By querying the local relational database for real-time inventory matching the user's budget and vibe, it generates verified, bookable custom itineraries.

4. First look

- Users can view destinations through pre-uploaded photos and videos.

5. Chatbot for Assistance

- AI chatbot providing conversational travel support, destination recommendations, and cultural answers. Powered by an external AI API, it contextually guides users toward relevant exploration and booking modules.

6. User Community and Reviews

- Users can post trip experiences, rate destinations, and review guides.
- All content is text and image-based (no live streams or social media integrations).

7. Admin Dashboard

- Admins can manage users, bookings, and content.
- Dashboard includes analytics and performance tracking tools.

8. Payment Gateway Integration

- Secure payments through Visa, Mastercard and InstaPay.
- No cryptocurrency or in-app wallet systems.

9. Responsive Web Design

- Works across desktops, tablets, and mobile browsers.
- No native mobile app development in this phase.

1.9.2 Out of Scope

1. Mobile Application Development

- Native Android/iOS apps are excluded in this phase; only web version included.

2. Third-Party Travel or Flight Booking Integration

- The system focuses on Egyptian tourism; no global travel or flight systems.

3. AI with Physical or Predictive Abilities

- The AI will not have voice commands, or act outside of programmed recommendations.
- No augmented reality (AR) or virtual reality (VR) expansion.

4. Human Resource and Payroll Management

- The system does not manage employee data, salaries, or agency HR systems.

5. Custom Package Creation by Users

- Users can suggest trips, but cannot create commercial packages or sell them through the system.

6. Offline Functionality

- The system requires an internet connection; offline booking or chatbot use is not supported.

7. Customer Support Beyond Chatbot Scope

- No 24/7 live agent or call center integration in this phase.

8. Dynamic Pricing Algorithms

- Pricing is managed manually or by admin inputs not through automated AI pricing or yield management.

9. Advanced Data Analytics Dashboard for External Clients

- Analytics are for internal admin use only no access for partner agencies or advertisers in this phase.

1.10 SWOT analysis

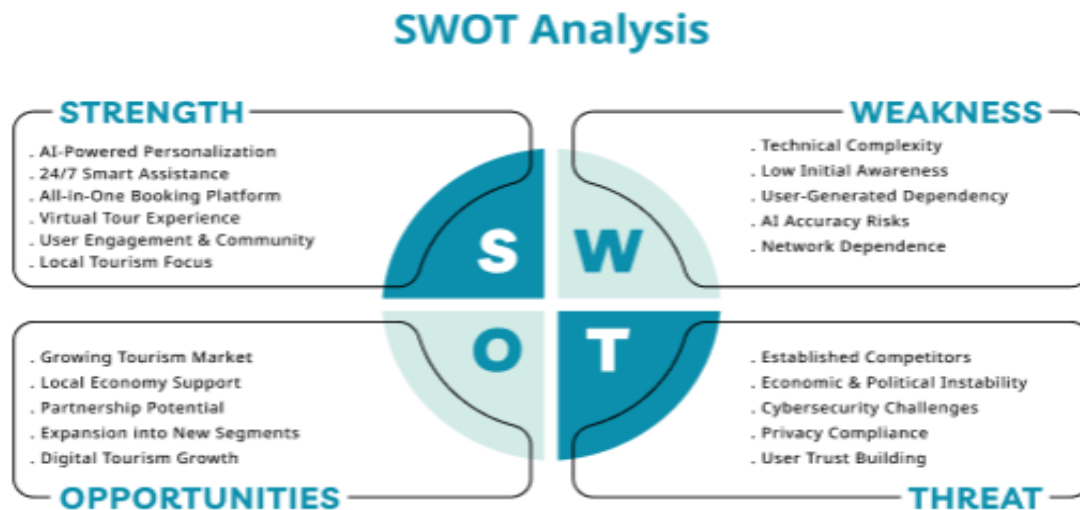


Figure 1.1 SWOT Analysis

1.10.1 Strengths

- **AI-Powered Personalization:**
Smart recommendation system adapts to each user's interests and travel style, creating a personalized and engaging planning experience.
- **24/7 Smart Assistance:**
Integrated AI chatbot offers instant help with bookings, trip info, and support improving reliability and user trust.
- **All-in-One Booking Platform:**
Centralized system where users can book hotels, camps, and trips easily from one place, simplifying the entire process.
- **First look Experience:**
Interactive visuals (photos & videos) allow users to explore destinations before booking, increasing confidence and excitement.

- **User Engagement & Community :**

User-generated trip suggestions and user posts build an active community and increase brand loyalty through shared experiences.

- **Local tourism focus:**

Highlights authentic Egyptian destinations and connects users directly with verified local guides, supporting Egypt's economy and cultural identity.

1.10.2 Weaknesses

- **Technical complexity:**

Integrating multiple systems (AI, chatbot, booking, community) demands strong technical coordination and maintenance.

- **Low initial awareness:**

Being a new brand, it will require solid marketing efforts to gain visibility and user trust.

- **User generated dependence:**

Community features rely on active participation; without engagement, early users may find limited content

- **AI accuracy risk:**

Poorly trained AI models may provide irrelevant or incorrect suggestions, potentially reducing credibility.

- **Network dependence:**

The website's interactive content (especially videos or tours) depends on a stable internet connection for smooth performance.

1.10.3 Opportunities

- **Growing Tourism Market:**

Egypt's tourism is expanding rapidly over 15.7M visitors in 2024 with a target of 30M by 2028 creating strong market potential.

- **Local Economy Support:**

By keeping booking and payment systems local, the platform helps retain foreign currency and supports Egyptian businesses.

- **Partnership Potential:**

Collaborations with verified hotels, guides, and eco-resorts can improve service quality and create exclusive packages.

- **Expansion into New Segments:**

Ability to include wellness, eco, and medical tourism (e.g., Siwa, Safaga) broadens the platform's audience.

- **Digital Tourism Growth:**

With over 72% of Egyptians online, the adoption of digital travel tools and AI-based systems is increasing.

1.10.4 Threats

- **Established Competitors:**

Competing with well-known platforms (TripAdvisor, Booking.com) that already have brand trust and loyal users.

- **Economic & Political Instability:**

Fluctuating currency rates, health crises, or political changes can directly affect tourism demand.

- **Cybersecurity Challenges:**

Managing user data, transactions, and AI systems safely requires advanced protection against cyber threats.

- **Privacy Compliance:**

Must follow strict data privacy regulations (like GDPR equivalents) to maintain user confidence.

- **User Trust Building:**

New platforms often face skepticism; success depends on consistent experience and transparent communication.

1.11 Feasibility study

1.11.1 Market Feasibility:

Tourism Market Overview in Egypt:

- Egypt is one of the most visited countries in the Middle East and Africa, with over 14 million tourists annually (pre-pandemic) and rapid recovery in recent years.
- The UN World Tourism Organization (UNWTO) ranks Egypt among the top African destinations due to its unique blend of historical, cultural, and natural attractions (e.g., Pyramids of Giza, Red Sea resorts, Nile cruises).

Target Market:

- International tourists (Europe, Asia, USA, GCC countries).
- Local Egyptian travelers seeking domestic tourism options.
- Travel agencies and operators wanting digital exposure.

Market Demand:

Growing trends in online travel booking, mobile tourism apps, and virtual cultural experiences indicate strong market demand for an integrated, trustworthy platform promoting Egypt.

1.11.2 Technical Feasibility:

Website Features:

- Interactive destination guide (maps, images).
- Online booking system (tours, hotels, camps).
- AI trip customizer
- Suggest trip system
- Community feature
- Chatbot support
- Payment gateway integration (Visa, Mastercard, InstaPay).
- Admin dashboard for partner agencies and user management.

Human Resources:

- Web developers
- UI/UX designer
- Digital marketer / SEO specialist
- Content creator
- Project manager

1.11.3 Legal Feasibility:

- Compliance with Egypt's e-commerce and data protection laws.
- Business registration and tourism license acquisition.
Copyright for photos, content, and brand identity.

Category	Amount (USD)	Explanation
ASSETS		
Cash	10,000	Initial cash available in the company's account.
Equipment	3,000	Computers, monitors, and basic office setup.
Website Development	15,000	Cost for website design, programming, and testing.
Licenses & Software	2,000	Domain, hosting, and software tools.
Content Creation	3,000	Photography, articles, and videos for marketing.
Total Assets	33,000	—
LIABILITIES		
Short-term Loans	5,000	Small loan or credit from partners or a bank.
Accounts Payable	2,000	Payments due to designers, agencies, or suppliers.
Total Liabilities	7,000	—
OWNER'S EQUITY		
Owner's Capital	25,000	Investment from founders or main investors.
Retained Earnings	1,000	Small profit or savings from early operations.
Total Owner's Equity	26,000	—

Table 1.1 balance sheet table

1.11.4 Operational Feasibility:

- Partnership with local tour operators and hotels.
- Collaboration with Egypt's Ministry of Tourism for verified data.
- 24/7 online customer support via chat and email.
- Continuous updates of travel regulations and events.

1.12 System methodology

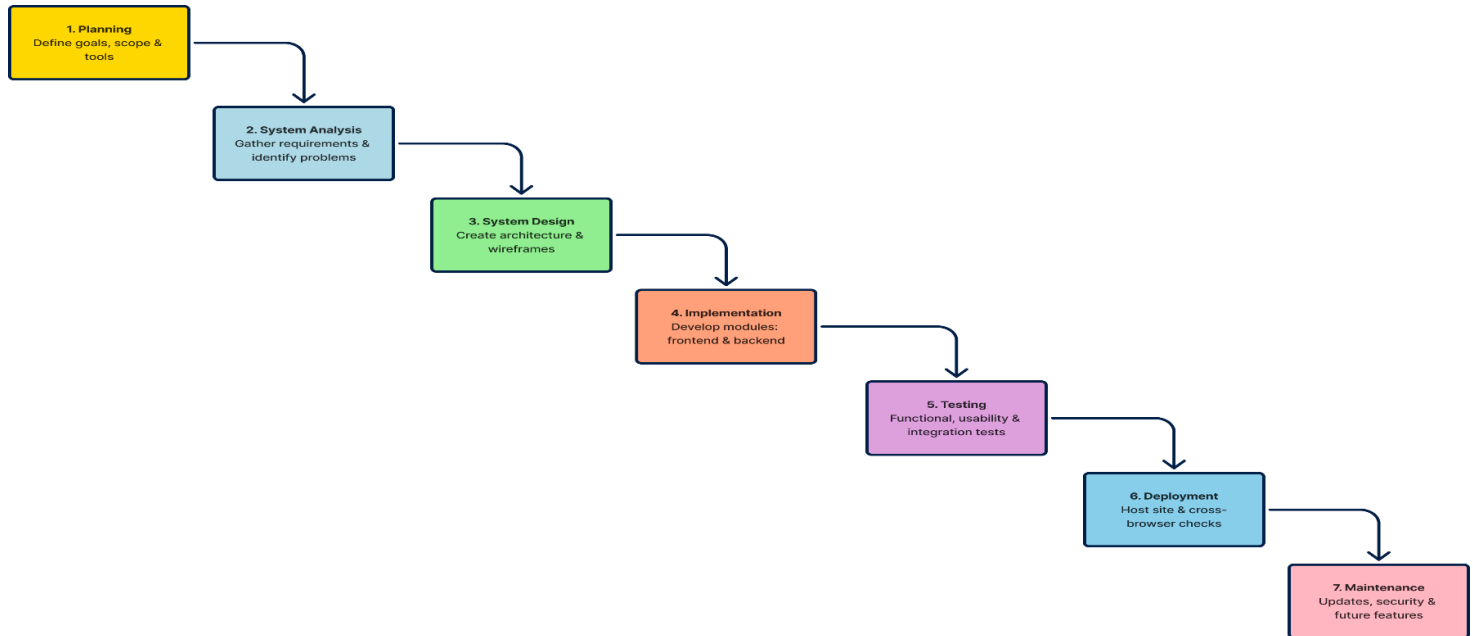


Figure 1.2 system methodology

The development of this tourism website follows the **Waterfall Model**, which is a linear and structured approach where each phase must be completed before moving to the next. This model is suitable for the project since the requirements are clearly defined and the objectives are stable from the beginning.

Planning Phase

In this phase, the goals, objectives, and scope of the system were defined. The planning involved understanding the tourism market in Egypt, identifying user needs (both local and international), and outlining the main features such as AI trip customization, hotel and camp booking, and the community.

Tools, technologies, and resources required for development were also selected, including HTML, CSS, JavaScript, PHP, and MySQL.

System Analysis Phase

This phase focused on gathering and analyzing system requirements through research and data collection.

It identified problems in the current manual and foreign-dependent systems, such as limited personalization, lack of local support, and slow booking processes.

Functional and non-functional requirements were defined to ensure that the new system meets user expectations in terms of performance, usability, and accessibility.

System Design Phase

In this phase, the overall structure of the system was designed. The UI/UX design was created to ensure an attractive and user-friendly interface. The system architecture was planned to define how data flows between users, the AI recommendation engine, and the database.

Wireframes and mockups were created to represent the key pages of the website, such as the home page, tour listings, and booking pages.

Implementation Phase

The implementation phase involved the actual development of the system.

Developers used PHP and MySQL for backend functionality and HTML, CSS, and JavaScript for the frontend interface.

Each module including the booking system, AI trip customizer, chatbot, and user account management was developed based on the design specifications.

Testing Phase

After implementation, the system underwent different testing stages such as:

- Functional Testing: To ensure each feature works correctly.
- Usability Testing: To verify that the website is easy to navigate for all users.

- Integration Testing: To confirm that all modules interact properly.
Bugs and errors found during this phase were fixed to improve performance and user experience.

Deployment Phase

In this phase, the website was hosted on a live web server, making it accessible to users. The deployment process also included checking the compatibility of the system across different browsers and devices.

Maintenance Phase

After deployment, the system enters a continuous maintenance cycle. Regular updates, security checks, and feature enhancements are carried out to improve functionality. Future updates may include adding mobile application integration and 360° photo to expand the platform's usability.

1.13 Business model



Figure 1.3 Business model

The Egypt Tourism & Trip Website follows a multi-sided platform business model that connects travelers with tourism service providers across Egypt. It creates value by offering users a seamless way to explore, plan, and book travel experiences, while enabling hotels, tour operators, and camp owners to promote their services and reach wider audiences.

Value Proposition:

- For travelers: A one-stop platform for booking hotels, camps, and trips with personalized recommendations based on interests, budget, and duration.
- For service providers: A digital marketplace to list, manage, and promote tourism services with increased visibility and customer reach.
- For Egypt's tourism sector: A system that supports local economic growth by keeping bookings and transactions within the country.

Customer Segments:

- International tourists interested in exploring Egypt's cultural, historical, and adventure destinations.
- Local travelers seeking affordable or customized experiences inside Egypt.

Channels:

- Main access through the website (desktop and mobile browsers).
- Future scalability to a mobile application for Android and iOS.
- Social media marketing for awareness and engagement.

Customer Relationships:

- Chatbot assistant and AI planner for real-time support and personalized recommendations.

Revenue Streams:

- Commission-based income from hotel, camp, and trip bookings.
- Partnerships with local businesses and tourism boards.

Key Resources:

- Web platform and database infrastructure.
- AI recommendation system.
- Secure payment gateway.
- Development and marketing teams.

Key Activities:

- Managing and updating listings, offers, and trip programs.
- Ensuring platform reliability, scalability, and user satisfaction.
- Promoting the platform through partnerships and digital marketing.

Key Partnerships:

- Hotels, camps, and travel agencies.
- Local tour guides and transportation agencies.
- Payment gateway providers and digital marketing partners.

Cost Structure:

- Website development and hosting.
- Marketing and advertising.
- Maintenance, staff salaries.
- Continuous improvement and AI model training.

1.14 GO TO MARKET strategy (GTM)

What is a Go-To-Market Strategy?

A go-to-market (GTM) strategy is a comprehensive plan that defines how a company introduces its product or service to the market, reaches target customers, and achieves competitive advantage.

For YOC (Your Only Chance) — an innovative tourism platform serving both Egyptian travelers and international tourists — the GTM strategy focuses on positioning the website as Egypt’s leading digital ecosystem for exploring, booking, and experiencing authentic trips, hotels, and camps.

It outlines how YOC will enter the tourism market, attract users through digital channels, build partnerships, and create a trusted, AI-driven platform that simplifies travel inside Egypt.

Executive:

YOC is an innovative platform designed to serve both local Egyptian travelers and international tourists. It provides users with an integrated experience to explore, book, and enjoy hotels, camps, and trips across Egypt. The platform combines AI-based trip customization, community engagement, and verified listings to offer a seamless travel experience. The main goal is to make traveling within Egypt easier, more interactive, and accessible for everyone — locals discovering their country and foreign visitors seeking authentic Egyptian adventures

Benefits of a Go-To-Market Strategy

A well-structured GTM strategy ensures YOC achieves efficient market entry and sustainable growth by:

- Clarifying market positioning: Defining YOC's unique place between traditional travel agencies and modern digital platforms.
- Reducing risk: Anticipating challenges such as competition, economic shifts, and user trust issues.
- Enhancing brand awareness: Building recognition across both the local Egyptian audience and international travelers.
- Improving ROI: Allocating marketing budgets smartly between digital ads, partnerships, and Public Relation campaigns.

Who Needs a Go-To-Market Strategy?

Any business introducing a new product or expanding into new markets needs a GTM strategy — especially startups in competitive industries.

For YOC, a GTM strategy is essential because it operates in Egypt's growing but crowded tourism market.

It helps the platform clearly communicate its value to:

- Local Egyptian travelers seeking affordable and unique experiences.
- International tourists looking for authentic, well-organized trips to Egypt.
- Local partners like hotels, camps, and tour guides who benefit from verified exposure.

The Components of a Go-To-Market Strategy

YOC's GTM framework includes the following components:

Market Analysis

Egypt's tourism industry is one of the most dynamic in the Middle East, attracting millions of visitors annually. While international tourists remain the primary source of revenue, domestic tourism has grown significantly as Egyptians increasingly seek affordable and unique local experiences. Current market players such as Visit Guide and Halla Travel focus mainly on hotel bookings and predefined packages, lacking personalized experiences and social interaction. This creates a strong opportunity for a hybrid platform that connects both domestic and international markets through digital innovation.

Key market opportunities

- Growing interest in cultural and eco-tourism.
- Increased use of mobile and online booking in Egypt.
- Desire for authentic, customizable travel experiences.
- Government support for digital tourism transformation.

Target Audience & Personas

- International Tourists:
 - Cultural Explorers interested in Egypt's heritage and landmarks.
 - Adventure Seekers drawn to safaris, diving, and desert camps.
- Local Egyptian Travelers:
 - Young Professionals looking for weekend getaways and affordable escapes.
 - Families seeking safe, enjoyable domestic vacations.
 - Students/Budget Travelers preferring economic group trips.

YOC stands out by merging AI-based trip personalization, verified local providers, and a social travel community all in one platform.

Unique Value Proposition (UVP)

YOC Website stands out by blending personalization, trust, and social interaction in one ecosystem. Unlike conventional travel sites, it will:

- Offer AI-based trip customization by interest, budget, and duration.
- Promote verified local operators and real reviews for credibility.
- Include a community feed where users share tips, experiences, and photos.
- secure payments.
- Social feed for experience sharing.

Marketing Goals

- Build awareness locally and globally.
- Gain 10,000+ users in the first year.
- Partner with hotels, tour operators, and eco-camps.
- Establish YOC as Egypt's most trusted travel community.

Marketing Channels & Launch Plan

Domestic (Egypt):

- Arabic social media ads (Facebook, Instagram, TikTok).
- Partnerships with local influencers and vloggers.
- Google Ads targeting "Egypt local trips."
- Collaborations with universities and youth programs.

International (Global)

- SEO optimization for keywords like “Egypt tours,” “Desert Safari Egypt,” and “Nile cruises.”
- Partnerships with travel bloggers and YouTube creators.
- Paid ads in Europe, GCC, and Asia.
- Cooperation with Egypt’s Ministry of Tourism.

Content & PR

- Storytelling campaigns with real travelers.
- Press features in Arabic and international outlets.
- Seasonal promotions for hidden Egyptian destinations

Budget & KPIs

Budget Allocation:

- 35% → Digital Advertising
- 25% → Content Creation
- 20% → Partnerships & Collaborations
- 15% → PR & Events
- 5% → Contingency

KPIs to Track:

- Number of new users (local/international)
- Website traffic and conversion rate
- Booking frequency and repeat usage
- Social feeding engagement

Risk Management

Challenges:

Potential challenges and mitigation plans:

- Competition: Focus on mixed-market positioning and community-based differentiation.
- Trust Issues: Highlight verified listings, secure payments, and transparent reviews.
- User Adoption: Provide clear guidance for all users.

Steps to Build a Go-To-Market Strategy

1. Identify the target audience – Segment local and international travelers.
2. Define YOC's value proposition Focus on trust, personalization, and community.
3. Analyze competitors Highlight YOC's AI and social engagement advantage.
4. Select marketing channels Use a mix of social, SEO, PR, and influencer marketing.
5. Develop partnerships Collaborate with hotels, camps, and government tourism bodies.
6. Set KPIs & budget allocation Track performance and ROI continuously.
7. Launch & optimize Gather feedback, analyze results, and improve campaigns dynamically.

Go-To-Market Strategy Examples

- Visit Guide: Focuses on hotel bookings but lacks personalization.
- Halla Travel: Offers local packages but no community engagement.
- YOC Combines both worlds through AI-driven customization, verified operators, and user-generated content representing the next evolution of Egypt's travel tech industry.
- Go-To-Market Strategy FAQs

Who is responsible for the GTM strategy?

For YOC, the GTM strategy is a shared responsibility between the Marketing Team, the Product Development Team, and the Partnerships Department. The Marketing Team handles audience targeting, campaigns, and PR. The Product Team ensures the platform delivers on the brand promise smooth booking, verified listings, and AI customization. The Partnerships Team builds collaborations with hotels, camps, and the Ministry of Tourism to expand reach and trust.

What is the difference between a product-based and customer-based go-to-market strategy?

A product-based GTM strategy focuses on introducing the features and technical aspects of the platform for example, YOC's AI-powered trip planner or bilingual interface. A customer-based GTM strategy, on the other hand, focuses on solving user problems and creating emotional connection like helping young Egyptians discover affordable adventures or making it easier for tourists to explore authentic Egypt. YOC combines both approaches: showcasing innovative technology and delivering personalized, human-centered travel experiences. Next Steps: Create a GTM strategy for your next big thing After YOC successfully establishes its brand and user base in Egypt, the next logical step is to scale the GTM model for expansion. Potential next moves include: Launching a mobile app version for iOS and Android.

- . Why does YOC need a GTM strategy?
Because the Egyptian travel market is competitive YOC needs a clear plan to stand out with innovation and trust.
- Q2. How is YOC different from existing platforms?
It merges AI-based personalization, verified listings, and a social community, unlike competitors that only offer standard bookings.
- Q3. What's YOC's biggest advantage?
Its dual-market focus (local + international) and smart customization features.
- Q4. What metrics will define success?
User acquisition, retention rate, booking value, and engagement across the community feed.
- Q5. What's the long-term vision?
To evolve from a booking website into a digital tourism ecosystem that supports sustainable travel and local Egyptian businesses.

1.15 Work plan

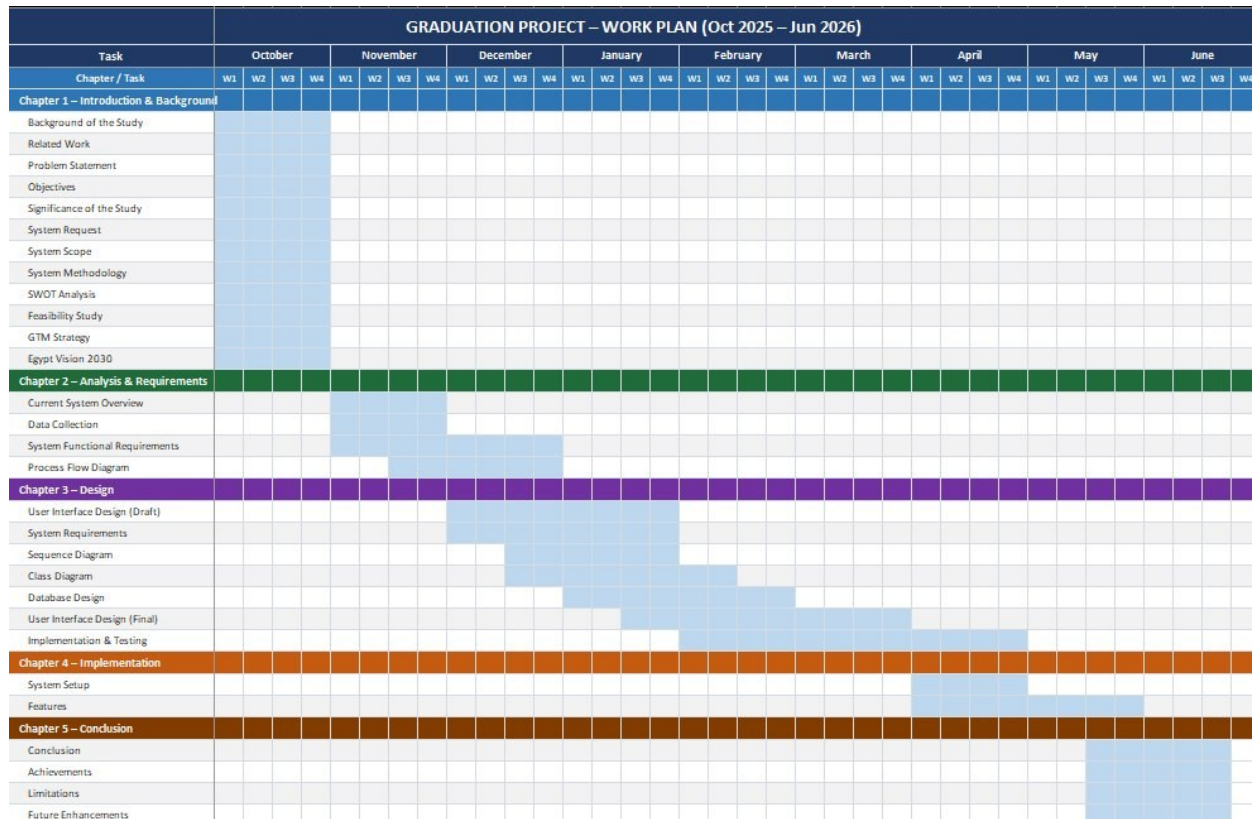


Figure 1.4 work plan

chapter 2

system analysis

2.1 Current System Overview

2.1.1 System Overview

The current system represents the initial implementation of the Your Only Chance (YOC) tourism web platform. It is a web-based system designed to support tourism activities inside Egypt by providing users with a centralized digital environment to explore destinations, discover travel options, and complete bookings online.

The platform targets both local and international travelers and focuses on simplifying the trip planning process by combining exploration, recommendation, and booking services into one unified system.

2.1.2 Current System Functionality

The current implementation of the YOC system provides the following core functionalities:

- **User Registration and Authentication**
Users can create personal accounts, log in securely, and manage their profiles.
- **Destination and Service Browsing**
Users can explore destinations, hotels, camps, and guided trips with detailed information such as pricing, ratings, and locations.
- **Search and Filtering**
The system supports searching and filtering travel options based on destination type, price range, and ratings.
- **Booking Management**
Users can book hotels, camps, and trips directly through the platform. The system verifies availability, calculates total cost, and confirms bookings.
- **Online Payment**
Secure online payment is supported through trusted payment gateways such as Visa, MasterCard, and InstaPay.

- **AI Planner**
The system includes an AI Trip Planner that suggests personalized trips and destinations based on user preferences such as budget, interests, and location. It generates recommendations using the database to help users plan suitable trips and explore relevant places.
- **Trip Suggestions**
users can submit trip suggestions through a form including details such as destination, price, pictures, and other relevant information then the admin reviews each suggested trip approves and publishes it on the website or rejects it.
- **Community Interaction**
Users can share posts, reviews, and travel experiences, contributing to a community-driven tourism environment.
- **Chatbot Assistance**
A chatbot is available to providing conversational travel support, destination recommendations, and cultural answers. Powered by an external AI API, it contextually guides users toward relevant exploration and booking modules.

2.1.3 System Users

The current system supports multiple user roles, including:

- **Travelers (Users):** Browse destinations, book services, and interact with the community.
- **Administrators:** Manage users, content, bookings, and validate platform data.

Current Technology Stack

The YOC platform is implemented using the following technologies:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL

This technology stack ensures compatibility with web browsers and supports future scalability.

2.1.4 Current System Limitations

Although the current implementation provides essential tourism services, it has several limitations:

- The system is limited to a web-based platform and does not currently include native mobile applications.
- Administrative processes such as content validation and booking approvals rely on partial manual intervention.
- Advanced analytics and decision-support dashboards are not fully implemented in the current version.

These limitations highlight opportunities for future system enhancement and expansion

2.2 Data Collection

2.2.1 Survey Responses

1-What is your Gender?

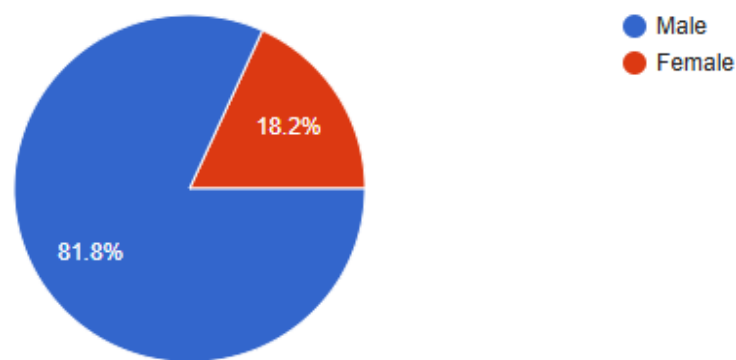


Figure 2.1 Gender Distribution

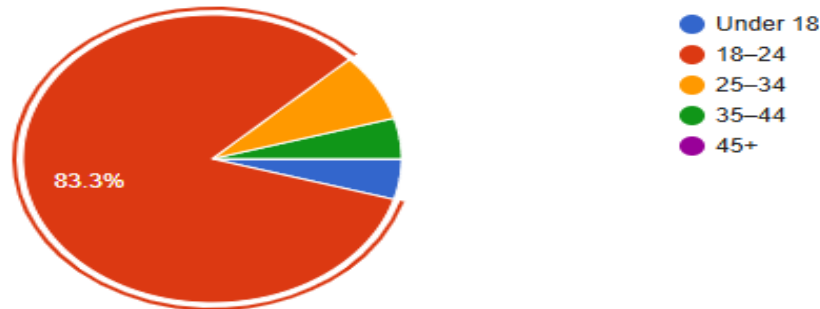
Most users belong to the 18–24 age group (83.3%), with very small percentages in other age ranges.

Insight: The system mainly targets young users, especially university-age people.

2- What is your age group?

Figure 2.2 Age Group distribution

Most users belong to the 18–24 age group (83.3%), with very small percentages in other age



ranges.

Insight: The system mainly targets young users, especially university-age people.

3-Have you traveled within Egypt in the past year?

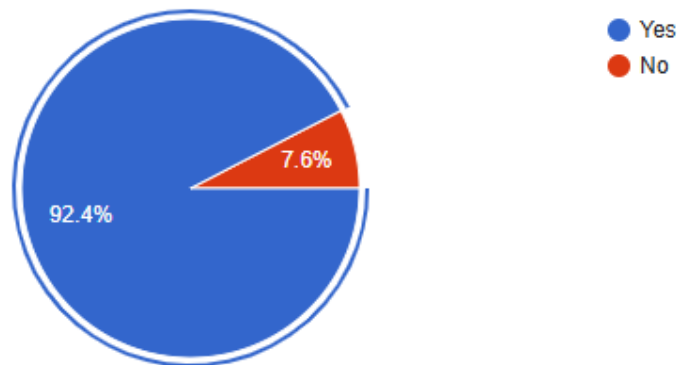


Figure 2.3 travel Frequency within Egypt

The majority of users (92.4%) have traveled within Egypt, while only 7.6% have not.

Insight: Domestic tourism is very common among users.

4-How many times do you travel inside Egypt per year?

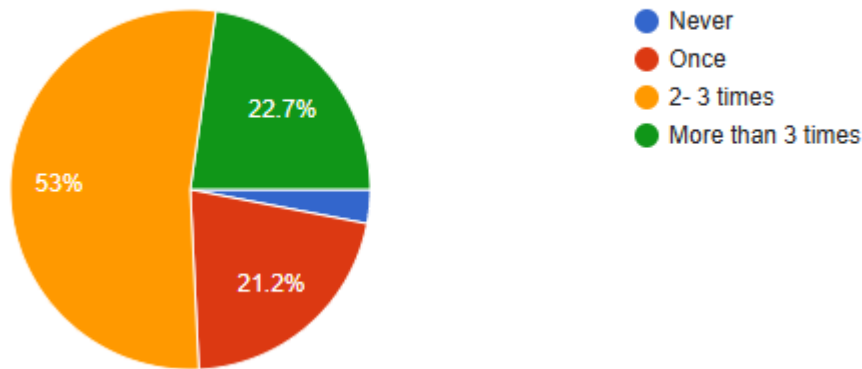


Figure 2.4 Preferred Destination Types

Most users travel 2–3 times per year (53%), followed by more than 3 times (22.7%).

Insight: Users are generally active travellers, not occasional ones.

5-What type of destinations do you usually travel to?

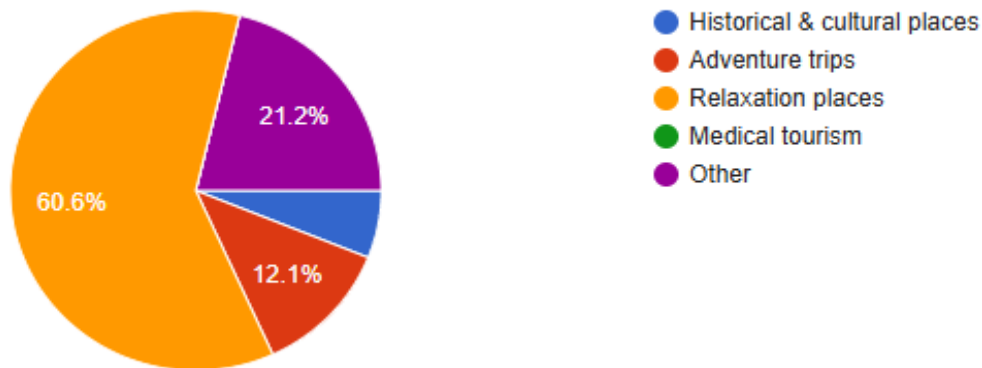


Figure 2.5 Methods of Searching for Destinations

Most users prefer relaxation places (60.6%), followed by medical tourism (21.2%) and adventure trips (12.1%).

Insight: Relaxation tourism is the most dominant preference.

6-How do you usually search for travel destinations?

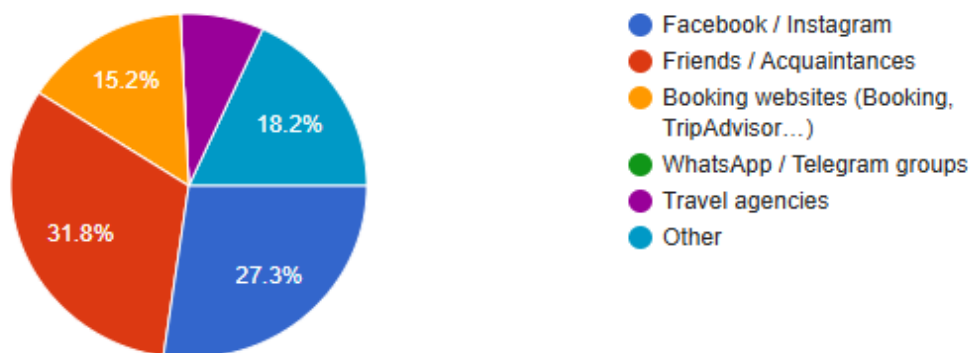


Figure 2.6 Methods of Searching for Destinations

Most users rely on friends/relatives (31.8%) and social media (27.3%), followed by booking websites (18.2%).

Insight: Word-of-mouth and social platforms are more trusted than official platforms.

7-Have you faced any problems while searching for trips or tourist places?

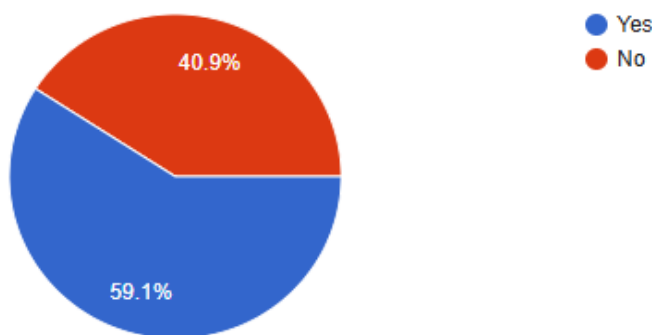


Figure 2.7: problems while search trip

Most users faced lack of information (51.5%) and trust issues with companies (43.9%).

Insight: The biggest problem is insufficient and unreliable travel information.

8-Which of these problems have you faced the most?

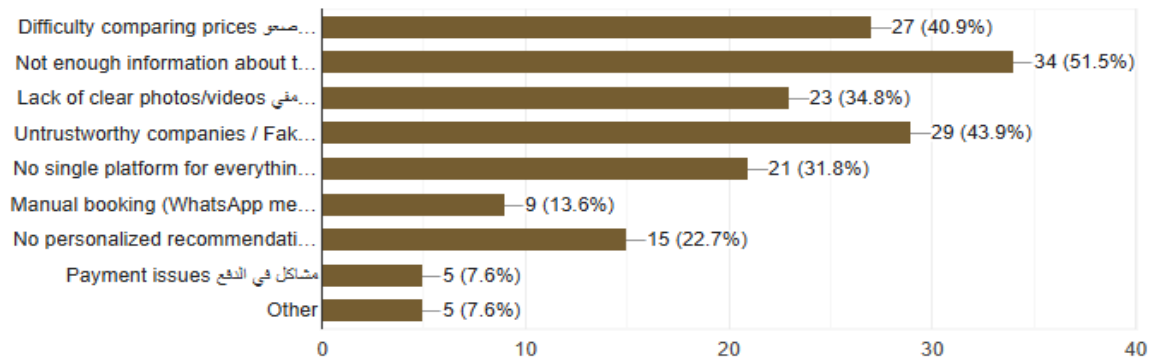


Figure 2.8 Problems Faced During Trip Planning

Most users complain about lack of information, trust issues, and difficulty comparing prices.

Insight : Users need a platform that is clear, trustworthy, and easy to compare options.

9-How satisfied are you with current platforms?

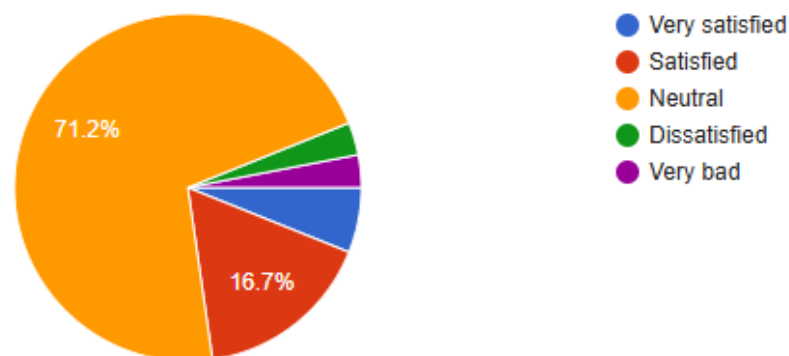


Figure 2.9 how satisfied with current platforms

- Most users feel neutral, and only a small number are satisfied.
- Insight: Current platforms are not good enough and need improvement.

10- Would you like an Egyptian platform that combines (Hotels + Camps + Trips + Guides) in one system?

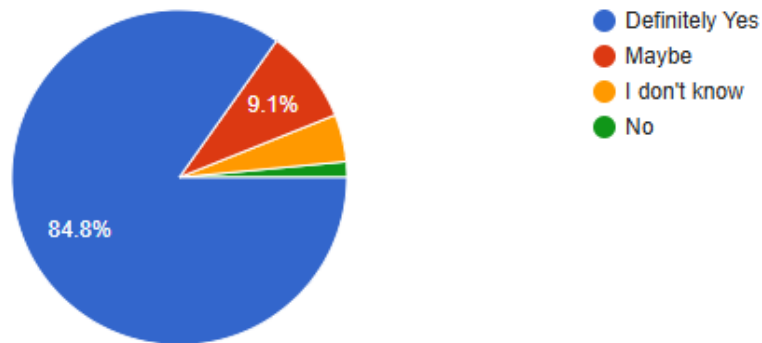


Figure 2.10 would you like an integrated platform

Most users said “Definitely Yes” to having one platform for everything.

Insight: There is a strong demand for an all-in-one Egyptian platform.

11- Is it important for you that the platform uses AI to recommend trips based on your interests?

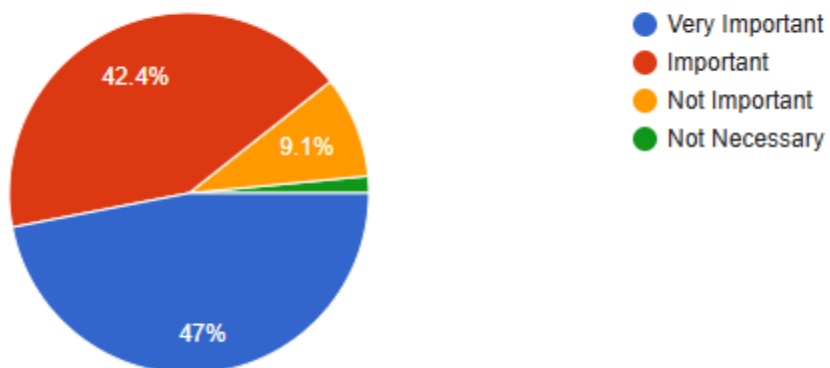


Figure 2.11 important for the platform uses AI to recommend trips

Most users think AI is very important or important.

Insight: AI recommendations are a key feature users expect.

12- What is the most critical feature you look for in travel platform?

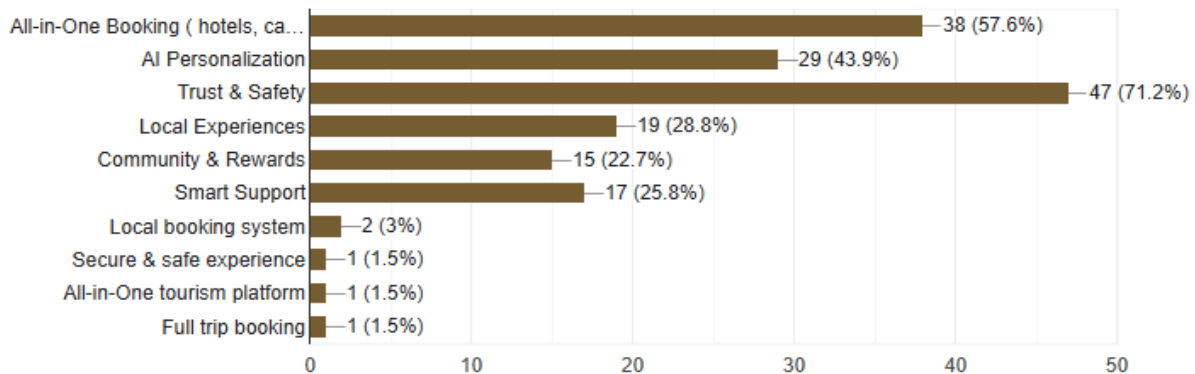


Figure 2.12 critical feature for travel platforms

- Top feature is Trust & Safety, then All-in-one booking and AI personalization.
- Insight: Users care most about feeling safe and having everything in one place.

13- Which features would encourage you to try a new Egyptian tourism platform?

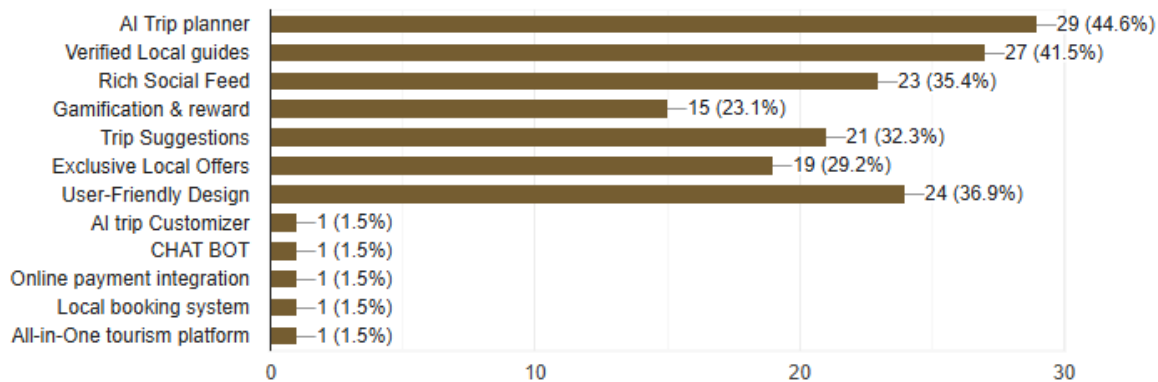


Figure 2.13 what encourage people to try Egyptian tourism platform

- Top features are AI trip planner, verified local guides, and easy design.
Insight: Users are attracted to smart tools + trusted services.

14- What makes you trust a booking website?

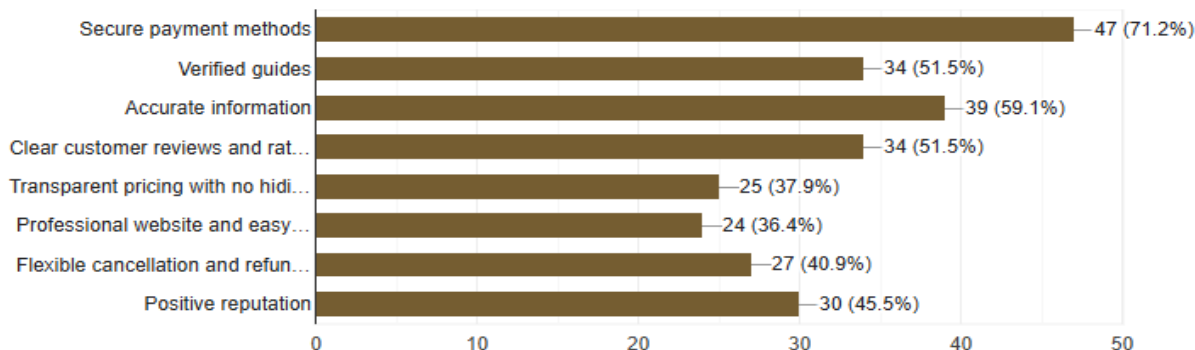


Figure 2.14 what makes people trust booking websites

Main factors are secure payment, accurate info, and verified guides/reviews.

Insight: Trust comes from security + transparency.

15- Do you prefer that the system suggests a ready-made itinerary based on (mood + budget + city + interests)?

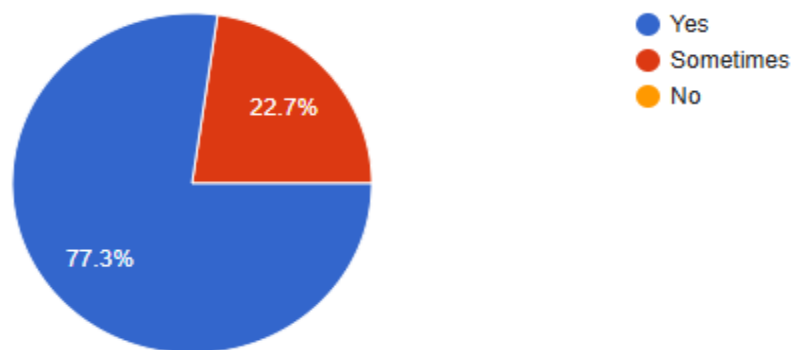


Figure 2.15 does people prefer suggests bases system

Most users said Yes.

Insight: Users like quick and easy planning.

16- Are you comfortable paying online at a trusted website?

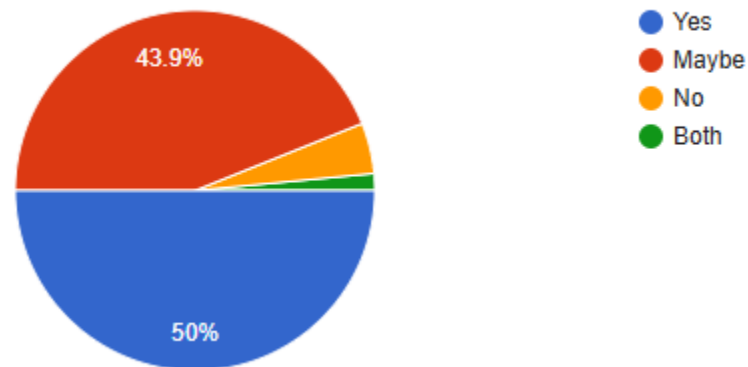


Figure 2.16 Trust to pay online on platforms

Most users are comfortable paying online.

Insight: Online payment is not a big barrier anymore.

17- Do you prefer to book through:

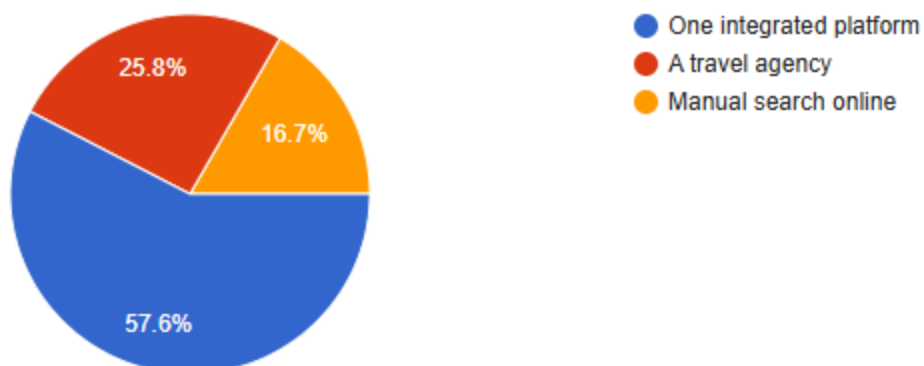


Figure 2.17 what would travelers to book through

Most prefer one integrated platform over agencies or manual search.

Insight: Users want everything in one place.

18- Would you like to share your experiences and photos in the community?

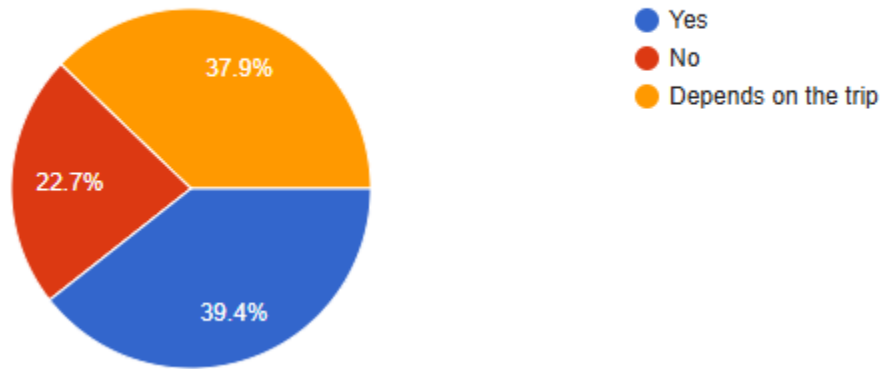


Figure 2.18 would like to share experiences in community

Users are divided, but many said Yes or depends.
Insight: Community features are useful but not essential.

2.3 Proposed system

YOC, is a fully integrated tourism web platform designed to modernize and streamline the travel experience in Egypt. The system provides a unified digital environment where users can explore destinations, receive personalized recommendations through AI, preview locations using photos and videos, and book hotels, camps, and guided tours—all within a single platform.

Unlike existing solutions, which are fragmented and limited in functionality, YOC combines multiple essential travel services into one cohesive system. It focuses on delivering a personalized, efficient, and trustworthy experience for both local and international travelers.

The system allows users to create personal accounts, set their travel preferences, and receive AI-generated itineraries tailored to their interests, budget, and travel mood. Users can also interact with a community space where they can share experiences, read reviews, and suggest trip programs. In addition, YOC supports secure online payments, booking management, and access to verified Egyptian tour guides to ensure credibility and safety.

An administrative dashboard is included for managing destinations, bookings, users, reviews, and partner submissions. This ensures that all information displayed to users is reliable, up-to-date, and validated by the system administrators.

By integrating personalization, online booking, community engagement, and AI-driven recommendations into one platform, the proposed system aims to overcome the major limitations of the current tourism environment in Egypt. YOC enhances decision-making, improves user trust, reduces manual communication, and promotes local tourism through a modern, user-friendly, and intelligent digital solution.

2.4 Functional requirements

User Registration

- The system shall allow users to create an account using email and password.
- The system shall validate all required fields before creating an account.

User Login

- The system shall authenticate users using valid email and password.
- The system shall prevent unauthorized access by rejecting invalid data.

Password Reset

- *The system shall provide a password reset flow via email token.*

Profile Management

- The system shall allow users to update personal information (name, phone, profile picture).
- The system shall allow users to view and update travel preferences.
- The system shall allow users to view their booking history and saved destinations

Destination Listings

- The system shall display a list of destinations, hotels, camps, and guided trips.
- The system shall show essential details for each destination including price, rating, and location.

Search Function

- The system shall allow users to search for destinations by name or type
- The system shall suggest popular keywords during search.

Sorting

- The system shall allow users to sort results by:
 - Price range
 - Rating

Filters

- The system shall allow users to filter results by:
 - Destination
 - Price range
 - Rating
 - category
- The system shall allow users to sort results by price, rating, and popularity.

Booking Creation

- The system shall allow users to book hotels, camps, or guided tours.
- The system shall check availability before confirming the booking.
- The system shall calculate the total cost (taxes, number of people, duration).

Payment

- The system shall allow users to pay through:
 - Visa
 - MasterCard
 - PayPal
- The system shall display a confirmation message after successful payment.
- The system shall record the transaction reference.

Booking History

- The system shall allow users to view all previous and current bookings.
- The system shall allow users to cancel bookings according to provider policies.

AI Recommendation System

Preferences Input

- The system shall allow users to input:
 - Budget
 - Category
 - Destination
 - Trip duration

Trip suggestions

- Users can suggest custom trip plans.
- Suggested trips are reviewed by admins and, if approved, displayed on the platform.

Chatbot Assistance

- The system shall provide an AI chatbot for:
 - Booking support
 - Destination information
 - Travel tips
 - Frequently asked questions

Save AI Plan

- authenticated users can save generated itineraries for later review.

User Posts

- The system shall allow users to publish posts including photos and travel stories.
- The system shall allow users to react and comment on posts.

Ratings

- The system shall allow users to rate hotels, camps, and guides.
- The system shall display average rating.

User Management

- The system shall allow admins to add, delete, and update user accounts.

Content Management

- Admins shall be able to add/edit/delete:
 - Destinations
 - Hotels
 - Camps
 - Trips
 - Photos/videos
 - Guide profiles

Booking Management

- Admins shall monitor, update, and verify bookings and payments.

Payment Gateway Integration

- The system shall integrate with secure and verified payment gateways.
- The system shall encrypt payment data.
- The system shall notify users of payment success or failure

2.5 System Analysis Tools

2.5.1 Use case diagram

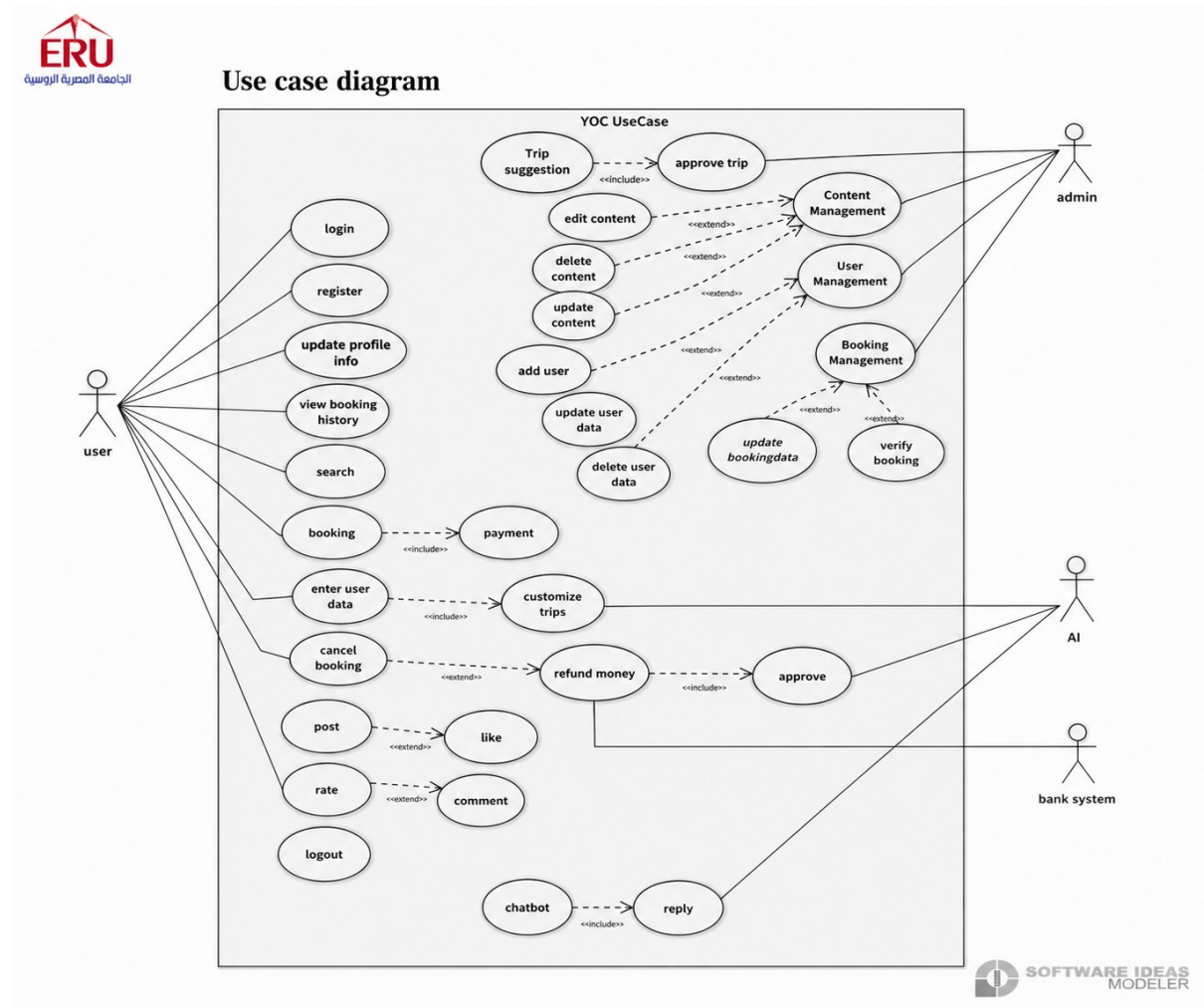


Figure 2.19 use case diagram

2.5.2 Use case description

Table 2.1 Use Case Description

Use case name: Registration	Id: 1		Level: High
Primary actor: User		Use case type: Essential	
Stakeholders and interests: • User: Wants to create an account quickly and securely to start booking and interacting. • System / Platform: Needs accurate user data, prevent duplicating accounts, verify identity, and maintain data integrity. • Admin: Wants valid user records and ability to moderate suspicious sign-ups.			
Brief description: A new user provides required information (name, email, password, phone). The system validates input, checks for duplicates, creates a user record, and (optionally) triggers an email verification step. After successful registration, the user can log in and use platform features. This aligns with the project’s Use Cases document.			
Pre-condition: • User does not already have an account with the same email/username. • System is available (internet + backend up).			
Post-condition: • New user record created in the database (status = Pending or Active depending on verification).			
Trigger: User clicks the “Register / Sign up” button after filling the registration form. Type: Primary — User-initiated			
Relationship: • Include: None (no required internal use case to complete registration). • Extend: Login (user will typically log in after confirming email or immediately if auto-activated). • Association: User • Generalization: None			
Normal flow of events: 1. User opens the registration page/form. 2. User fills required fields (e.g., full name, email, password, confirm password). 3. User submits the form. 4. System validates input format (email, password rules) and checks for existing account with same email. 5. System creates a new user record (status = Active or Pending). 6. If verification is required, system sends a verification email/SMS. 7. System displays success message (“Registration successful — please verify your email” or “Account created — you can now log in”). 8. User proceeds to Login (or completes verification then logs in).			
Sub flows / business rules: • Password must meet complex rules (min length, mix of characters). • Optional: captcha or bot check to prevent automated signups. • Optional fields: phone number (may require SMS OTP), profile preferences. • If the system uses email confirmation → account remains Pending until token confirmation or token expiry.			

Use case name: login	Id: 2		level: high
Primary actor: user		use case type: essential	
- Stakeholder and interests: - User: Wants to access their personal account and system features securely. - System: Ensures the user is authenticated and protects the platform from unauthorized access.			
Brief description: The user enters their login credentials (email/username and password). The system verifies the data and logs the user into their account to access available services such as booking, updating profile information, and interacting with platform content.			
- Pre-condition: The user must already be registered in the system. - The system must be online and accessible.			
- Post condition: The user is successfully authenticated. - A valid session is created for the user.			
Trigger: The user clicks the “Login” button. Type: Primary – User-initiated			
Relationship: Extend: Logout (Logout can only happen after a successful login) Association: User Include: None Extend: Logout Generalization: None			
- Normal flow of events: - The user opens the login page. - The user enters their email/username. - The user enters their password. - The system validates the provided credentials. - If valid, the system creates a new session. - The user is redirected to the home/dashboard page.			
- Sub flow: The system checks if the email format is valid. - The system verifies that the account is active and not blocked.			
Alternate flow: A1 – Invalid password: - The system detects incorrect credentials. - An error message “Invalid email or password” is displayed. The user re-enters correct credentials.			

Use case name: Logout	Id: 3	Level: High
Primary actor: User	Use case type: Essential	
Stakeholders and interests: - User: Wants to securely end their session and protect their account on shared devices. - System: Needs to terminate the session, clear sensitive caches/tokens, and prevent unauthorized access. - Admin/Security team: Ensure sessions are ended properly for auditing/security.		
Brief description: User requests to end their active session. The system invalidates the session token(s), clears any client-side session artifacts (cookies/local storage), and redirects the user to the public landing or login page.		
Pre-condition: User is already authenticated (logged in) and has an active session.		
Post-condition: - The user session is terminated and cannot be reused. - Sensitive session data/tokens removed from server and client as applicable. - User is redirected to the login/home page.		
Trigger: User clicks the “Logout” button or link (or session expires and system triggers server-side logout). Type: Primary — User-initiated (also system-initiated for session expiration)		
Relationship: - Extend: Logout extends Login (Logout depends on an active Login/session). - Include: None. - Association: User - Generalization: None		
Normal flow of events: - The user clicks “Logout” (or chooses “Sign out”). - The client clears local session data (cookies, local Storage/session Storage). - The system redirects the user to the login or public landing page and shows confirmation (e.g., “You have been logged out”).		
Sub flows / business rules: - If “Remember me” was enabled, clear persistent login tokens. - If user has multiple active sessions/devices, optionally offer “Log out from all devices” as an extra action. - Log the logout event for auditing (timestamp, user id, IP).		
A1 — User decides to log out from inside the app - The user opens the menu/profile section. - The user selects “Logout” from the available options. - System shows a confirmation message: “Are you sure you want to log out?” - User confirms. - System terminates the active session and redirects the user to the login page.		

Use case name: Update Info	Id:4	Level: High
Primary actor: User	Use case type: Essential	
Stakeholders and interests: - User: Wants to edit personal information to keep profile updated and accurate. - System: Needs to ensure data validity, prevent duplicates (email/phone), and maintain consistent records. - Admin: Gains accurate user data for support and analytics.		
Brief description: The user updates their personal profile information such as name, email, phone number, or other optional details. The system validates the inputs and saves the updated data into the database.		
Pre-condition: - User must be logged in. - User profile exists in the system.		
Post-condition: - User information is successfully updated in the database. - System logs the update for auditing.		
Trigger: User selects “Edit Profile” or “Update Info” from the account settings. Type: Primary — User-initiated		
Relationship: - Include: None - Extend: None - Association: User - Generalization: None		
Normal Flow of Events: - User opens the profile settings page. - User selects “Edit” to modify personal details. - User updates fields (e.g., name, email, phone, profile picture). - User clicks “Save Changes”. - System validates all inputs. - System updates user data in the database. - System displays a success message: “Your information has been updated successfully.”		
Sub Flows: - SF1: Validate email format System checks that the email follows correct formatting. - SF2: Validate phone number System ensures phone number is numeric and follows country format. - SF3: Duplicate email check System verifies that the new email is not already used by another account.		
Alternate Flows: A1 — Invalid or incomplete data - User enters invalid data (wrong email format, empty required fields). - System highlights wrong fields and shows an error message. - User corrects data and resubmits. A2 — Email already exists - User enters an email used by another account. - System shows: “Email already in use.”		

Use case name: Change Password	Id: 5	Level: High
Primary actor: User	Use case type: Essential (user-initiated account management)	
Stakeholders and interests: - User: Wants to securely update their password (routine or after suspicion of compromise). - System / Security team: Must ensure password strength, prevent brute-force, and protect account integrity. - Admin: Wants audit trail for security incidents and optionally the ability to force a reset.		
Brief description: An authenticated user requests to change their account password. The system verifies the user's current password (or uses verification via email/SMS if required), validates the new password against complexity rules, updates the credential store (hashing the new password), invalidates existing sessions/tokens if applicable, and confirms the change to the user.		
Pre-condition: - The user is authenticated (logged in) OR follows a verified password-recovery flow (if changing via “forgot password”). - The system is online and reachable.		
Post-condition: - The user's password is updated in the database (stored hashed). - Existing sessions may be invalidated depending on policy. - User receives confirmation (on-screen and/or email).		
Trigger: User selects “Change Password” from account settings or clicks a “Reset password” link (different flow for forgot password). Type: Primary — User-initiated (also supports system/admin-initiated resets via forgot-password or forced reset)		
Relationship: - Include: None (the flow may use an authentication sub-process but not a separate include use case). - Extend: None - Association: User - Generalization: None		
Normal Flow (Main Success Scenario) — simplified - User navigates to Account Settings → Change Password. - System prompts the user to enter current password, new password, and confirm new password. - System verifies that the current password is correct. - System checks that the new password meets complexity rules and is different from the old password. - System updates the password in the database (hashed). - System shows a success message: “Password changed successfully.”		
Sub Flows — only the important ones SF1 — Password complexity check System checks the new password meets minimum requirements. SF2 — Current password verification System compares current password with stored hash.		
Alternate Flows A1 — Incorrect current password - System rejects the request and shows: “Current password is incorrect.” - User tries again. A2 — Weak or invalid new password - Svstem detects weak passwords or passwords identical to old ones.		

- System shows inline error. - User enters a stronger password.
A3 — Password confirmation mismatch
- System shows: “Passwords do not match.” - User corrects and resubmits.

Use case name: Search	Id: 6		Level: High
Primary actor: User		Use case type: Essential	
Stakeholders & interests: - User: Wants fast, relevant results for trips, hotels, camps, or other content. - System: Provide accurate, filtered results with good performance and UX. - Admin/Product: Wants analytics on popular queries and ability to tune results.			
Brief description: User enters keywords and optional filters (dates, location, price, type). The system executes a search, ranks results, and displays matching items. User can open details, refine filters, or start booking from results.			
Pre-condition: - System search index/service is available. - At least one searchable item exists (trips, hotels, camps).			
Post-condition: - Search results are displayed to the user. - User can proceed to view details or start booking.			
Trigger: User enters a query and presses search or uses quick filters/autocomplete.			
Relationship: - Include: None (Search is a standalone use case but may call internal ranking/search services). - Extend: None - Association: User - Generalization: None			
Normal flow (main success scenario) - User opens the search interface or homepage search box. - User types keywords and selects optional filters (location, date range, price, category). - System retrieves and ranks matching items (relevance + business rules). - System returns paginated results with highlights/snippets. - User reviews results and either: - Opens an item to view details			
Sub flows / Business rules (key ones) - SF1 — Autocomplete & suggestions: provide instant suggestions and recent queries. - SF2 — Filters & sorting: support filters (date, price, rating) and sorts (recommended, price low high, newest). - SF4 — Availability check: hide or mark out-of-stock/unavailable trips.			
Alternate flows: A1 — No results found - System returns zero matches. - Show friendly message: “No results found.” - Offer suggestions: relax filters, check spelling, try nearby dates/locations, or show popular alternatives. A3 — Invalid query / input - User submits empty or malformed query. - Svstem shows hint: “Please enter search terms” or auto-suggests queries.			

Use case name: View Booking History	Id: 7		Level: High
Primary actor: User		Use case type: Essential	
Stakeholders & interests: - User: Wants to view previous and upcoming bookings clearly. - System: Must retrieve accurate booking records and display them efficiently. - Admin: Uses this data for support inquiries and audits.			
Brief description: The user opens the booking history page to view all previous, current, or upcoming bookings. The system retrieves the data from the database and displays it in an organized list.			
Pre-condition: - User is logged in. - User has at least one booking (or the system handles empty state).			
Post-condition: - Booking history is displayed to the user. - User can choose to open booking details or cancel eligible bookings.			
Trigger: User selects “Booking History” from the menu or profile section.			
Relationship: - Include: None - Extend: None - Association: User - Generalization: None			
Normal Flow - User navigates to Booking History. - System fetches all bookings belonging to the user. - System organizes results into sections (Past / Upcoming / Canceled). - System displays booking items with key details (date, status, price, trip name). - User may click a booking to view full details.			
Sub Flows (Business Rules) SF1 — Sorting: System sorts of bookings by newest → oldest or groups by status. SF2 — Pagination / Load more: If user has many bookings, system loads results in pages. SF3 — Status labels: System shows labels such as Confirmed, Completed, Canceled, Pending Approval, etc.			
Alternate Flows (Only the important ones) A1 — User has no booking history - System finds no records. - Shows message: “You have no bookings yet.” - Suggestions browsing available trips. A2 — System fails to fetch booking data - Database/server error occurs. - System shows: “Unable to load your booking history. Please try again.” - Users stay on the same page. A3 — User filters history to a range with no results - User selects date range, category, or status filter. - System returns zero results. - System shows: “No bookings match your filters.”			

Use case name: Booking	Id: 8		Level: High
Primary actor: User		Use case type: Essential	
Stakeholders & interests: - User: Reserve a trip quickly and securely. - System: Ensure availability, handle payments, and store booking records. - Admin: Manage bookings and handle exceptions. - Bank/Payment Gateway: Process transactions.			
Brief description: User selects a trip and completes the booking process. The system collects traveler information, processes payment, verifies availability, approves the booking, and returns a confirmation/receipt.			
Pre-condition: - User is logged in (or provides required identity info). - The selected trip has availability. - Payment gateway is reachable.			
Post-condition: - A booking record is created (status = Confirmed/Approved). - User receives confirmation (on-screen and optionally email/SMS). - Inventory/seats updated.			
Trigger: User clicks “Book Now” on a trip detail page.			
Relationship: - Include: Enter User Data , Payment , Approve - Extend: None - Association: User			
Normal flow (main success scenario) - User clicks Book Now on a trip. - System opens booking form (dates, travelers, extras). - Include: Enter User Data — user fills traveler information and contact details. - System validates entered data and checks availability. - Include: Payment — user selects payment method and completes payment. - Payment gateway returns success. - Include: Approve — system finalizes booking, marks it Approved/Confirmed. - System saves booking record and updates availability. - System shows booking confirmation and sends receipt/notification.			
Sub flows - SF1 — Availability check: Realtime seats. - SF2 — Price calculation: base price + taxes + fees + discounts/promo codes. - SF3 — Input validation: traveler names, contact info. - SF4 — Idempotency: prevent duplicate bookings on retries. - SF5 — Notification: send email/SMS confirmation with booking details and cancellation policy.			
Alternate flows (important) A1 — Trip just became unavailable - System detects no availability during validation. - System informs user: “This trip is no longer available.” A2 — Invalid or incomplete user data - Validation fails (missing name, invalid phone). A3 — Payment failed - Payment gateway returns failure. - System shows: “Payment failed — try another method.” - User retrieves payment or cancels booking. A4 — User cancels before payment - User abandons or cancels the form before payment. - System discards temporary booking and releases any inventory held.			

Use case name: Payment	Id: 9	Level: Low (Included in Booking)
Primary actor: User - User: Wants to complete payment securely and confirm the booking. - System: Needs to verify payment and record the status. - Bank System: Processes the transaction and returns final payment status. - Admin: Needs accurate transaction records for refunds or support.		Use case type: Essential
Stakeholders and interests User: Wants to complete payment securely and confirm the booking.		
Brief description The user enters or selects a payment method to complete the booking. The system sends a payment request to the Bank System, receives the response, and updates the payment status accordingly.		
Pre-condition - User has completed Enter User Data. - Booking details and total price are calculated. - Trip is still available. - Payment gateway is reachable.		
Post-condition - Payment status saved as Success or Failed. - On success → booking continues to Approved. - On failure → user is asked to retry or choose another method.		
Trigger User clicks “Proceed to Payment” in the booking process. Type Primary — User initiated		
Relationship - Included by: Booking - Include: None - Extend: None - Association: User, Bank System		
Normal flow of events : User chooses a payment method and enters required payment details.		
Sub flows - SF1 — Payment validation: system checks required fields (card details/wallet ID). - SF2 — Security: system must not store raw card data.		
A1 — User cancels payment - User closes the payment window or presses cancel. - System marks payment as Canceled and returns the user to the previous screen.		

Use case name: Approved	Id:10		Level: Low (included in Booking)
Primary actor: System (Booking subsystem) Secondary actor: Bank System / Payment Gateway		Use case type: Essential (system-initiated)	
Stakeholders and interests - User: Wants confirmed booking after successful payment and checks. - System: Must finalize booking, ensure availability and data integrity. - Admin: Needs reliable status for support and reconciliation. - Bank System: Confirms payment outcome.			
Brief description After a successful payment and final checks, the system finalizes the booking by setting its status to Approved (or flags it for manual review). The system updates inventory and notifies the user.			
Pre-condition - Payment completed successfully - Enter User Data completed and validated - Booking record exists and initial availability check passed.			
Post-condition - Booking status updated to Approved / Confirmed. - Inventory/seats decremented accordingly. - Confirmation notification (email/in-app/SMS) sent to the user. - Approval event logged for audit.			
Trigger Payment success callback or internal event indicating payment & validations passed. Type System-initiated (automated internal step)			
Relationship - Included by: Booking (UC-08) - Include: None - Extend: None - Association: System, Bank System			
Normal flow of events (main success scenario) - System receives payment success callback from Payment (UC-10). - System re-checks real-time availability for the booked slots. - System validates payment transaction details (amount, transaction id) against booking. - System applies business rules (pricing finalization, promos, fraud checks). - System updates booking status to Approved and decrements inventory. - System logs the approval event and sends confirmation notification to the user.			
Sub flows / business rules - SF1 — Final availability check: ensure no overbooking before finalizing. - SF2 — Fraud & validation checks: run lightweight anti-fraud heuristics; flag suspicious cases. - SF3 — Idempotency: handle duplicate callbacks safely (one approval only). - SF4 — Notification content: include booking reference, dates, cancellation policy, contact info. - SF5 — Accounting hooks: trigger downstream accounting or reporting processes if needed.			
Alternate flows (errors & edge cases) A1 — Inventory conflict detected after payment - Final availability check fails (no slots left). - System sets booking to Pending and notifies user with alternatives. - If payment already captured → trigger Refund Money or queue admin action.			

Use case name: Cancel Booking	Id:11		Level: High
Primary actor: User		Use case type: Essential	
Stakeholders and interests • User: Wants the ability to cancel a booking and (if eligible) receive a refund. • System: Must update booking status accurately and trigger refund processes when needed. • Admin: Manages exceptions and handles policy disputes. • Bank System: Processes refunds when required.			
Brief description The user requests to cancel a completed booking. The system verifies eligibility according to cancellation policies, updates the booking status to Canceled, and if applicable, triggers Refund Money automatically.			
Pre-condition • User is logged in. • Booking exists in the system. • Booking is eligible to be canceled (before the cutoff time or allowed by admin).			
Post-condition • Booking status updated to Canceled. • Refund process triggered (Refund Money UC-12) if applicable. • System notifies the user of the cancellation outcome.			
Trigger User clicks “Cancel Booking” from booking details or booking history.			
Type Primary — User initiated			
Relationship • Extend: Refund Money (UC-12) • Include: None • Association: User			
Normal flow of events 1. User navigates to booking details and selects Cancel Booking. 2. System verifies that the booking can still be canceled based on policy and timing. 3. System updates booking status to Canceled. 4. If booking is eligible for refund → system triggers Refund Money (UC-12). 5. System displays confirmation message to the user and updates booking history.			
Sub flows / business rules • SF1 — Policy check: cancellation allowed only before cutoff time (e.g., 24 hours before trip). • SF2 — Refund eligibility: full, partial, or no refund based on policy. • SF3 — Automatically trigger refund: Refund Money use case starts if refund applies. • SF4 — Logging: cancellation action logged for auditing and dispute resolution.			
A1— Forced cancellation by admin 1. Admin cancels the booking on behalf of user (system or admin initiated). 2. System marks booking as Canceled. 3. Refund (if applicable) is triggered automatically or manually reviewed.			

Use case name : refund money	Id:12	Level: low
Primary actor: User Secondary actor: Bank System	Use case type: Essential	
Stakeholders and interests - User: get refunded according to policy. - System: compute and execute refunds reliably. - Bank System: process refund transactions. - Admin/Support: handle disputes and manual refunds.		
Brief description After a booking is canceled and eligible, the system computes refundable amount, initiates refund with bank/gateway, updates records, and notifies the user.		
Pre-condition - Booking canceled (Cancel Booking done). - Original payment captured and recorded. - Refund eligibility confirmed by policy or admin.		
Pre-condition - Booking canceled (Cancel Booking done). - Original payment captured and recorded. - Refund eligibility confirmed by policy or admin.		
Trigger User requests refund after cancellation OR system/admin initiates refund. Type Primary — User/System initiated		
Relationship - Extends: Cancel Booking (UC-11) - Association: User, Bank System		
Normal flow - System receives refund trigger. - System verifies payment and computes refundable amount (apply fees/taxes). - System creates refund record and sends refund request to Bank System.		
Sub flows / business rules - SF1 — Refund calculation: apply fees, taxes, promos, currency conversion. - SF2 — Refund method: try original payment method; fallback to store credit.		
Alternate flows: A1 — Partial refund: compute partial amount and process.		

Use case name: Post	Id: 13	Level: High
Primary actor: User	Use case type: Essential	
Stakeholders and interests - User: Wants to share posts in the community (text/media). - System: Needs to store posts, moderate content, and display it to other users. - Admin: Reviews inappropriate posts and handles reports.		
Brief description The user creates a new post within the platform’s community section. The system validates the content, saves the post, and displays it to other users. This use case is extended by Like and Comment.		
Pre-condition - User is logged in. - Community/posting feature is available.		
Post-condition - Post is saved in the database. - Post appears in the community feed.		
Trigger User clicks “Create Post” or taps on the post input field. Type Primary — User initiated		
Relationship - Extend: None - Include: None - Extended by: Like (UC-14), Comment (UC-15) - Association: User		
Normal flow of events User opens the posting interface. - User enters text and/or selects image/video to upload. - System validates the content (size, format, empty text check). - System stores the post in the database. - System displays the post in the community feed and confirms success.		
Sub flows / business rules - SF1 — Media validation: check image/video size, type, resolution limits. - SF2 — Text rules: enforce max character count, prevent empty posts. - SF3 — Moderation: apply automatic filters (bad words detection), flag suspicious content.		
Alternate flows (important only) A1 — Invalid or empty content - Users submit empty posts or unsupported media. - System shows error messages and highlights the problem. - User corrects and resubmits. A2 — Upload failure - Media upload fails due to network/server error. - System shows: “Could not upload media, please try again.” - Users retries or removes media.		

Use case name: chatbot	Id: 14	Level: Low
Primary actor: User Secondary actor: AI (Chatbot)	Use case type: Essential	
Stakeholders and interests - User: wants fast help / suggestions without waiting for human support. - System / Product: reduce support load, improve engagement, surface quick answers. - Admin: monitor chatbot performance and handover flows.		
Brief description User asks a question or requests help; the chatbot receives the input, sends it to the AI engine, and returns an automated reply or hands over to human support when needed.		
Pre-condition - User is on the platform (logged in or guest depending on scope). - Chatbot service is available and integrated with the NLP engine.		
Post-condition - User receives an automated reply or a handover message to support. - Interaction logged for analytics and improvement.		
Trigger User opens chat widget and submits a question / message. Type Primary — User initiated		
Relationship - Includes: Reply (UC-19) - Association: User, AI		
Normal flow - User types a question in the chat widget and sends it. - AI generates an answer (Reply) and returns it. - System displays the reply to the user and logs the interaction.		
Sub flows / business rules - SF1 — Confidence threshold: if AI confidence $\geq$ threshold $\rightarrow$ show automated reply; else go to SF2. - SF2 — Handover: low confidence $\rightarrow$ offer “transfer to human support” or escalate to admin. - SF3 — Context: keep short session context to handle follow-ups (limit memory window). - SF4 — Rate limiting / abuse protection: throttle repeated messages and block abusive content.		
Alternate flows (important) A1— User asks for sensitive/personal request: system refuses and prompts contact support / follow policy.		

Use case name: Trip Suggestions	Id: 15		Level: High
Primary actor: AI System Secondary actor: User		Use case type: Essential	
Stakeholders and interests - User: wants personalized trip recommendations that match their preferences. - System / AI engine: needs accurate data to generate meaningful suggestions. - Admin/Product team: uses results to improve relevance and engagement.			
Brief description The AI analyzes user behavior, preferences, and history to generate a list of recommended trips. Results are displayed to the user on the home page or suggestions section.			
Pre-condition - User has profile data or interaction history (search, bookings, preferences). - AI model/service is available. - System has active trips available for recommendation.			
Post-condition - Personalized trip suggestions generated. - Suggestions displayed to the user. - Interaction logged for improving future recommendations.			
Trigger User opens home page, “Suggested Trips” section, or requests recommendations.			
Type System-initiated / User-triggered (depending on entry point)			
Relationship - Include: None - Extend: None - Association: User, AI system			
Normal flow of Events User opens the suggestions section or home feed. - System sends user profile & behavior data to the AI engine. - AI analyzes the data (preferences, searches, past bookings, ratings). - AI generates a list of recommended trips. - User views or selects a trip to open its details.			
Sub flows / business rules - SF1 — Ranking: AI ranks trips by relevance score. - SF2 — Filtering: remove unavailable or restricted trips. - SF3 — Cold start: if user has no history → use popular trips or trending categories. - SF4 — Diversity rule: avoid repetitive suggestions; mix locations/types.			
Alternate flows (important) A1 — No personalized data available - AI cannot personalize results. - System shows default recommended trips (popular, trending, seasonal).			

Use case name: Content Management	Id: 16		Level: High
Primary actor: Admin		Use case type: Essential	
Stakeholders and interests - Admin: manage platform content (trips, hotels, media) to keep catalogue fresh and correct. - Product: ensure quality and business rules are enforced. - Users: see accurate, moderated content.			
Brief description Admin performs high-level content operations (add, edit, delete, update, approve trips). This is a parent use case that groups content lifecycle actions.			
Pre-condition - Admin is authenticated with proper role/permissions. - CMS / admin dashboard is available.			
Post-condition - Content changes are saved and reflected on the platform (or queued for approval). - Audit log entry created for the action.			
Trigger Admin opens the admin dashboard and selects a content operation. Type Primary — Admin initiated			
Normal flow (main success scenario) - Admin opens Content Management dashboard. - Admin selects an operation (Add / Edit / Delete / Update / Approve). - System presents the appropriate form or list. - Admin submits changes. - System validates and persists changes, updates indexes/cache, and logs the event. - System displays success confirmation.			
Sub flows / business rules (key ones) - SF1 — Role & permission checks: only authorized admins can perform actions. - SF2 — Validation rules: required fields, media constraints, pricing rules.			
Alternate flows (important) A1 — Validation failed: system shows errors; admin corrects and resubmits.			

Use Case: User Management	Id: 17		Level: High
Primary actor: Admin		Use case type: Essential	
Stakeholders and interests - Admin: manage user accounts. - System: maintain accurate user data and access rights. - Security: ensure proper permissions and control.			
Brief description Admin performs operations related to user accounts (add, update, delete). This is the parent use case grouping user control actions.			
Pre-condition Admin logged in with proper privileges.			
Post-condition - User accounts are created, updated, or removed correctly. - Audit logs are created.			
Trigger Admin opens the User Management section.			
Relationship - Includes: Add User (UC-28), Update User (UC-29), Delete User (UC-30) - Association: Admin			
Normal flow - Admin opens User Management dashboard. - Admin chooses an action (add/update/delete). - System loads the necessary form or list. - Admin performs the operation. - System validates and commits changes.			
Sub flows - Permission checks. - Validation of user fields. - Logging and auditing.			
Alternate flows - Permission denied. - Invalid input. - Conflicts (duplicate email/user).			

Use Case: Booking Management	Id:18		Level: High
Primary actor: Admin		Use case type: Essential	
Stakeholders and interests - Admin: manage all booking information and resolve issues. - System: ensure booking data stays accurate and consistent. - User: relies on correct booking details and status			
Brief description Admin oversees platform bookings, updates incorrect data, and verifies bookings when needed. This use case groups booking-related admin actions.			
Pre-condition - Admin logged in with booking-management permissions. - Booking exists.			
Post-condition - Booking details updated or verified successfully. - Changes logged for audit.			
Trigger Admin opens Booking Management from the dashboard.			
Relationship - Includes: Update Booking Data ,Verify Booking . - Association: Admin			
Normal flow - Admin opens Booking Management page. - Admin selects an operation (update or verify). - System loads the target booking. - Admin performs the selected action.			
Sub flows - Permission checks. - Logging changes. - Conflict detection if booking was modified by another admin.			
Alternate flows - Booking not found. - Admin lacks permission. - Validation error → changes not saved.			

2.5.2 Data flow diagram (DFD)

2.5.2.1 Context diagram

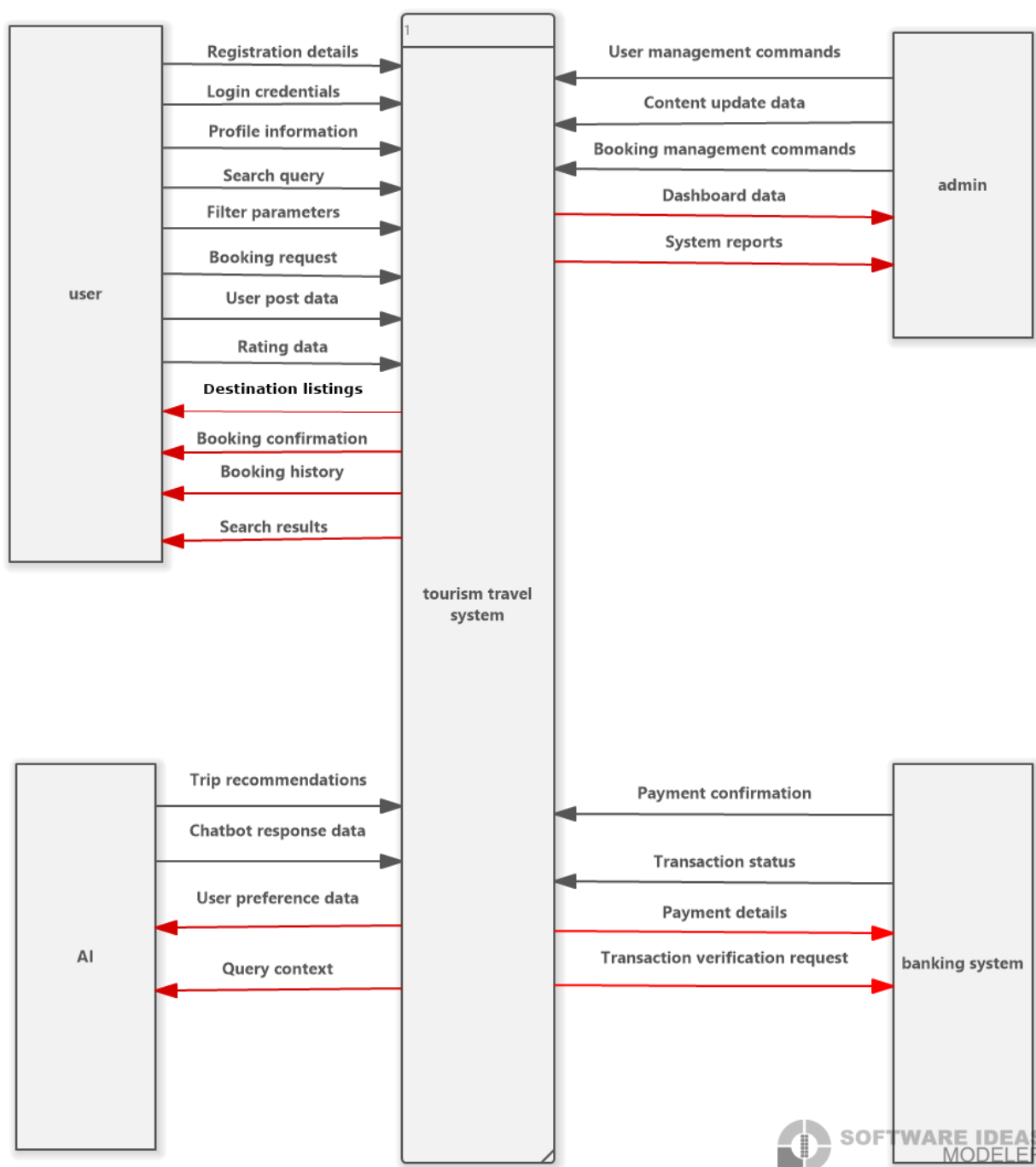


Figure 2.20 context diagram

2.5.2.2 Level 1

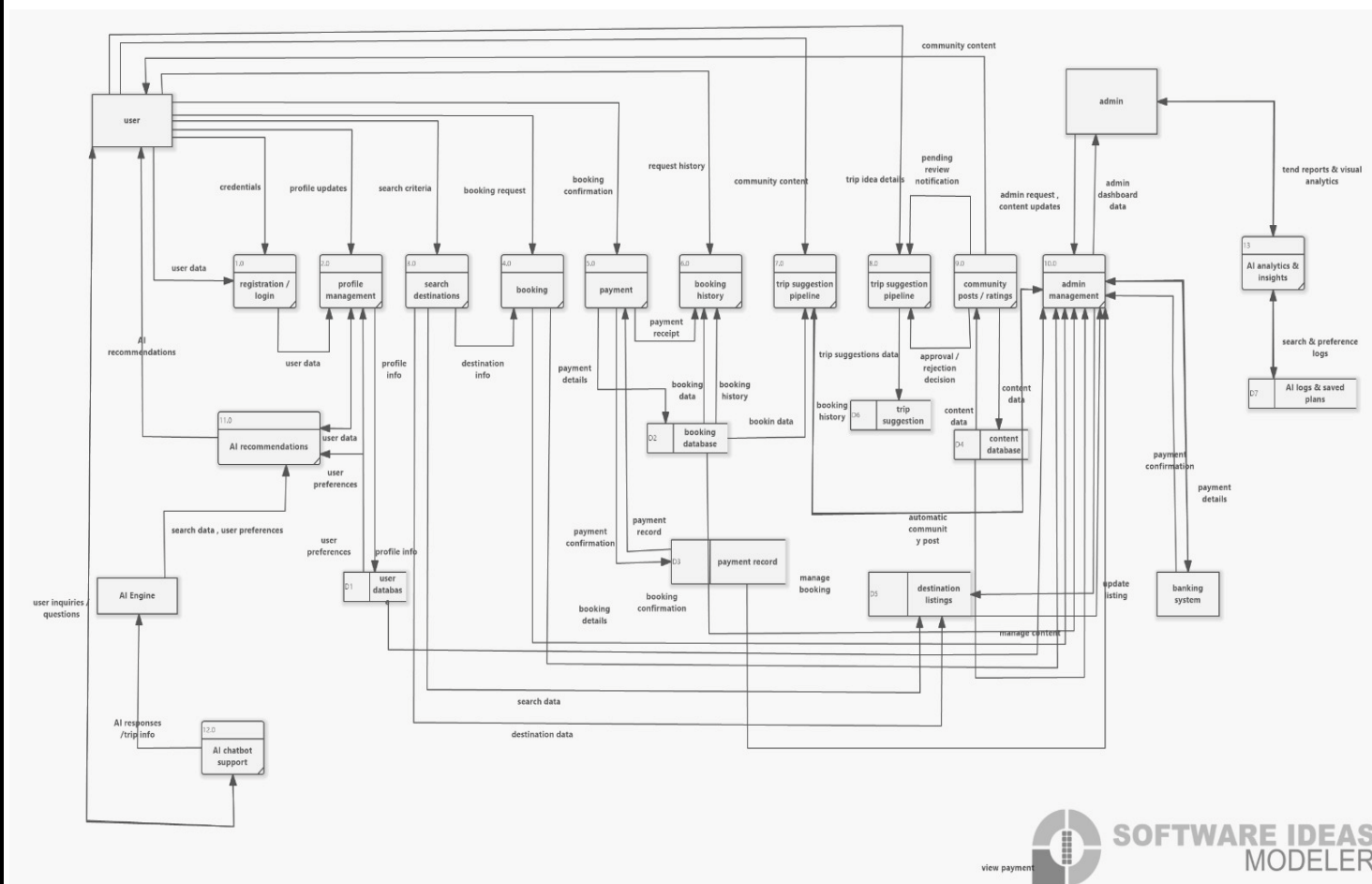


Figure 2.21 DFD

2.6 Data dictionary

2.6.1 Entities

2.6.1.1 User entity

Table 2.2 user entity

Field	Details
Entity name:	User
Description:	The main system user who browses destinations, makes bookings, interacts with content, and completes payments.
Alternate name:	Customer, Client
Input dataflow:	• Registration data • Login credentials • Search requests • Booking data • Payment data • Posts / reviews
Output dataflow:	• Booking requests • Payment confirmations • Reviews and ratings • User feedback
Notes:	The user interacts with search, booking, payment, community content, and AI recommendations.

2.6.1.2 Admin entity

Table 2.3 admin entity

Field	Details
Entity name:	Admin
Description:	The system administrator responsible for managing content, users, and system operations.
Alternate name:	System Manager
Input dataflow:	• Admin login credentials • Reports request • Content updates • Approval decisions
Output dataflow:	• Approved/rejected content • System updates • Dashboard data • Reports and analytics
Notes:	Admin manages listings, monitors system performance, and controls system data.

2.6.1.3 Ai engine

Table 2.4 Ai engine entity

Field	Details
Entity name:	AI Engine
Description:	An intelligent component that provides recommendations, chatbot responses, and analytics.
Alternate name:	Recommendation System, AI Module
Input dataflow:	• User preferences • Search data • Booking history • User queries
Output dataflow:	• Personalized recommendations • Chatbot responses • Insights and analytics
Notes:	Enhances user experience through automation and personalization.

2.6.1.4 Banking system

Table 2.5 Banking entity

Field	Details
Entity name:	Banking System
Description:	External system responsible for processing payment transactions.
Alternate name:	Payment Gateway
Input dataflow:	• Payment request • Transaction details
Output dataflow:	• Payment confirmation • Transaction status
Notes:	External entity; does not store internal system data.

2.6.2 Data stores

2.6.2.1 User Database

Table 2.6 user data store table

Field	Details
Data Store name:	User Database
Description:	Stores all registered user information and account details.
Stored data:	• User ID • Name • Email • Password • Profile info • Preferences
Input dataflow:	• Registration data • Profile updates
Output dataflow:	• User profile data • Credentials validation
Notes:	Used by login, profile management, and AI recommendation processes.

2.6.2.2 Booking Database

Table 2.7 booking data store table

Field	Details
Data Store name:	Booking Database
Description:	Stores all booking transactions and trip reservations.
Stored data:	• Booking ID • User ID • Destination • Dates • Status
Input dataflow:	• Booking details
Output dataflow:	• Booking history • Booking confirmation data
Notes:	Connected with booking process and booking history module.

2.6.2.3 Payment Record

Table 2.8 payment data store table

Field	Details
Data Store name:	Payment Record
Description:	Stores all payment transaction details.
Stored data:	• Payment ID • Booking ID • Amount • Payment status • Transaction date
Input dataflow:	• Payment details • Payment confirmation
Output dataflow:	• Payment receipt • Payment history
Notes:	Linked with payment process and banking system.

2.6.2.4 Trip Suggestion

Table 2.9 trip data store table

Field	Details
Data Store name:	Trip Suggestion
Description:	Stores generated travel recommendations and trip ideas.
Stored data:	• Suggestion ID • Destination • Activities • Estimated cost
Input dataflow:	• AI generated suggestions
Output dataflow:	• Trip suggestions data
Notes:	Used by recommendation and trip suggestion processes.

2.6.2.5 Destination Listings

Table 2.10 destination data store table

Field	Details
Data Store name:	Destination Listings
Description:	Stores available travel destinations and related details.
Stored data:	• Destination ID • Name • Location • Description • Price
Input dataflow:	• Admin updates
Output dataflow:	• Destination data
Notes:	Used in search and booking processes.

2.6.2.6 Content Database

Table 2.11 content data store table

Field	Details
Data Store name:	Content Database
Description:	Stores community posts, reviews, and listings.
Stored data:	• Post ID • User ID • Content • Ratings • Comments
Input dataflow:	• User posts • Admin updates
Output dataflow:	• Community content
Notes:	Supports community features and moderation.

2.6.2.7 AI Logs & Saved Plans

Table 2.12 Ai logs and saved plans data store table

Field	Details
Data Store name:	AI Logs & Saved Plans
Description:	Stores user interactions, search logs, and saved travel plans.
Stored data:	• Search logs • User preferences • Saved trips
Input dataflow:	• User activity • AI processing data
Output dataflow:	• Analytics data • Personalized insights
Notes:	Used by AI engine and analytics system.

2.6.3 Processes

Table 2.13 Processes table

Process ID	Process Name	Description	Input Dataflow	Output Dataflow	Related Data Stores
1.0	Registration / Login	Handles user authentication and account creation	Credentials, registration data	Auth status, user session	User Database
2.0	Profile Management	Manages user profile updates	Profile data	Updated profile info	User Database
3.0	Search Destinations	Allows users to search for destinations	Search criteria	Destination results	Destination Listings
4.0	Booking	Handles booking requests and reservations	Booking request, destination data	Booking details	Booking Database
5.0	Payment	Processes user payments through external banking system	Payment details (User), transaction processing (Banking System)	Payment confirmation, receipt	Payment Record
6.0	Booking History	Retrieves user booking history	Booking data (from Booking Database)	Booking history	Booking Database
7.0	Trip Suggestion Pipeline	Generates travel recommendations based on user behavior	User preferences, booking history	Trip suggestions	User Database, Booking Database, Trip Suggestion
8.0	Trip Suggestion Review	Displays and manages suggestion details	Trip suggestions data	Trip details	Trip Suggestion
9.0	Community Posts	Manages user-generated posts and reviews	Posts, reviews, content data	Published content	Content Database

Process ID	Process Name	Description	Input Dataflow	Output Dataflow	Related Data Stores
10.0	Admin Management	Controls system operations, users, and content	Admin requests, updates, approvals	System updates, approved content	Content Database, Destination Listings, User Database
11.0	AI Recommendations	Provides personalized AI-based recommendations	Search data, user preferences, booking history	AI recommendations	User Database, Booking Database, AI Logs & Saved Plans
12.0	AI Chatbot	Handles user inquiries and support	User questions	AI responses	AI Logs & Saved Plans
13.0	AI Analytics	Generates reports and insights from system data	Logs, saved plans	Reports, analytics	AI Logs & Saved Plans

Chapter Three

System Design

3.1 Introduction

This chapter presents the system design phase of the YOC Website, which focuses on how the system will operate and meet user needs. It includes the system's nonfunctional requirements, along with design models such as the activity diagram, sequence diagram, Enhanced Entity Relationship Diagram (EERD), and the database schema. These components collectively define the system's structure, behavior, and data organization, forming the foundation for a functional and efficient implementation.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

- The system shall load main pages within **3 seconds**.
- Search results shall appear within **2 seconds**.
- The system shall support **1,000+ concurrent users**.
- The system shall optimize images and videos for fast loading.

3.2.2 Security Requirements

- All communication shall use **HTTPS encryption**.
- User passwords shall be stored using hashing algorithms.
- Credit card details shall not be stored on the server.

3.2.3 Reliability Requirements

- In case of failure, the system shall recover within **24 hours**.
- The system shall handle unexpected errors without crashing.

3.2.4 Usability Requirements

- The interface shall be user-friendly, modern, and easy to navigate.
- The system shall provide a consistent design across all pages.

- The Chabot shall support both Arabic and English.
- The platform shall follow accessibility standards:
 - Proper contrast
 - Alt-text for images
 - Clear fonts

3.2.5 Compatibility Requirements

- The system shall work on major browsers:
 - Chrome
 - Firefox
 - Safari
 - Edge
- The system shall run on:
 - Desktop
 - Tablet
 - Mobile browsers

3.2.6 Maintainability Requirements

- The system shall use modular architecture for easy updates.
- Admin dashboard allows system monitoring and content updates.

3.2.7 Legal & Compliance Requirements

- The system shall comply with Egyptian data protection regulations.
- The system shall not share user data without consent.
- All media and content must respect copyright laws.
- Payment protection.

3.3 System design tools

3.3.1 Activity diagram

3.3.1.1 User Login, registration and reset password

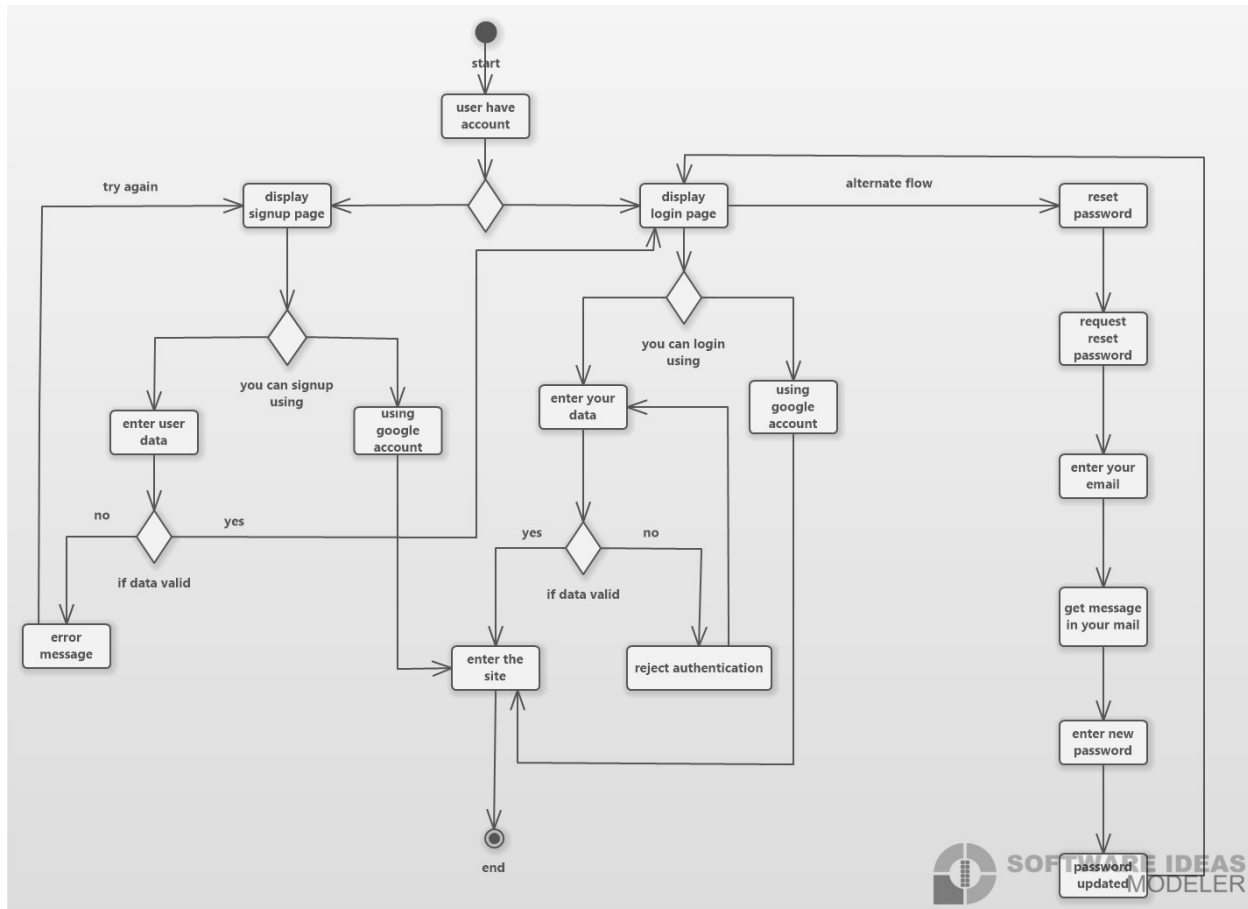


figure 3.1 user login activity diagram

This diagram maps out a website's authentication system, starting with whether a user has an account. New users are directed to a signup page to register manually or via a Google account, while existing users go to a login page. On the login page, users can log in using their credentials, quickly authenticate with Google, or choose an alternate password reset path. If manual data entry fails during signup or login, the system shows an error or rejects authentication so the user can try again. Once registration, login, or a password reset is successfully completed, the user is granted access to enter the site.

3.3.1.2 User Booking and payment

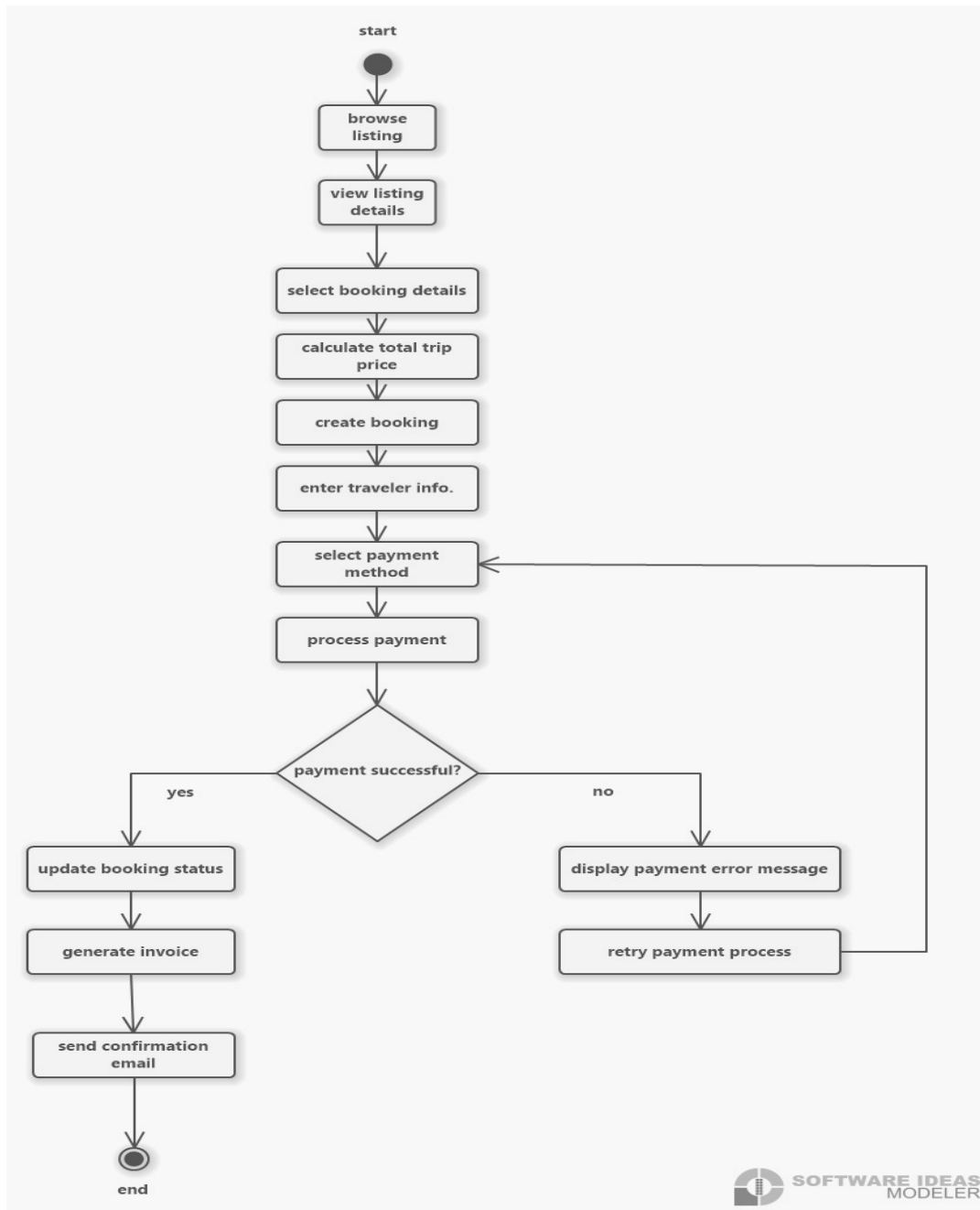


Figure 3.2 user booking activity diagram

This diagram shows the process of booking and paying for a trip online. The user browses a listing, chooses their details, enters traveler information, and submits a payment. If the payment fails, the system shows an error message and asks them to try again. If the payment succeeds, the system updates the booking, creates an invoice, and sends a confirmation email to finish the process.

3.3.1.3 Ai Planner

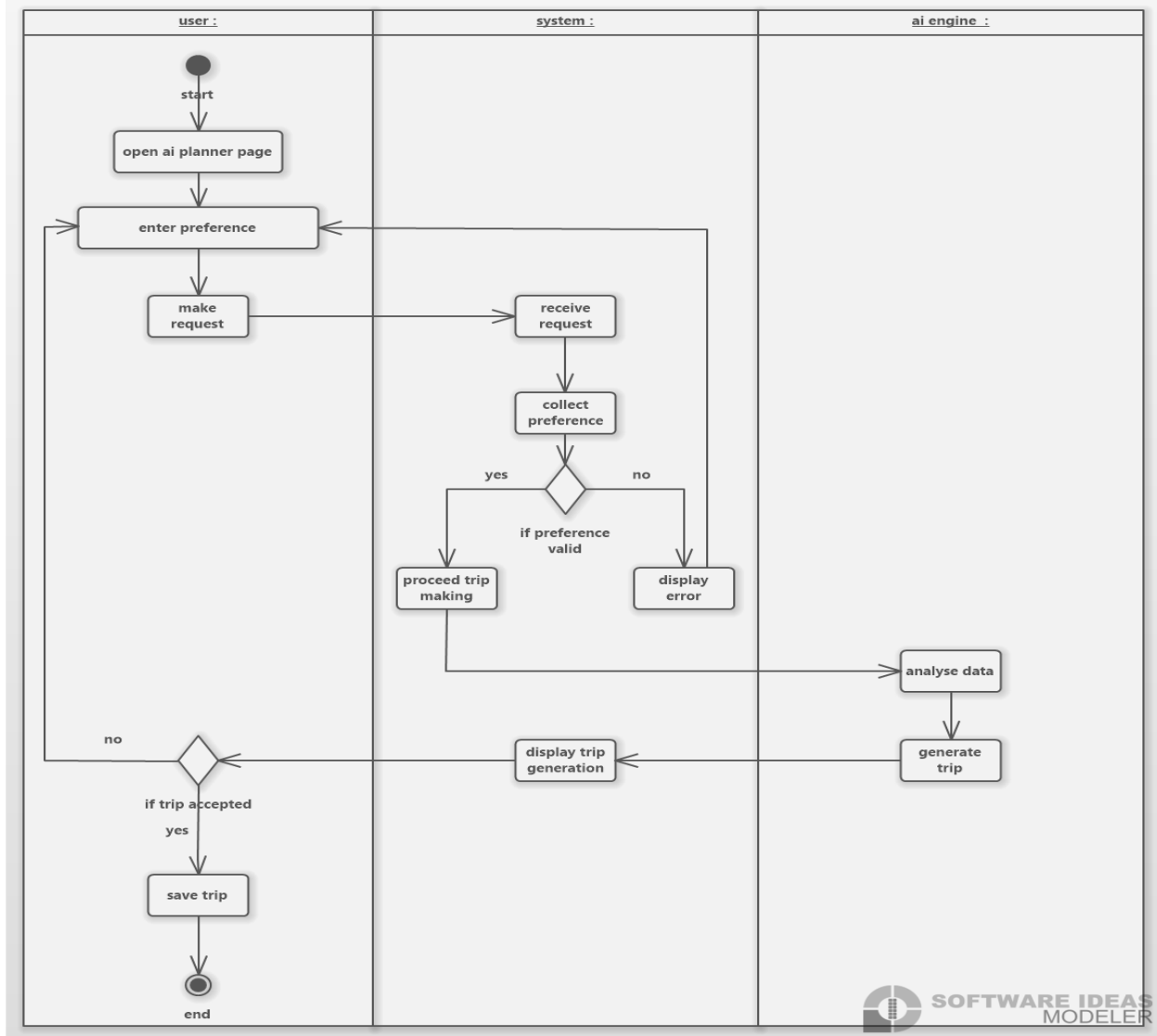


Figure 3.3 Ai planner activity diagram

This diagram shows how a user, a system, and an AI engine work together to create a travel plan. The user starts by entering their preferences and submitting a trip request. The system checks the request, displays an error if something is wrong, or sends it to the AI engine if it looks good. The AI engine then processes the data and generates a customized itinerary. Finally, the user reviews the generated trip, choosing to either save it and finish or reject it to try again.

3.3.1.4 suggest trip

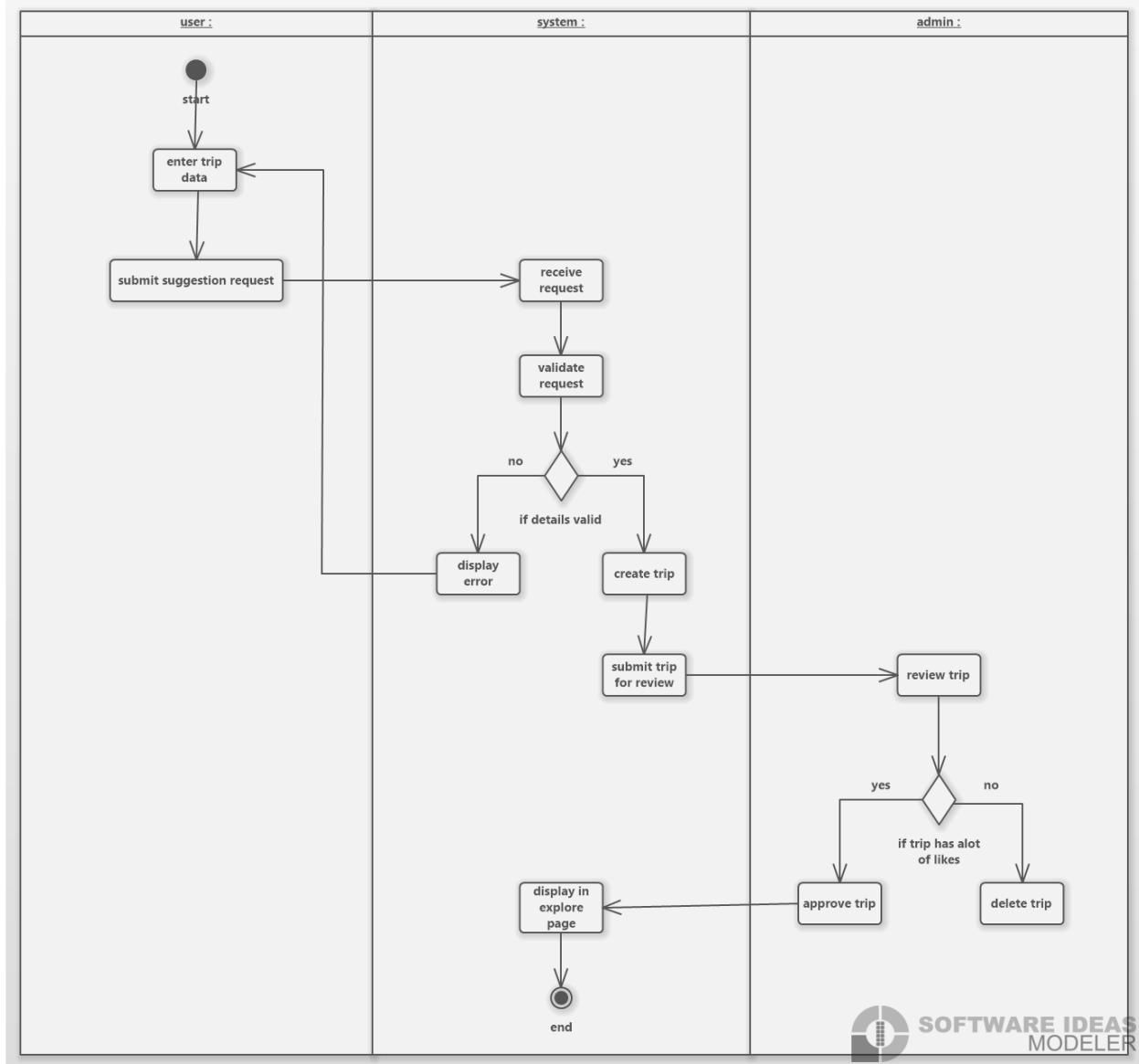


Figure 3.4 suggest trip activity diagram

This diagram shows how a user, a system, and an admin handle trip suggestions. The user starts by entering trip data and submitting a suggestion request. The system validates the request, displaying an error if it is invalid or creating the trip and sending it for review if it looks good. Finally, the admin reviews the trip, approving it to be displayed on the explore page if it has enough likes, or deleting it if it does not.

3.3.2 Sequence Diagram

3.3.2.1 user login

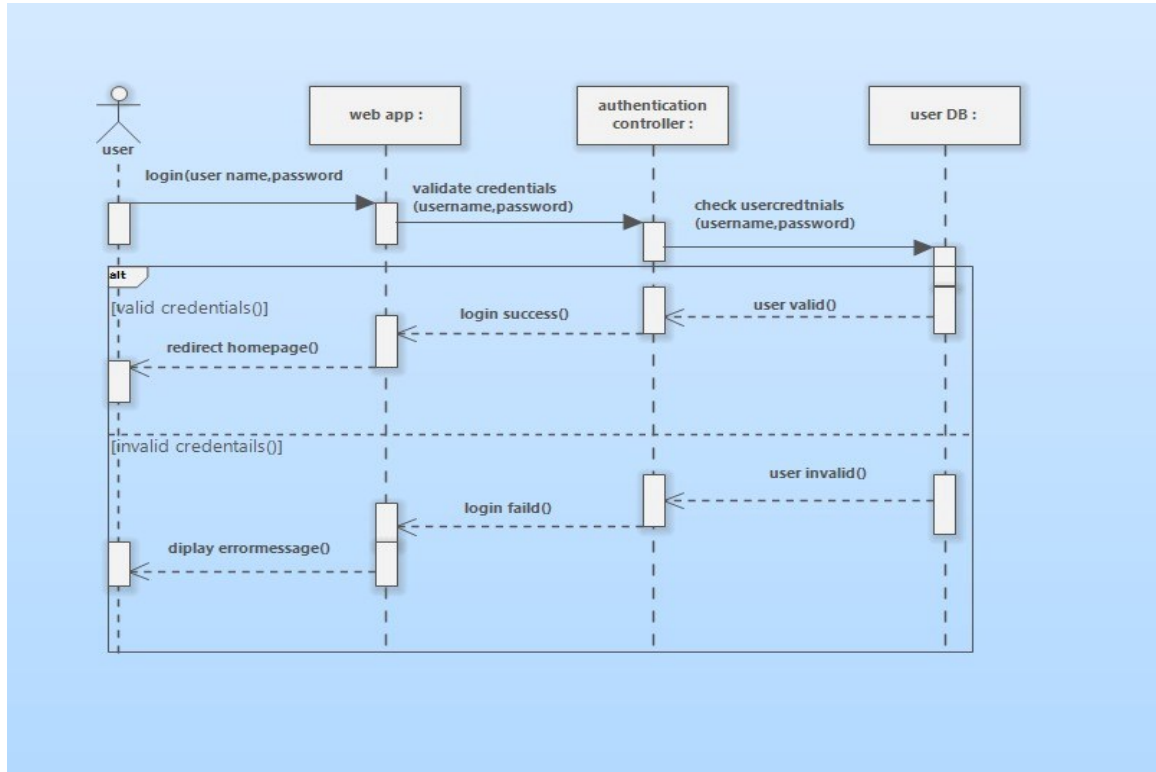


Figure 3.5 user sequence diagram

The sequence diagram illustrates the step-by-step process of a user logging into a web application. First, the user submits their username and password, which the web app passes to an authentication controller to check against the user database. The diagram then splits into an "alt" box to show the two possible outcomes of this check. If the credentials match, the user is successfully validated and redirected to the homepage; if they are incorrect, the system triggers a failure response and displays an error message.

3.3.2.2 community

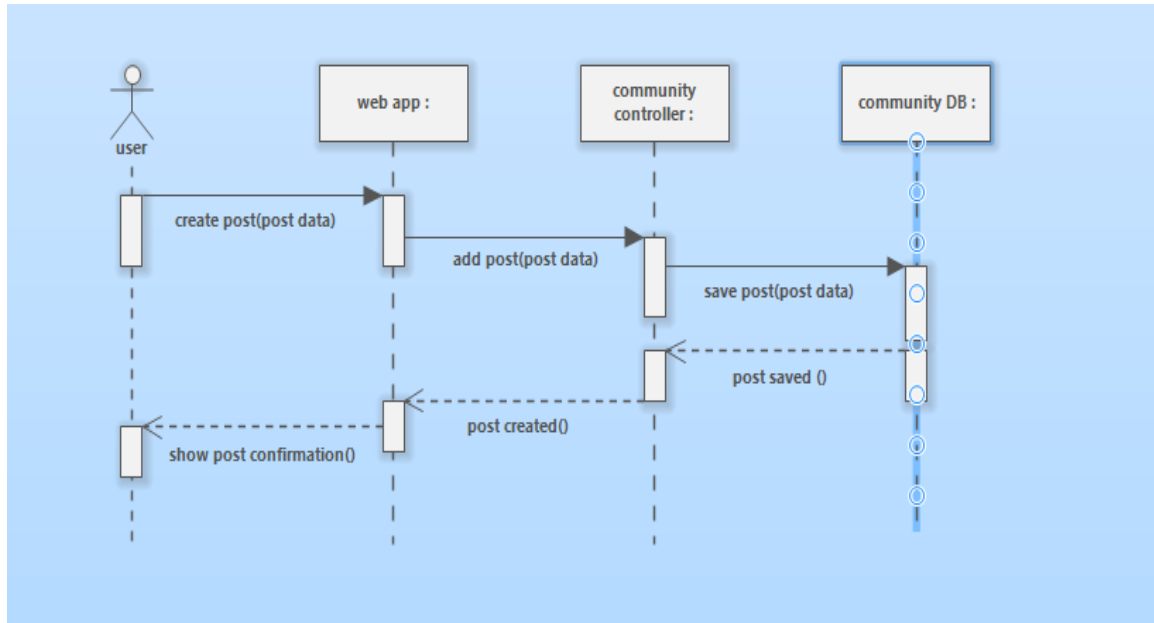


Figure 3.6 community sequence diagram

The sequence diagram shows the step-by-step process of a user publishing a new post on a platform. First, the user submits their post content through the web app, which quickly forwards the data to the community controller backend. Next, the controller sends this information to the community database to securely store and save the new post. Finally, once the database confirms the post is successfully saved, the success message travels back up the chain until the web app displays to the user.

3.3.2.3 payment

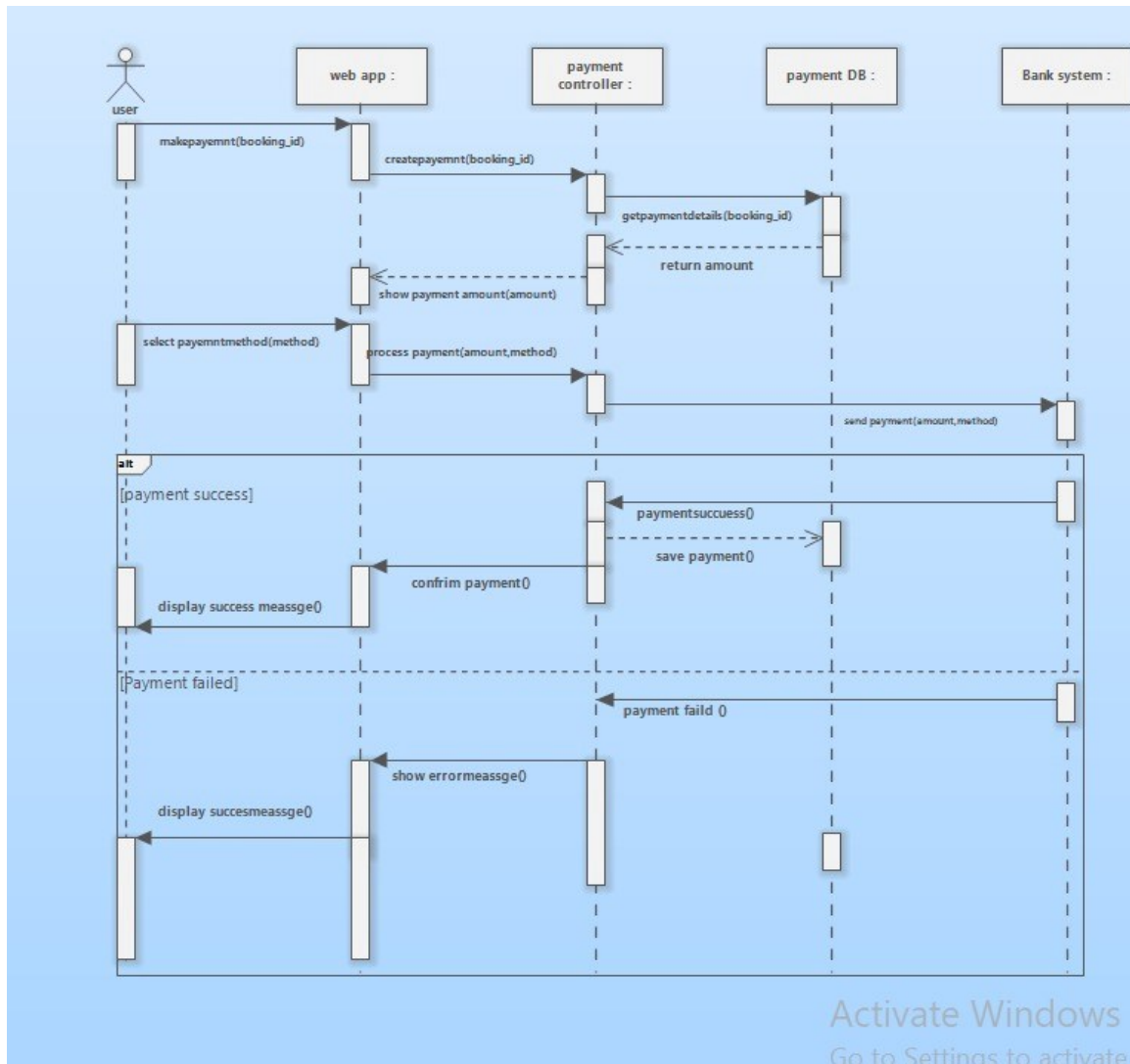


Figure 3.7 payment sequence diagram

The sequence diagram details the workflow of a user completing a booking payment through an online system. Initially, the user initiates a payment, prompting the web app and payment controller to retrieve the required billing amount from the payment database and display it to the user. Once the user selects their payment method, the controller forwards the transaction request directly to the external bank system for processing. Finally, an "alt" conditional box handles the bank's response: a successful transaction saves the payment data to the database and displays a confirmation message, while a failed transaction routes an error message back to the user.

3.3.2.4 booking system

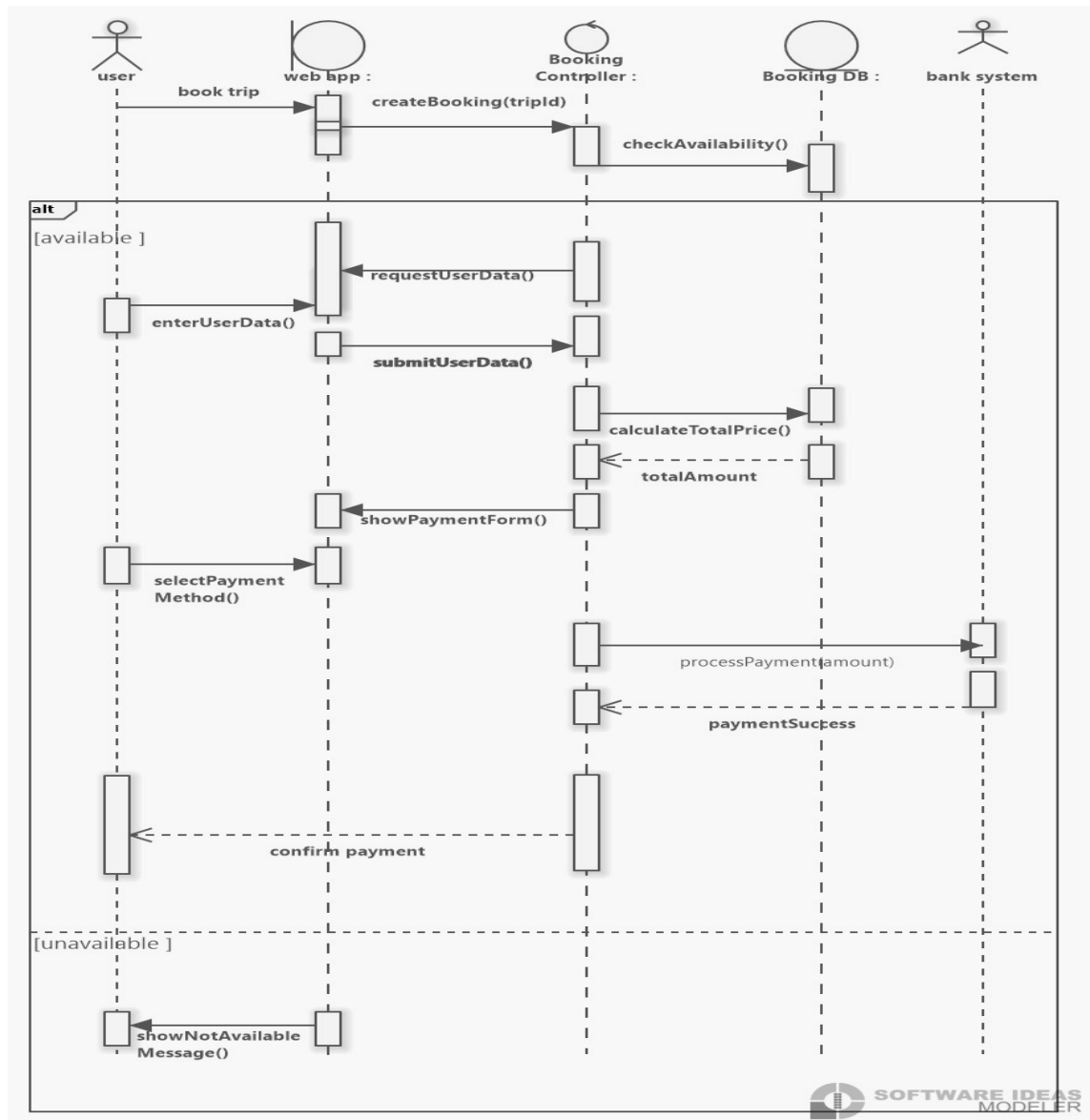


Figure 3.8 booking sequence diagram

The sequence diagram maps out the user flow for booking a trip through an online application. When a user requests a booking, the web app triggers the booking controller to check trip availability with the booking database. An "alt" conditional box separates the two possible outcomes: if the trip is unavailable, the web app simply displays a "not available" message to the user. If the trip is available, a longer sub-process is triggered where the user submits their data, the database calculates the total price, a payment form is generated, and the transaction is successfully processed through an external bank system before a final payment confirmation is sent to the user.

3.3.2.5 Admin

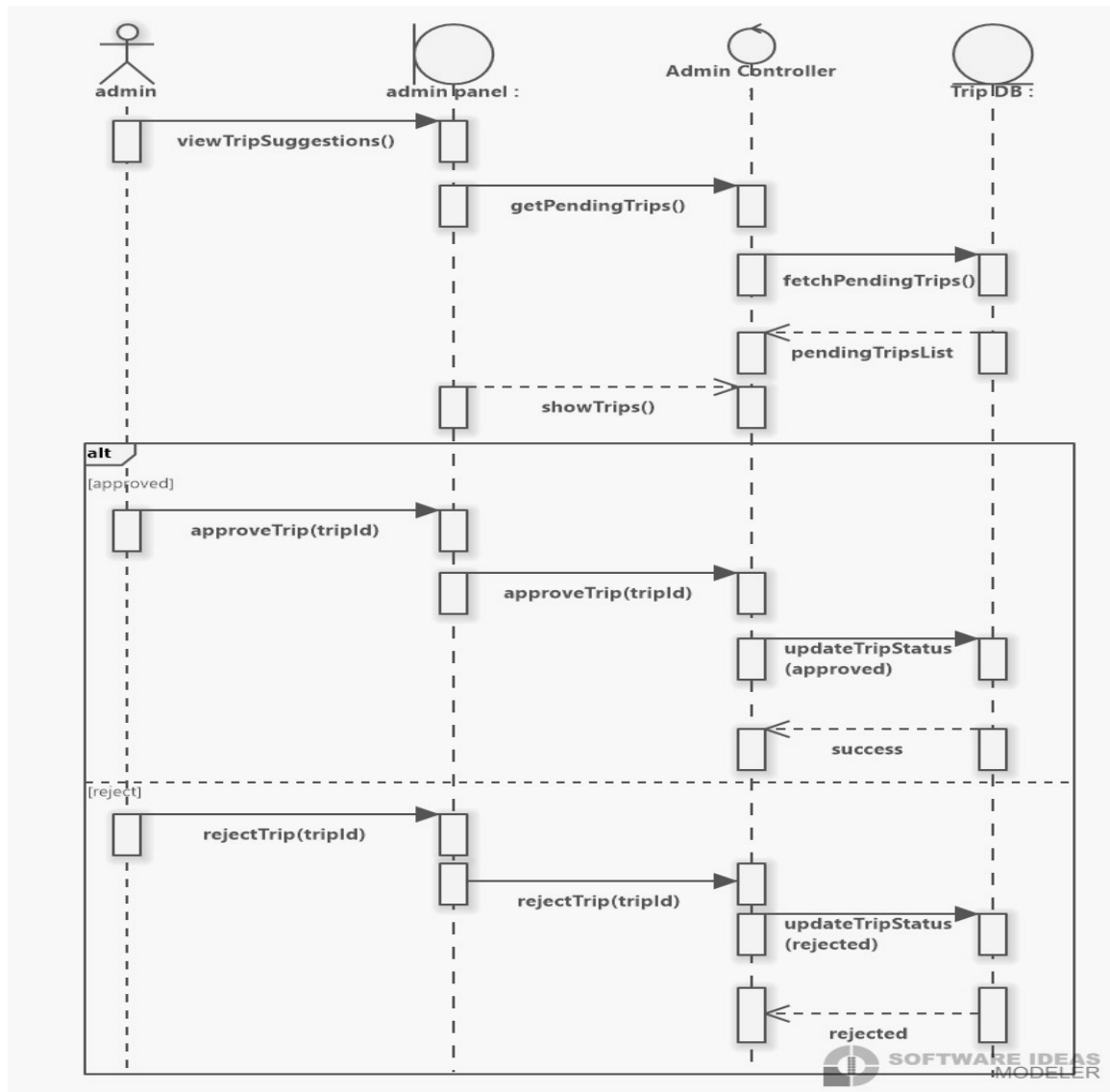


Figure 3.9 admin sequence diagram

The sequence diagram models an administrative workflow for reviewing and managing trip suggestions. Initially, when an administrator requests to view trip suggestions, the admin panel calls upon the Admin Controller to fetch a list of pending trips from the Trip DB database and display them. The workflow then enters an "alt" conditional block to execute one of two decision pathways chosen by the administrator. If the admin decides to approve a trip, the controller updates its database status to "approved"; conversely, if the admin chooses to reject it, the system updates the database entry to reflect that the trip has been "rejected".

3.3.2.6 AI-Planner

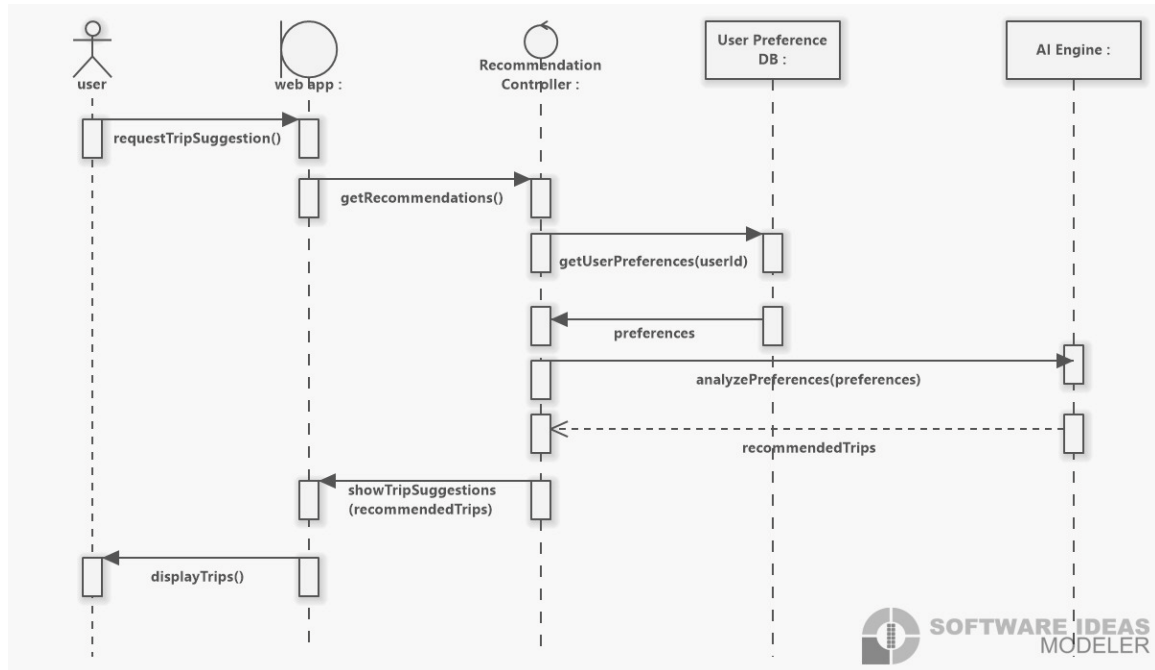


Figure 3.10 Ai trip sequence diagram

The sequence diagram illustrates the system process for generating personalized travel recommendations for a user. When the user requests a trip suggestion, the web app asks the recommendation controller to fetch tailored options. The controller first pulls the user's saved choices from the User Preference DB, then forwards those preferences to an AI Engine to analyze them and generate matching trips. Finally, the AI Engine returns the custom trip results to the controller, which passes them back up to the web app to be displayed on the user's screen.

3.3.2.7 Admin user management

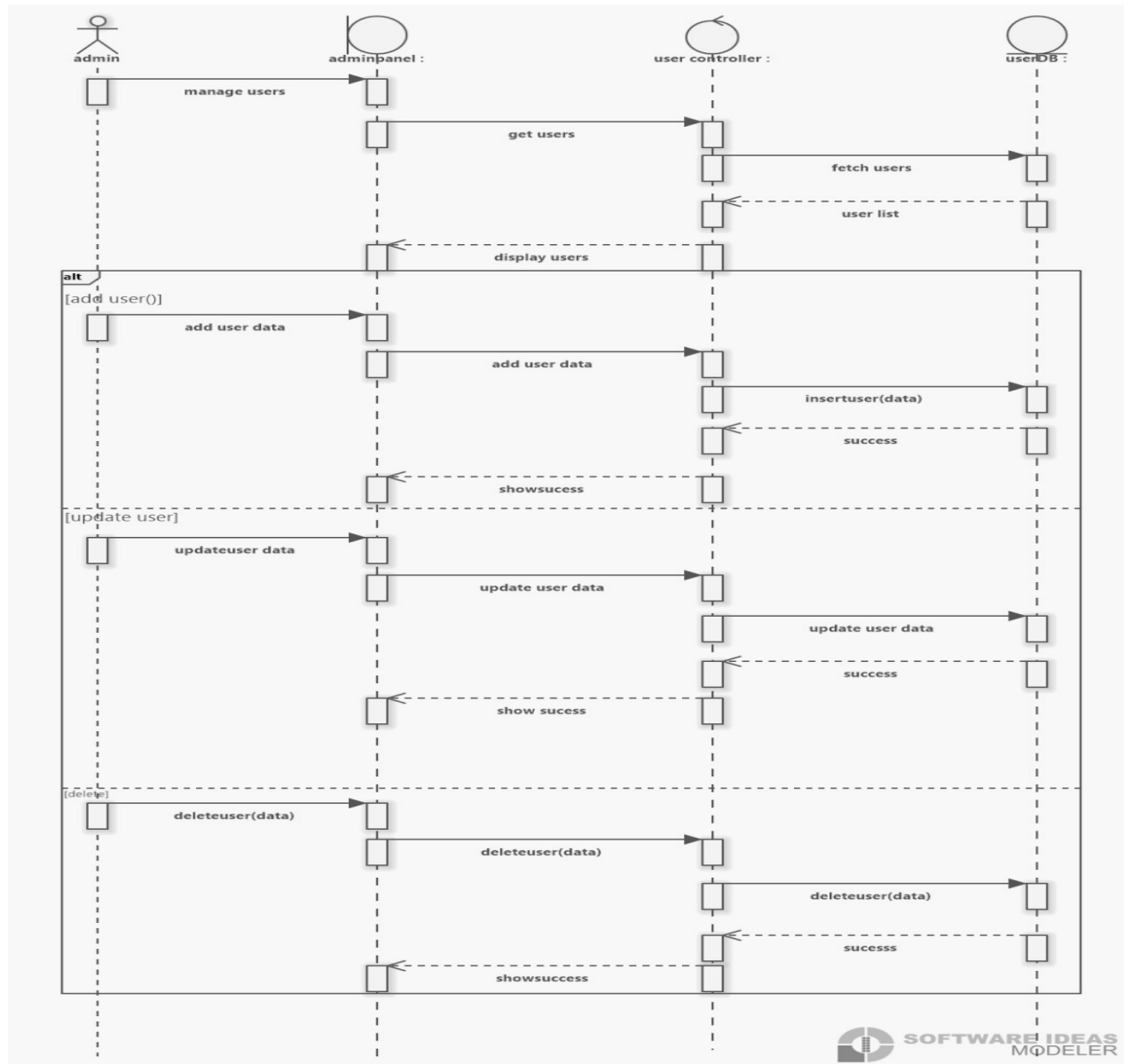


Figure 3.11 admin user management sequence diagram

The sequence diagram maps out how an administrator manages user accounts within a system. Initially, when the admin opens the user management dashboard, the admin panel calls the user controller to fetch and display the current registered accounts from the user database. The diagram then enters a large "alt" box that branches into three distinct account-management operations: adding a new user, updating an existing user's details, or deleting a user entirely. For whichever action the admin chooses, the data is routed through the controller to update the user database, which then sends back a success confirmation to be displayed on the admin panel.

3.3.3 Class Diagram

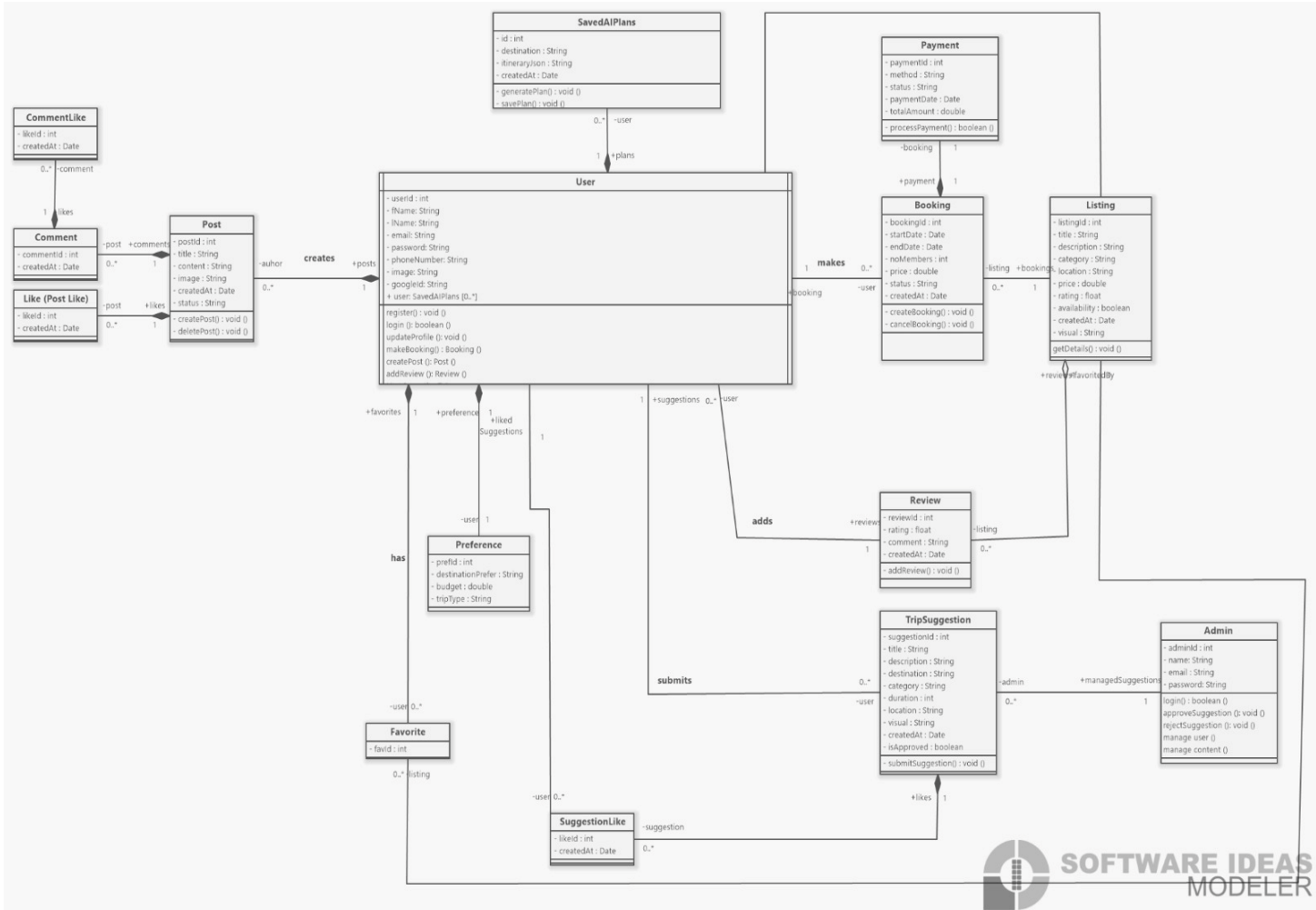


Figure 3.12 Class diagram

3.3.4 Database Design

3.3.4.1 EERD

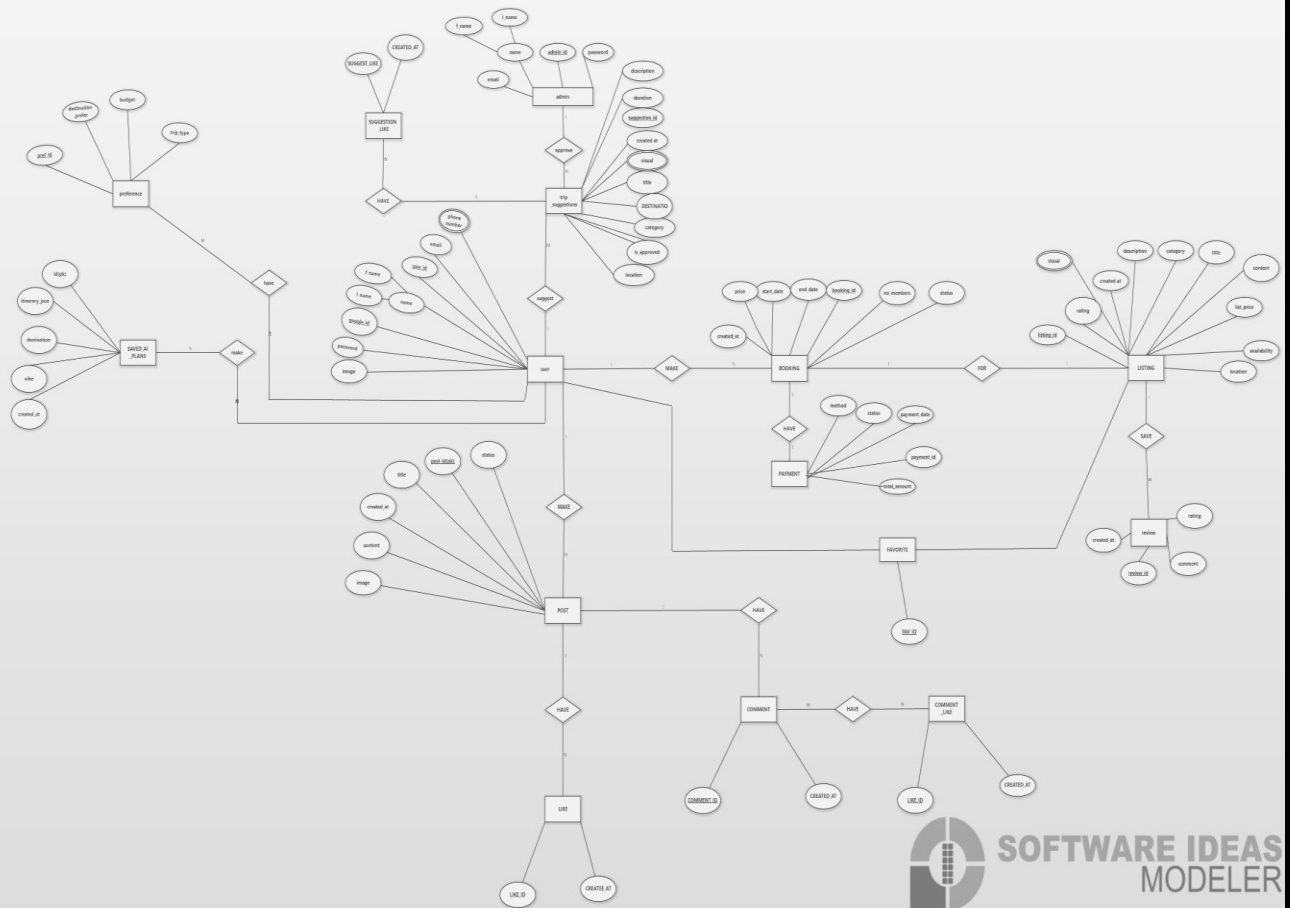


Figure 3.13 EERD diagram

3.3.4.2 Database Schema

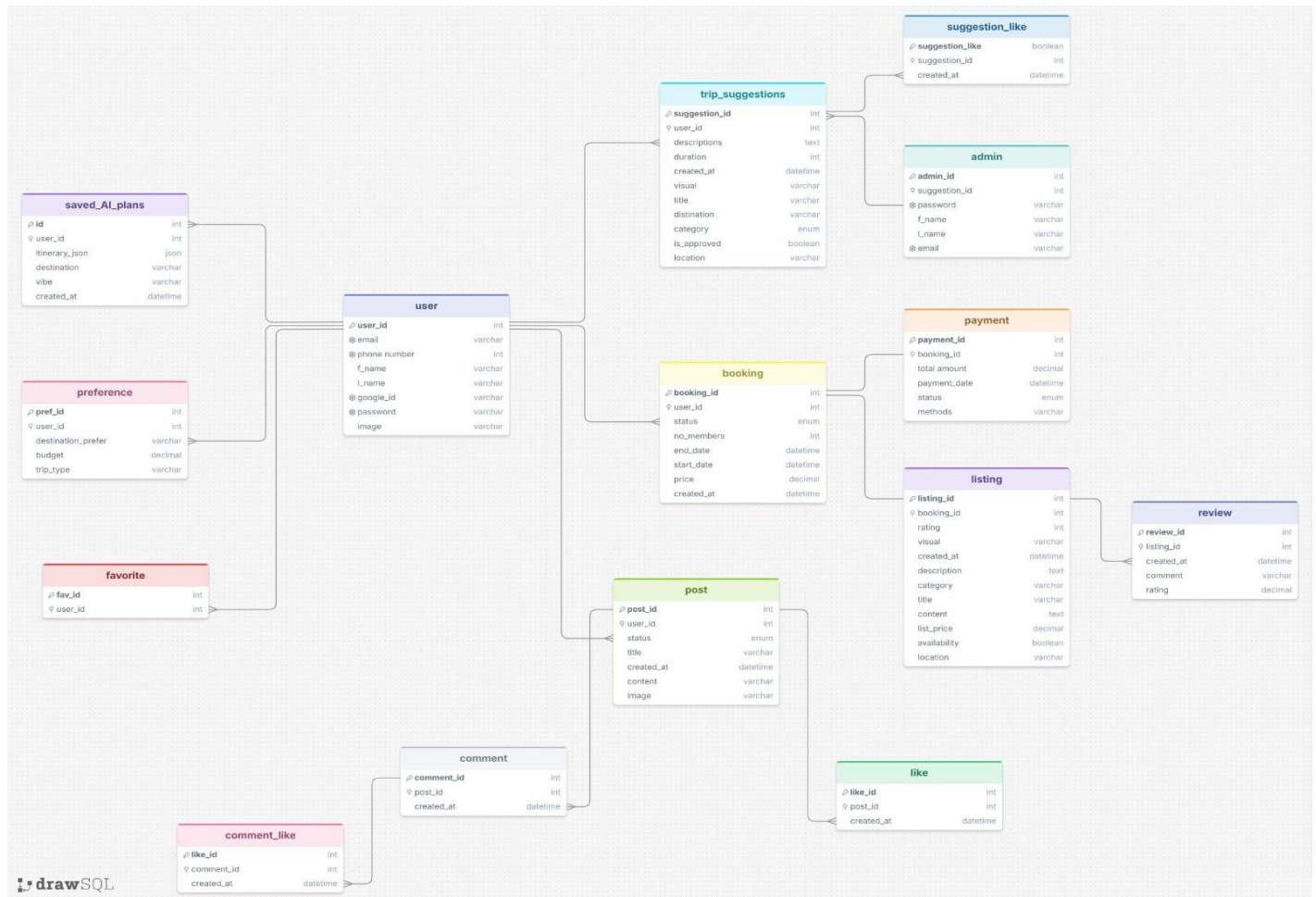


Figure 3.14 Schema diagram

Chapter Four System Implementation

4.1 System Setup

4.1.1 Development Tools and Environment

- **Visual Studio Code:**
Used as the main development environment for writing, organizing, and debugging the website source code.
- **XAMPP:**
Used as a local server environment that includes Apache, MySQL, and PHP to run and test the system locally.
- **Software Ideas Modeler:**
Used for designing system diagrams such as UML diagrams and database structure.
- **Microsoft Word and Excel:**
Used for documentation, planning the project, and organizing requirements.
- **Google Gemini API:**
Integrated as the main AI service to power the **Nefertari Chatbot** and the **AI Trip Planner**, helping in generating personalized travel suggestions.

4.1.2 Technologies Used

Front-End

- **HTML5:**
Used to structure the content and layout of all web pages.
- **CSS3 (Custom Styling):**
Used to design the user interface with modern styles such as smooth effects and a clean visual identity for the platform.
- **JavaScript :**
Used to handle interactive features and communicate with the server dynamically.
- **Font Awesome:**
Used to add icons that improve the visual appearance of the interface.

4.1.3 Back-End

- **PHP:**
Used to handle server-side logic, user sessions, and data processing securely.
- **MySQL:**
Used as the main database to store system data such as users, bookings, travel suggestions, payments, and community posts.

4.2 Features – Main Functions

4.2.1 Front-end Interface

Login page

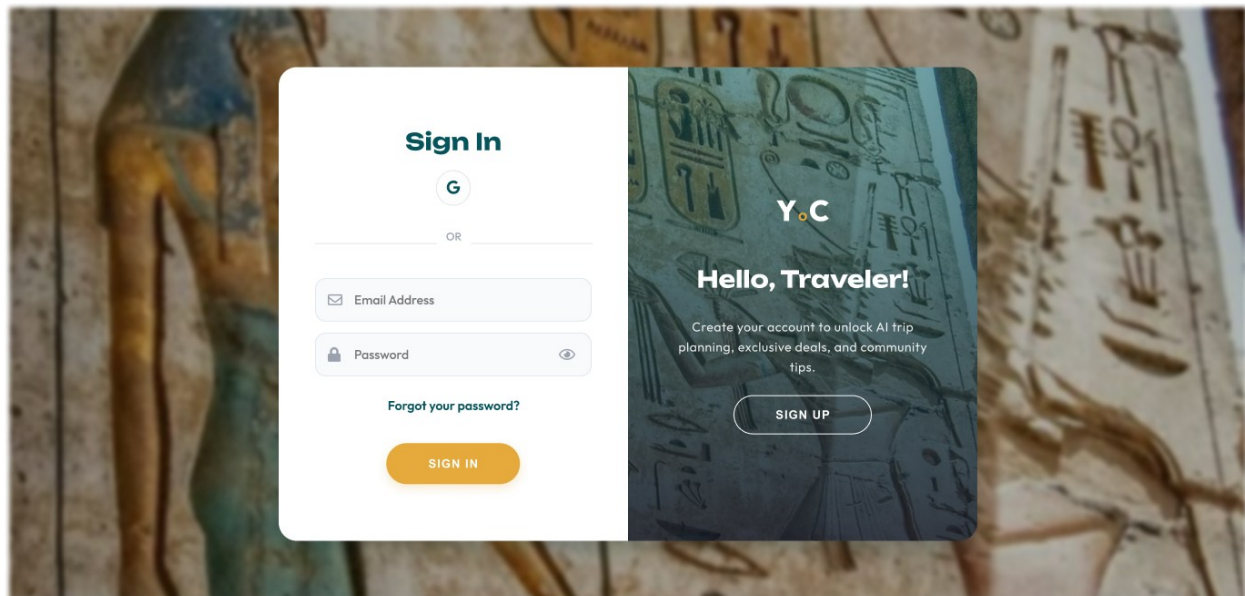


Figure 4.1 Login page

Description: This is the entry point for users. It features a clean interface where travelers can sign in using their email or social accounts. New users can also use the "Sign Up" button to join the community.

Home page

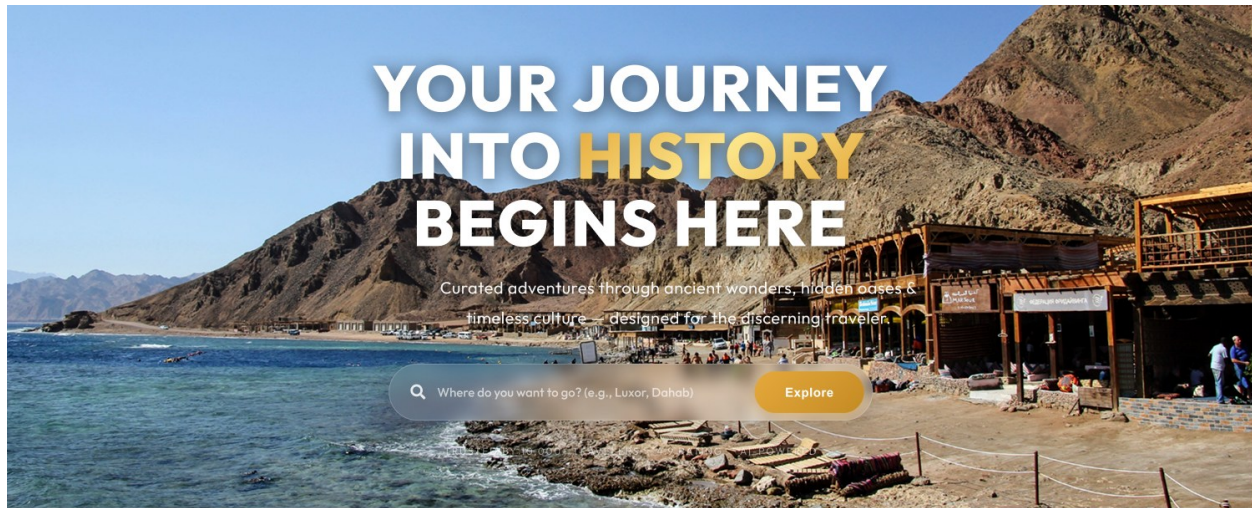


Figure 4.2 home page

Description: The main landing page welcoming users with the slogan "Your Journey Into History Begins Here." It includes a prominent search bar to explore specific Egyptian destinations like.

Suggest Trip

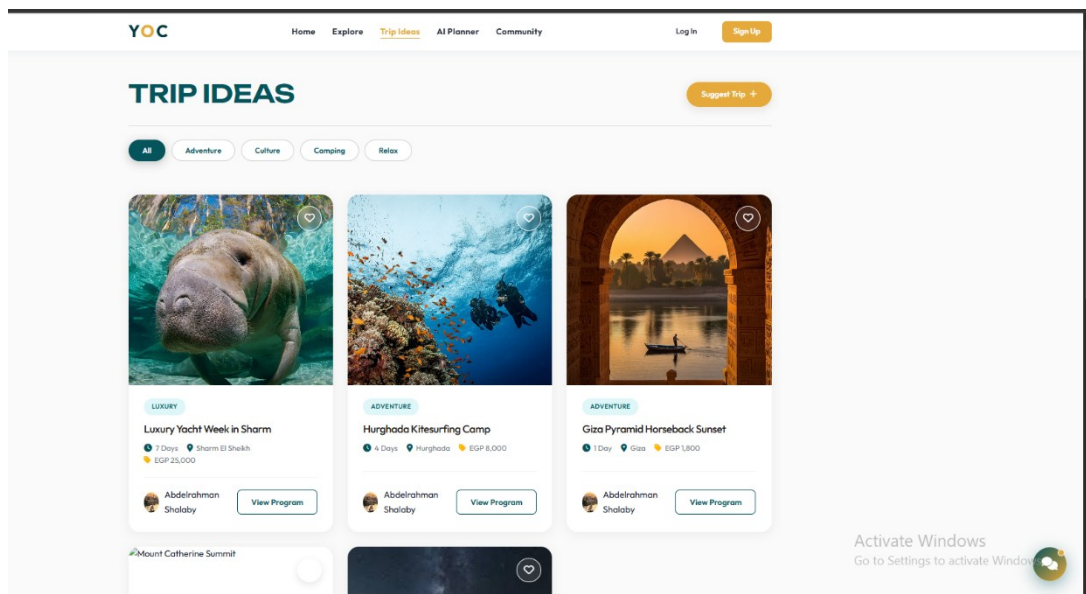


Figure 4.3 Suggest Trip page

Description: A community-driven section where users can propose new itineraries. It allows filtering trip ideas by categories such as Adventure, Culture, Camping, or Relax. Admin Confirmation

Ai Planner (3- Seamless Steps) The AI Planner helps users create a customized it

Step 1: Discover Your Vibe

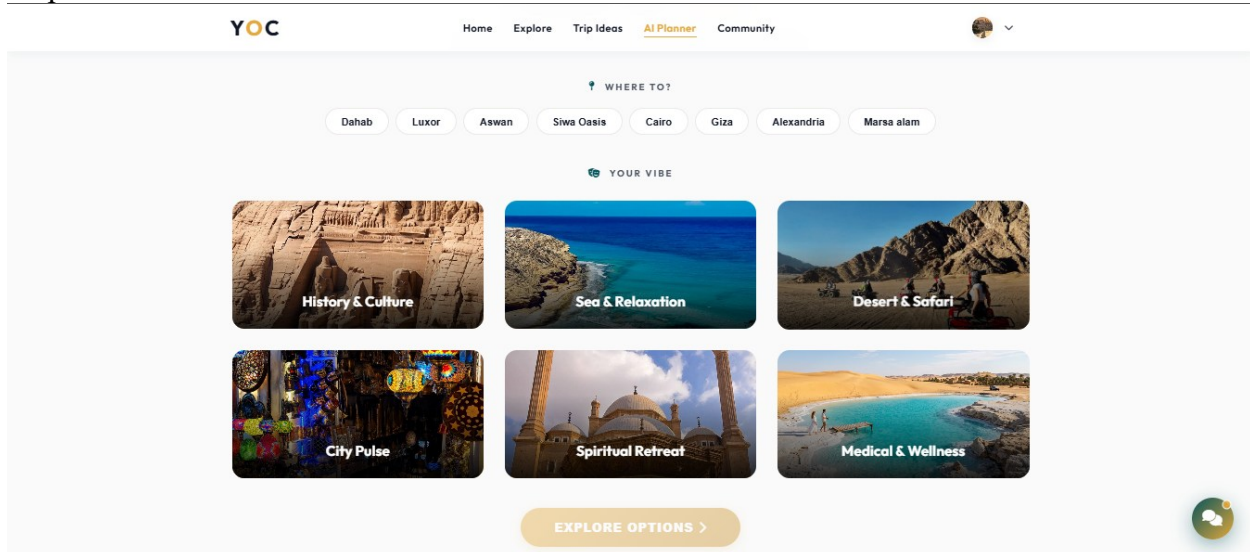


Figure 4.4 user chose the vibe

Description: Users select their destination (e.g., Aswan) and the type of energy they want for the trip, such as "History & Culture" or "Sea & Relaxation".

Step 2: Tailor Your Stay

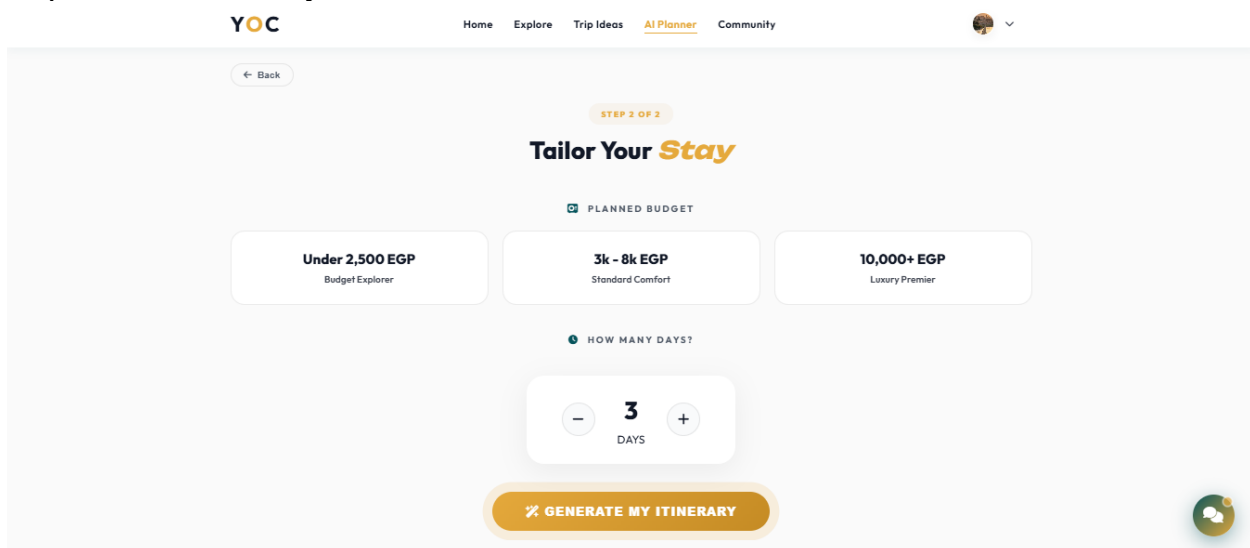


Figure 4.5 user customize his plan

Description: Users define their budget (Budget, Standard, or Luxury) and the number of days they plan to stay.

Step 3: Generated plan

YOC Home Explore Trip Ideas AI Planner Community

We have curated a 2-day journey in Dahab tailored for a Sea & Relaxation vibe and a 3k - 8k EGP budget.

DAY 1

Morning
mena mousa hotel
Check into your cozy base right by the beach. Super chill vibes to start your trip. If you want a quick bite first, Ralph's German Bakery is just around the corner.
3500.00 EGP
[BOOK NOW](#)

Afternoon
Lighthouse Beach Chill
The ultimate spot to swim and lounge. Grab a fresh mango juice at Everyday Cafe. If it's too crowded, find a cozy cushion at Friends Cafe nearby.
[View on Map](#)

Evening
Seafood Dinner at Ali Baba
Feast on the freshest catch of the day right by the water. If you're not in the mood for seafood, King Chicken is just a short walk away for the best grilled chicken in town.
[View on Map](#)

DAY 2

Morning
deep diving
Time to explore Dahab's legendary underwater world! If you prefer a more relaxed snorkel, the Lighthouse reef is an easy alternative right off the shore.
2550.00 EGP
[BOOK NOW](#)

Afternoon
Snorkeling at Eel Garden
A beautiful, quiet spot to watch the sand eels sway in the current. If the tide is too low, head over to Coral Coast for a relaxed beachside lunch.
[View on Map](#)

Evening
Bedouin Tea at Sheikh Salem House
Wind down your trip with authentic Bedouin tea by the campfire. If you want a livelier vibe to end the night, grab a cold drink at Yalla Bar instead.
[View on Map](#)

Figure 4.6 AI generate plan

Description: The system produces a detailed daily plan, highlighting activities like visiting Elephantine Island and discovering "Hidden Gems."

Community Features

The community section allows travelers to interact and share their experiences.

Step 1: Share Your Adventure

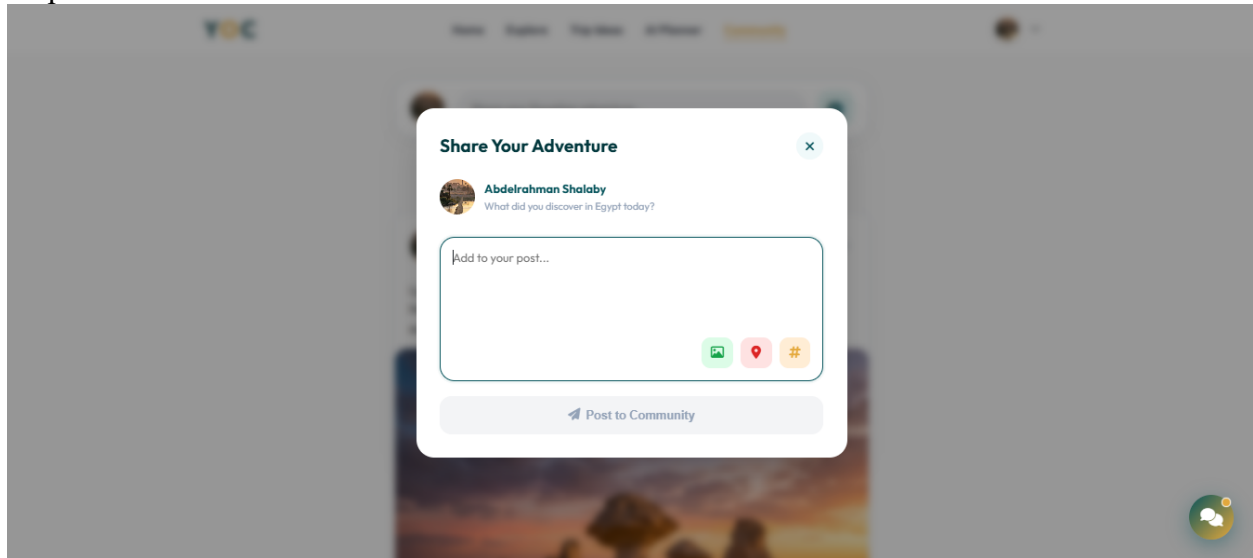


Figure 4.7 user share post

Description: A pop-up interface where users can write about their discoveries, add photos, and tag locations.

Step 2: Community Feed

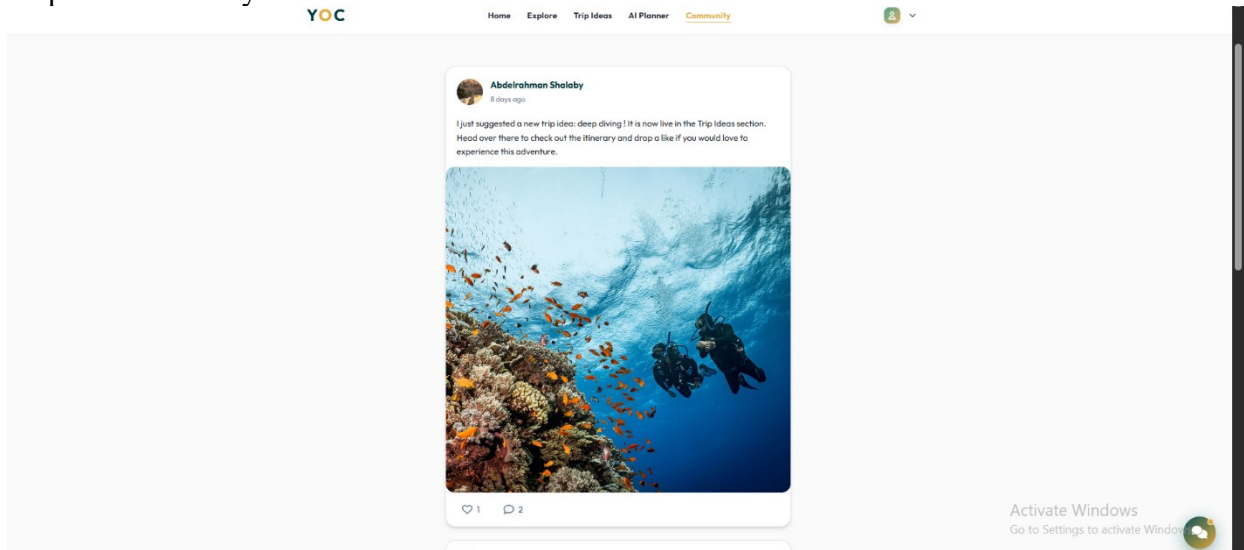


Figure 4.8 user browse posts

Description: A visual feed where users can browse posts from others, such as diving adventures or local tours.

Step 3: Post Interaction

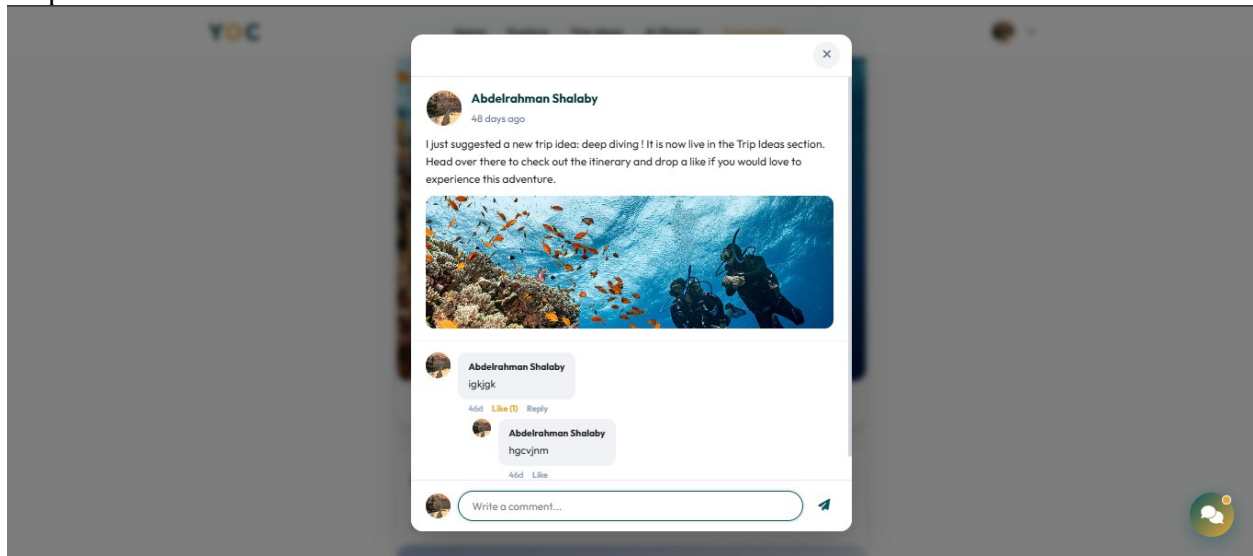


Figure 4.9 user interact with post

Description: Users can open specific posts to see detailed descriptions, view photos, and engage through likes and comments.

User Profile

Personal Profile

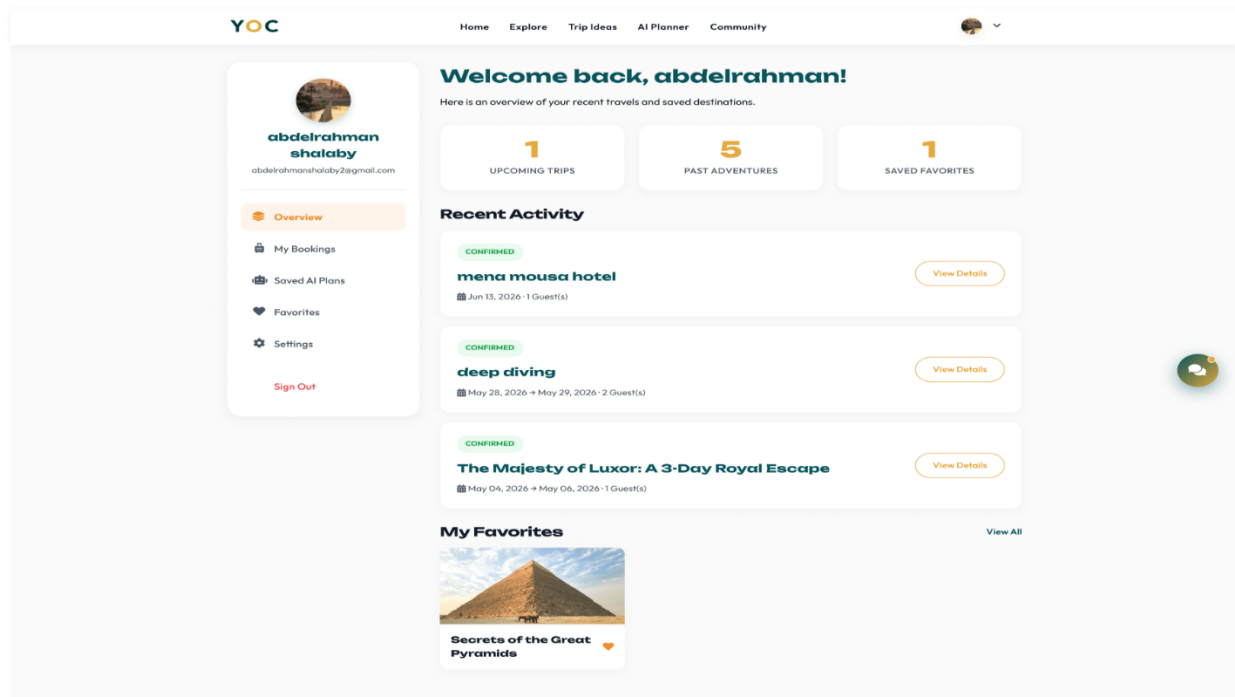
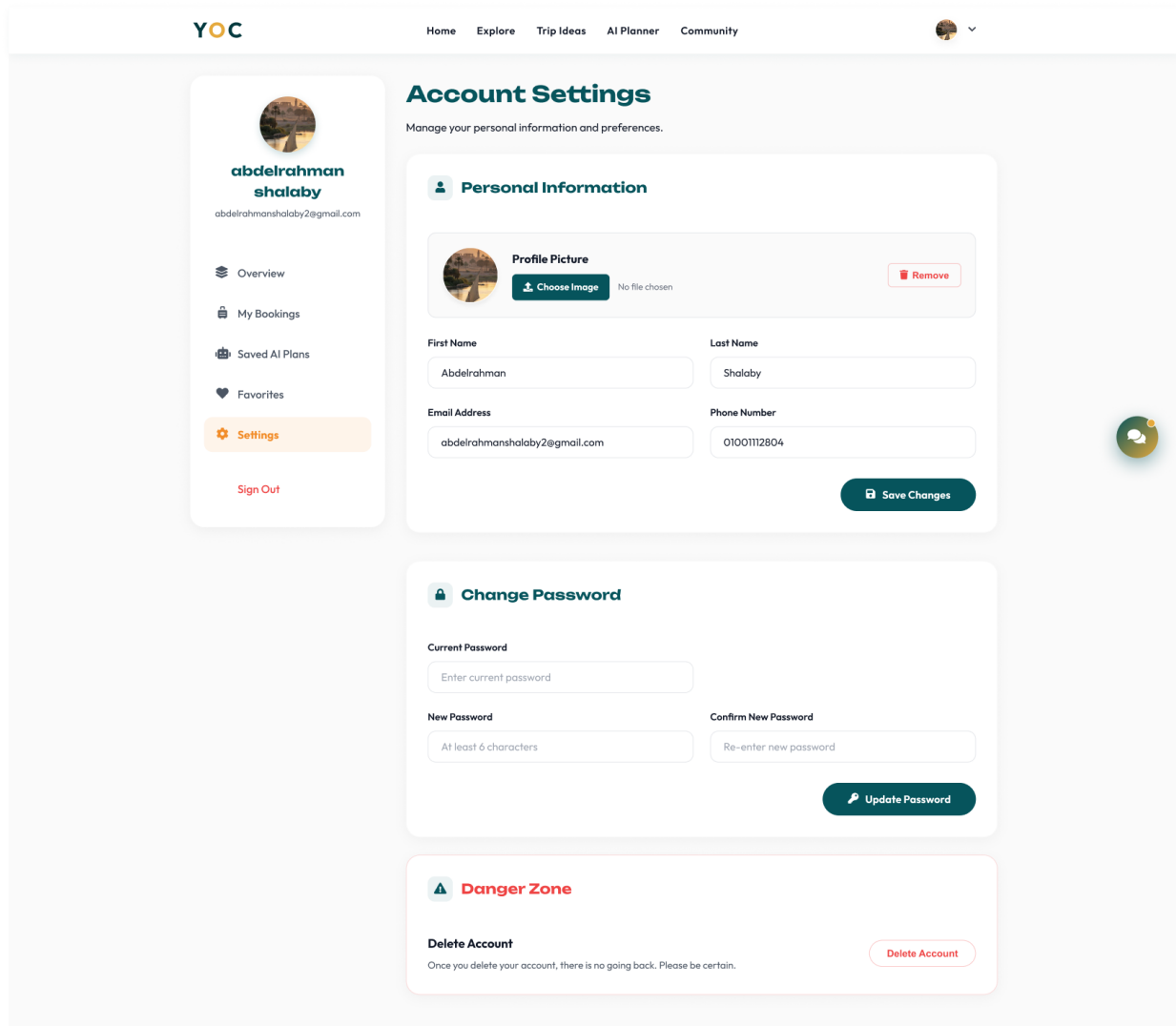


Figure 4.10 profile page

A private space for every traveller to view their journey and upcoming trips.

- **Overview:** View upcoming trips, past adventures, and saved favourites at a glance.

Account Control (Settings)



The screenshot shows the 'Account Settings' page of the YOC application. The page is divided into three main sections: Personal Information, Change Password, and Danger Zone. The user's profile is shown as 'abdelrahman shalaby' with the email 'abdelrahmanshalaby2@gmail.com'. The Personal Information section includes fields for First Name (Abdelrahman), Last Name (Shalaby), Email Address (abdelrahmanshalaby2@gmail.com), and Phone Number (01001112804). The Change Password section has fields for Current Password, New Password (with a hint 'At least 6 characters'), and Confirm New Password (with a hint 'Re-enter new password'). The Danger Zone section contains a 'Delete Account' button with a warning message: 'Once you delete your account, there is no going back. Please be certain.'

Figure 4.11 account settings page

Users can easily manage their account details to keep them up to date and secure.

- **Personal Information:** Update your name, email, picture and phone number.
- **Security:** A dedicated section to change your password and protect your account

The Booking Journey (6 Seamless Steps)

We designed a smooth booking experience to help users secure their trips in minutes.

Step 1: Selecting the Date & Guests

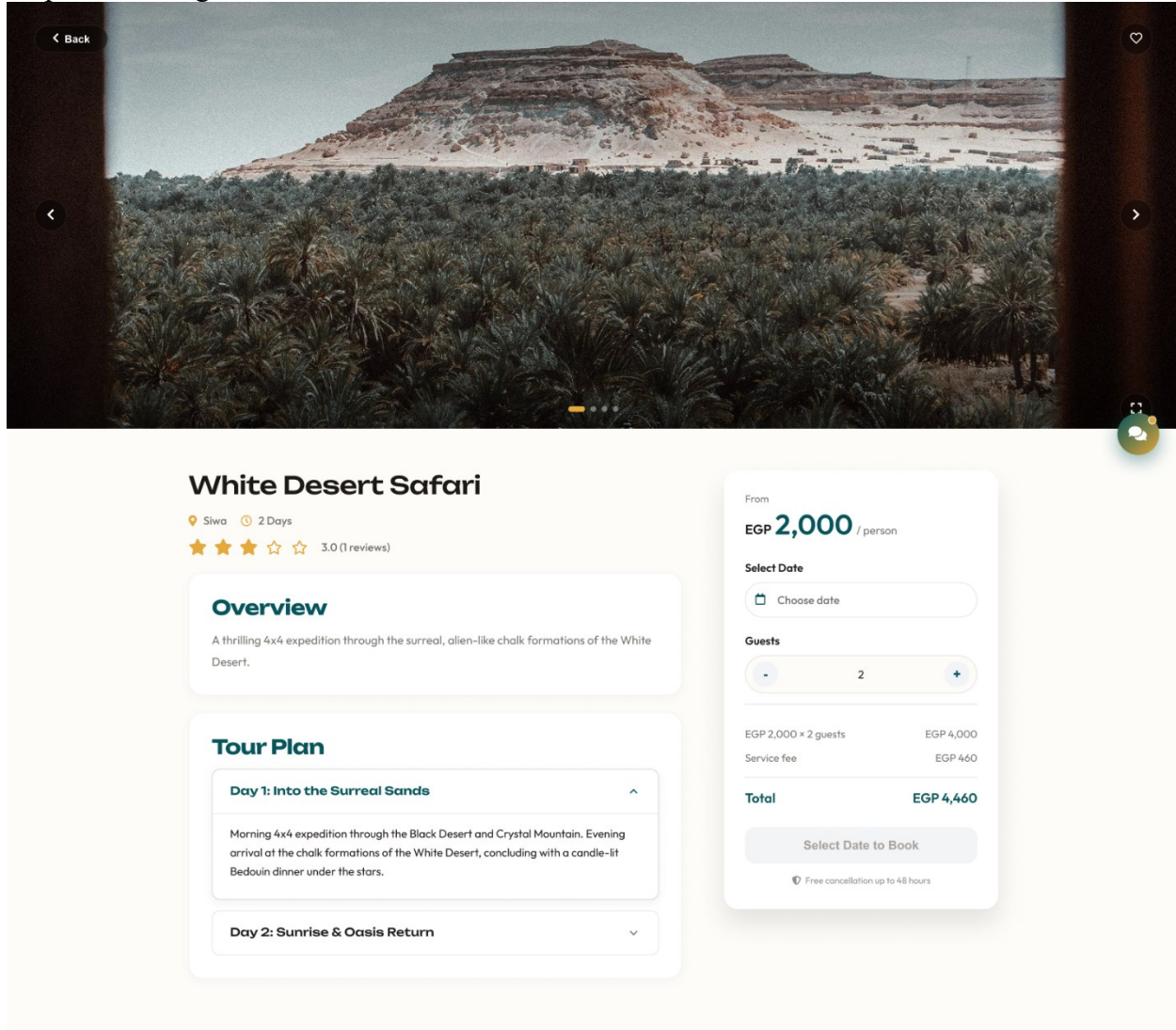


Figure 4.12 booking details page

Pick your preferred travel date and choose the number of guests. The system automatically calculates the total price.

Step 2: Reviewing Trip Details

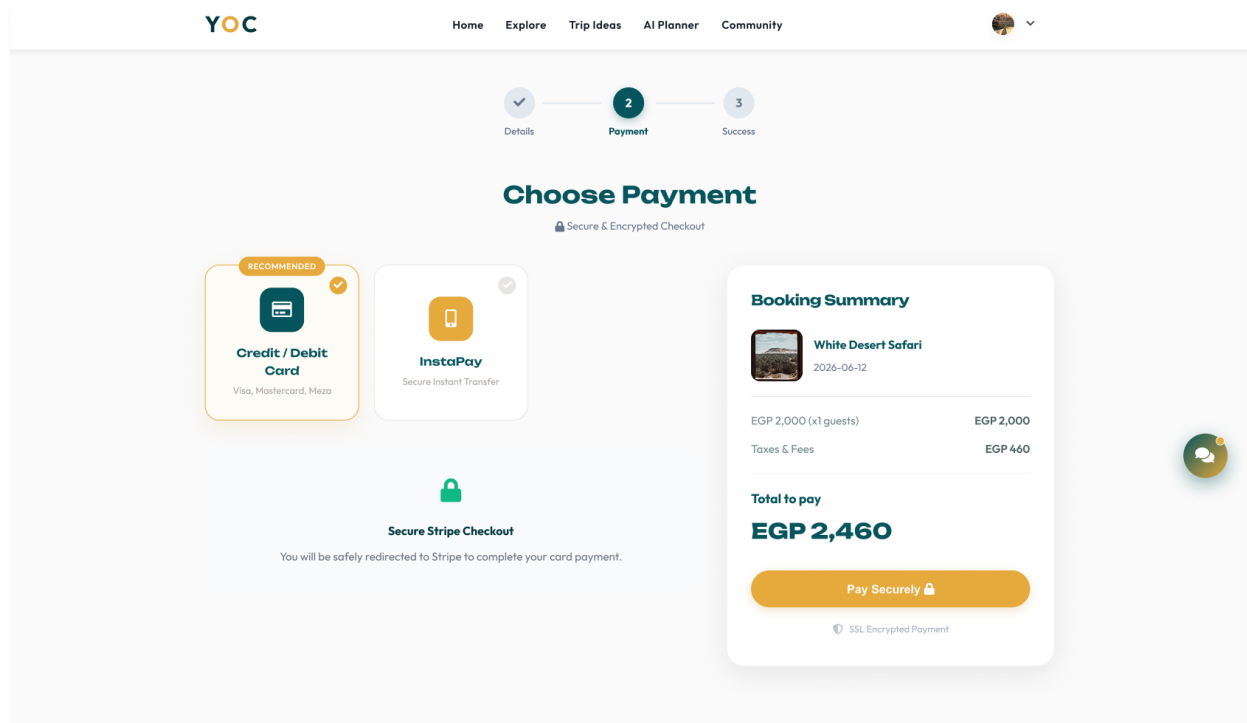
The screenshot shows the 'Reviewing Trip Details' screen in the YOC app. The top navigation bar includes 'Home', 'Explore', 'Trip Ideas', 'AI Planner', and 'Community'. A progress indicator at the top shows three steps: 1. Booking details (active), 2. Payment, and 3. Success. The main content area is divided into four sections:

- Trip details:** Features a photo of the White Desert, the title 'White Desert Safari', location 'Siwa', duration '2 Days', and a rating of '0.0 (0 reviews)'.
- Travel information:** Includes a 'DATE' field with 'Friday, June 12, 2026' and a 'GUESTS' field with '1'.
- Guest information:** Includes fields for 'FULL NAME' (Abdelrahman Shalaby), 'EMAIL' (abdelrahmanshalaby2@gmail.com), and 'PHONE NUMBER' (01001112804).
- Price summary:** Shows 'EGP 2,000 x 1 guests' for 'EGP 2,000' and a 'Service Fee' of 'EGP 460'. The total is 'EGP 2,460'. A 'Proceed to Payment' button is present, along with a note about 'Free cancellation within 48 hours' and 'Secure payment processing'.

Figure 4.13 user review booking details

A final look at the destination, date, and price summary before moving to the payment stage.

Step 3: Choosing Payment Method

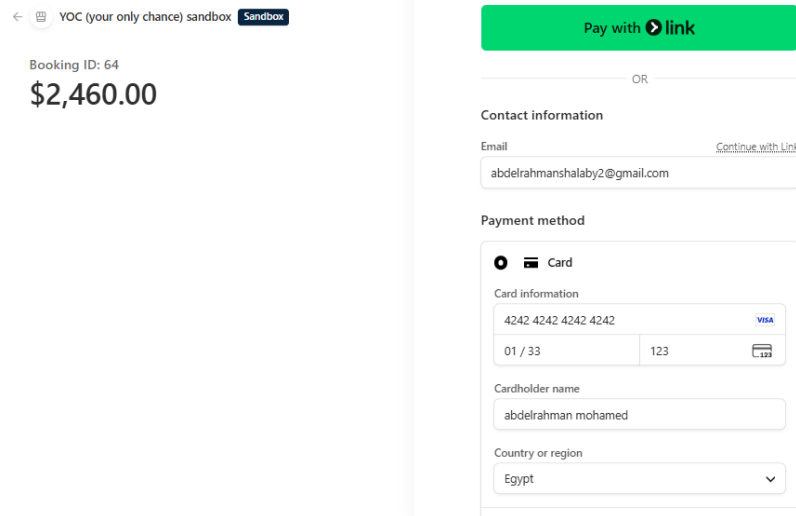


The screenshot shows the 'Choose Payment' screen of the YOC (You On Campus) system. At the top, there's a navigation bar with 'YOC' and links for 'Home', 'Explore', 'Trip Ideas', 'AI Planner', and 'Community'. A progress indicator shows three steps: 'Details', 'Payment' (current step), and 'Success'. The main heading is 'Choose Payment' with a sub-note 'Secure & Encrypted Checkout'. Two payment options are presented: 'Credit / Debit Card' (marked as 'RECOMMENDED' and 'Secure Stripe Checkout') and 'InstaPay' (marked as 'Secure Instant Transfer'). Below these, a 'Secure Stripe Checkout' section states: 'You will be safely redirected to Stripe to complete your card payment.' To the right, a 'Booking Summary' box for 'White Desert Safari' (2026-06-12) shows a total of EGP 2,460, including a base price of EGP 2,000 and taxes/fees of EGP 460. A 'Pay Securely' button is at the bottom of the summary box.

Figure 4.14 user choose payment method

Users can choose between secure "Credit/Debit Card" via Stripe or "InstaPay" for instant transfers.

Step 4: Secure Stripe Checkout

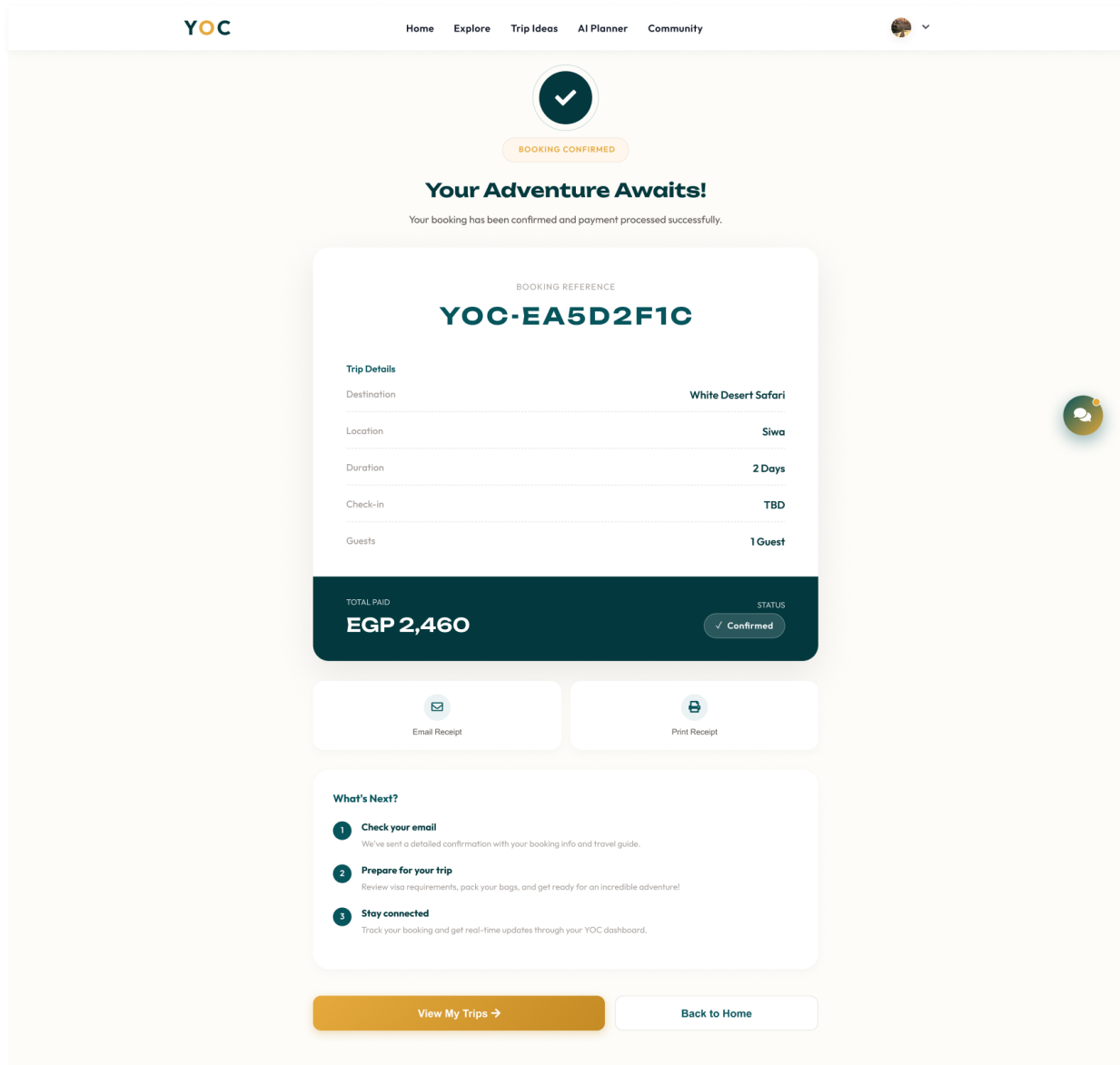


The image shows a Stripe Checkout form in a sandbox environment. On the left, it displays 'YOC (your only chance) sandbox' and 'Sandbox' buttons, along with 'Booking ID: 64' and a total amount of '\$2,460.00'. On the right, there is a green 'Pay with link' button. Below it, the 'Contact information' section includes an email field with 'abdelrahmanshalaby2@gmail.com' and a 'Continue with Link' link. The 'Payment method' section shows 'Card' as the selected method. Under 'Card information', there is a card number field with '4242 4242 4242 4242', an expiration date field with '01 / 33', and a CVV field with '123'. The 'Cardholder name' field contains 'abdelrahman mohamed'. The 'Country or region' dropdown is set to 'Egypt'.

Figure 4.15 secure payment checkout

A professional and encrypted payment gateway to enter card details safely.

Step 5: Booking Confirmation



The screenshot shows the 'Booking Confirmation' page for YOC. At the top, there's a navigation bar with 'YOC' logo and links: Home, Explore, Trip Ideas, AI Planner, and Community. A user profile icon is on the right. The main content area features a large green checkmark icon and a 'BOOKING CONFIRMED' badge. Below this, the text 'Your Adventure Awaits!' is displayed, followed by a confirmation message: 'Your booking has been confirmed and payment processed successfully.' The central part of the page is a white card with a 'BOOKING REFERENCE' section showing 'YOC-EA5D2F1C'. Below this is a 'Trip Details' section with a table:

Trip Details	
Destination	White Desert Safari
Location	Siwa
Duration	2 Days
Check-in	TBD
Guests	1 Guest

Below the table, a dark green bar shows 'TOTAL PAID' as 'EGP 2,460' and 'STATUS' as 'Confirmed' with a checkmark icon. Underneath are two buttons: 'Email Receipt' and 'Print Receipt'. A 'What's Next?' section follows, with three steps: 1. Check your email (We've sent a detailed confirmation with your booking info and travel guide.), 2. Prepare for your trip (Review visa requirements, pack your bags, and get ready for an incredible adventure!), and 3. Stay connected (Track your booking and get real-time updates through your YOC dashboard.). At the bottom, there are two buttons: 'View My Trips →' and 'Back to Home'.

Figure 4.16 user Confirmation booking

Once the payment is successful, the user gets a "Booking Reference" number and a summary of their confirmed adventure.

Step 6: Gmail Notification

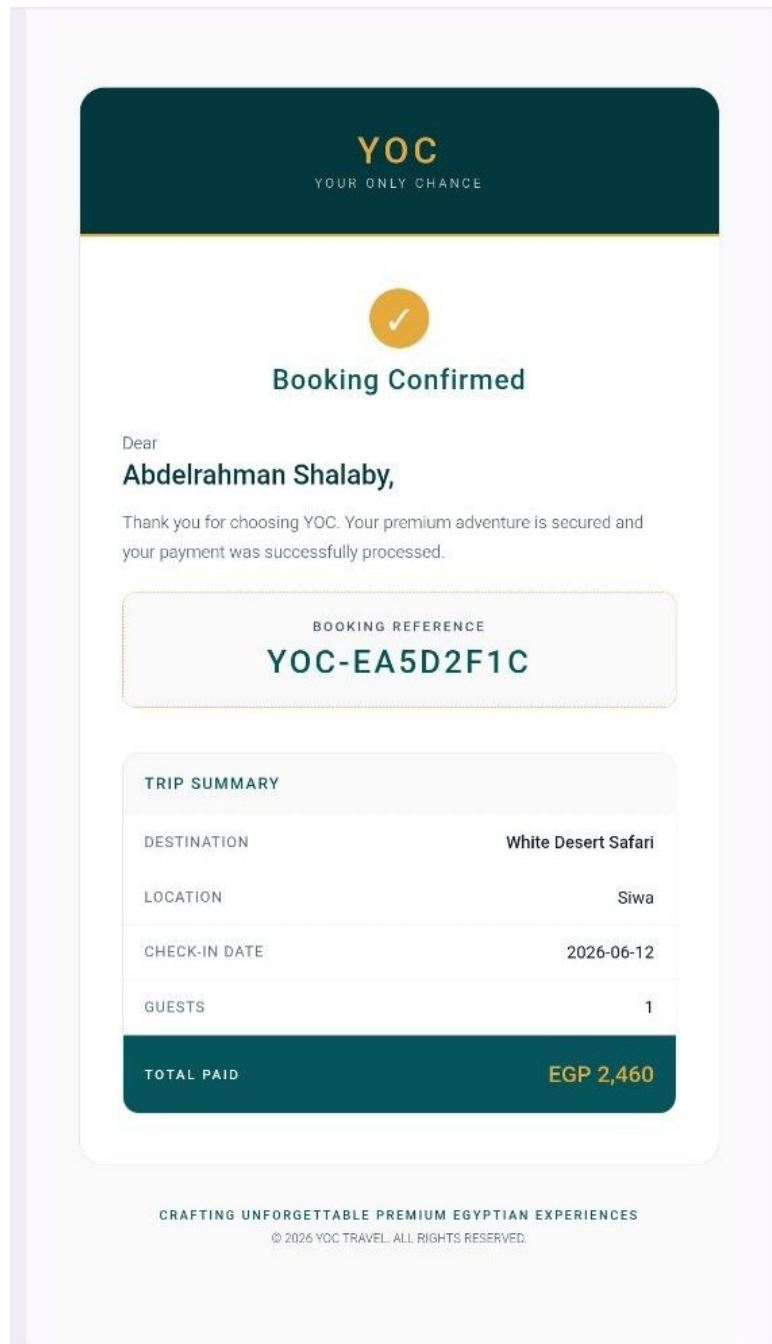


Figure 4.17 Gmail Confirmation message

To ensure peace of mind, a professional confirmation email is automatically sent to the user's Gmail with all the trip details.

Step7:AI Chatbot Nefertari (Egypt Guide)

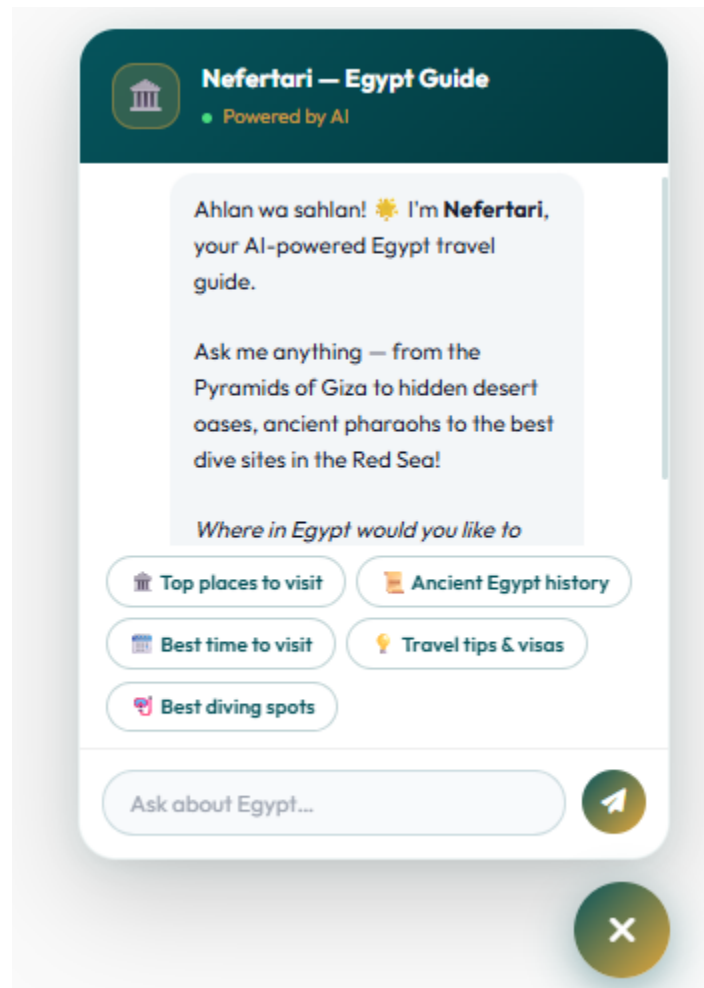


Figure 4.18 chatbot chat page

This page shows an AI chatbot called Nefertari, designed to help users explore Egypt and ask for information.

4.2.2 Admin Interface

Admin (Management Center)

This is the control center for the website team to manage everything.

Admin Login

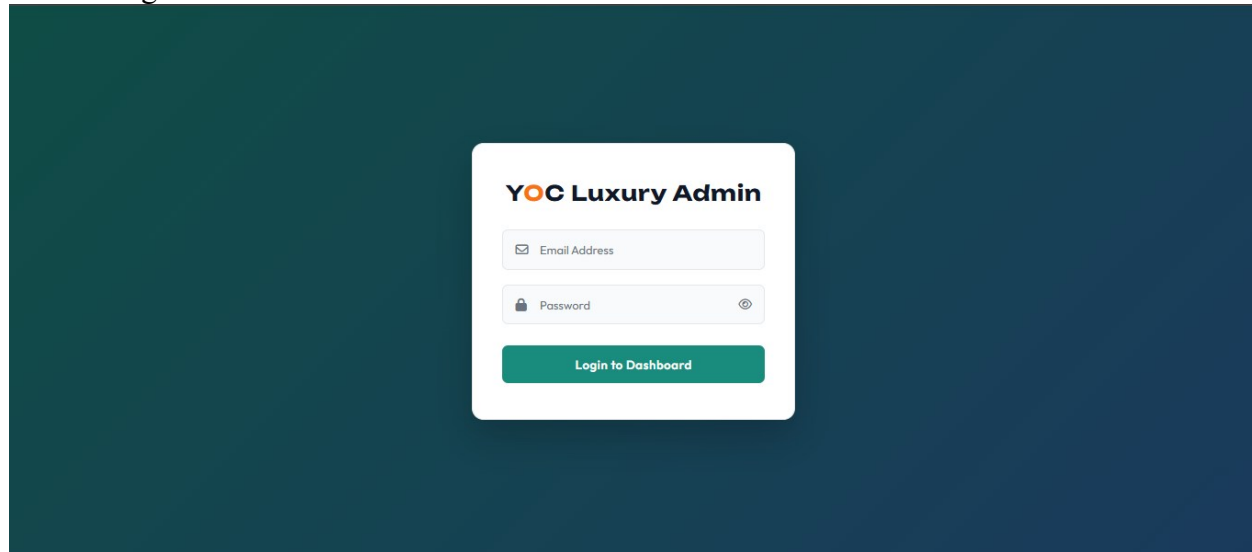


Figure 4.19 admin login page

A secure page for the admin to enter their email and password to start managing the site.

Dashboard Overview

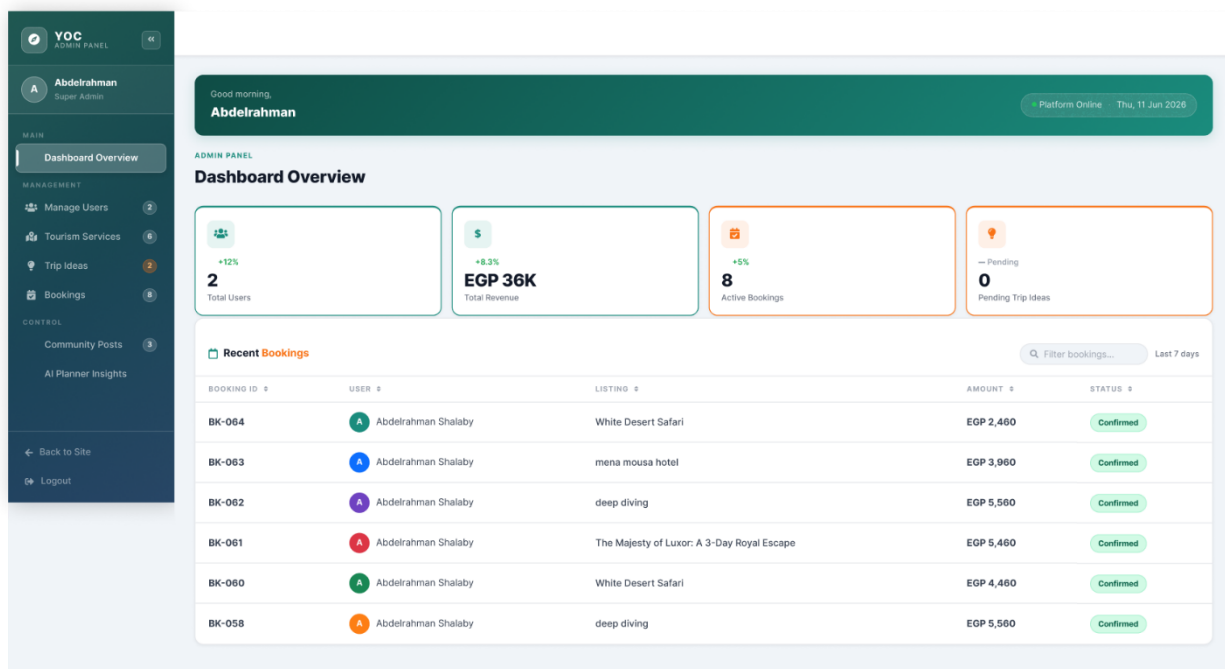


Figure 4.20 dashboard page

A quick look at the website's numbers:

- See how many users are joined.
- Check the total money earned (Revenue).
- View a list of the newest bookings.
- Approve or declined suggest trip.

Manage Users

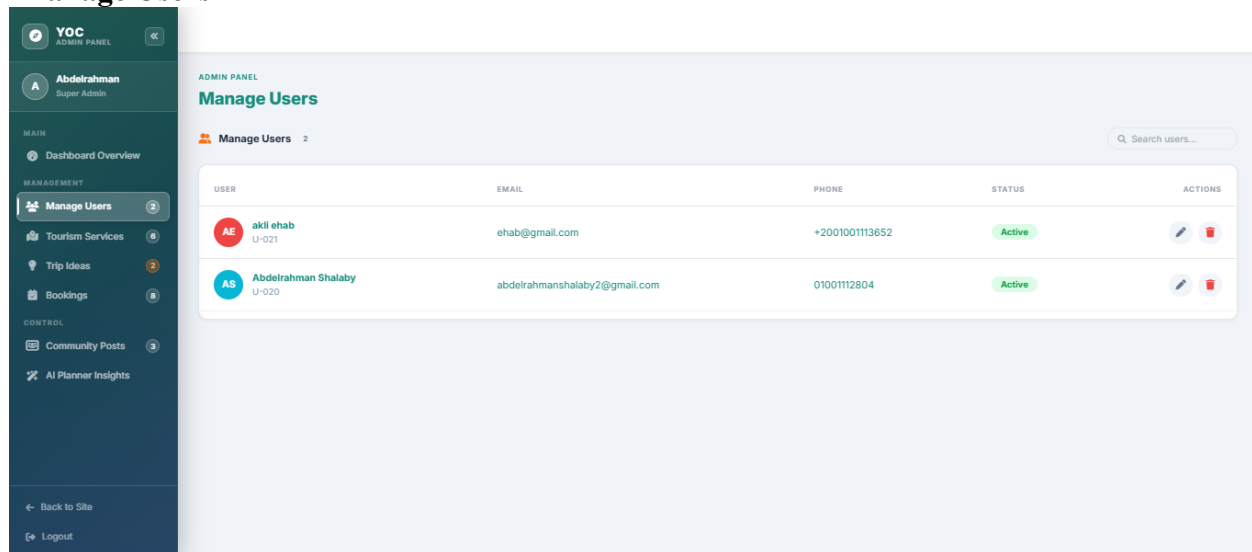


Figure 4.21 mange user page

A list of everyone using the platform. The admin can: See usernames and emails. Edit or delete users if needed.

Tourism Services

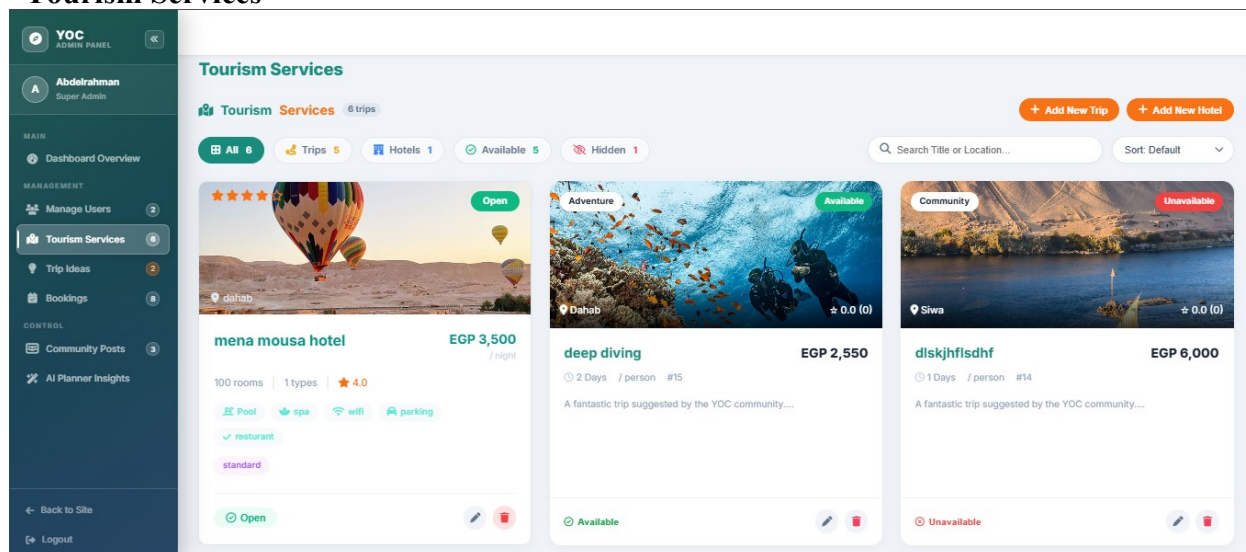


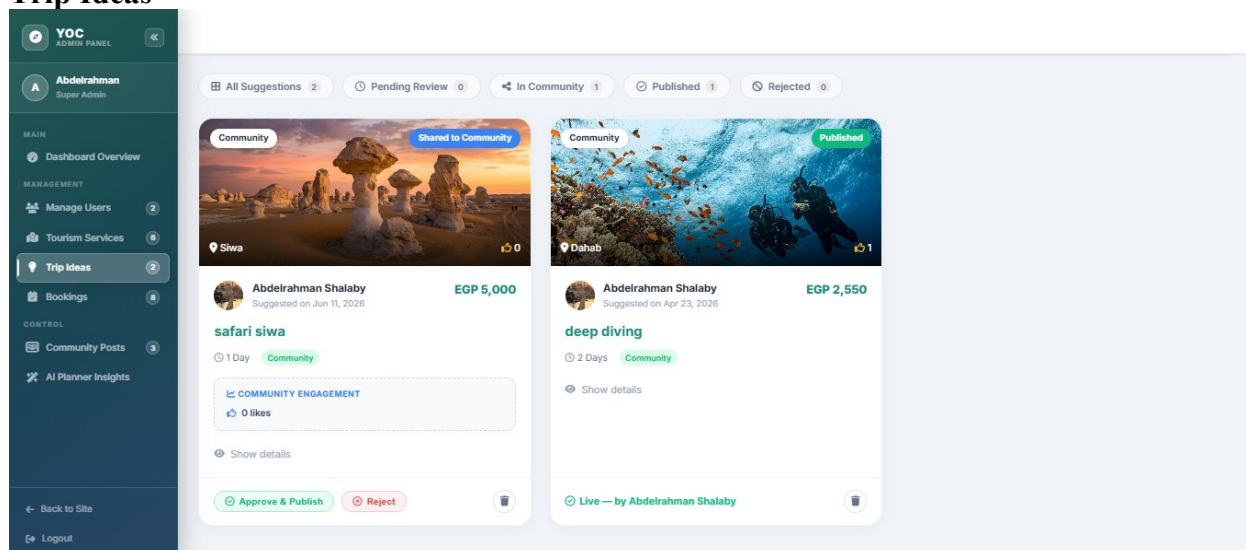
Figure 4.22 manage trips page

The admin manages the main trips here:

Add New Trip: Create a new adventure for users.

Edit or Remove: Update trip prices or delete old trips.

Trip Ideas



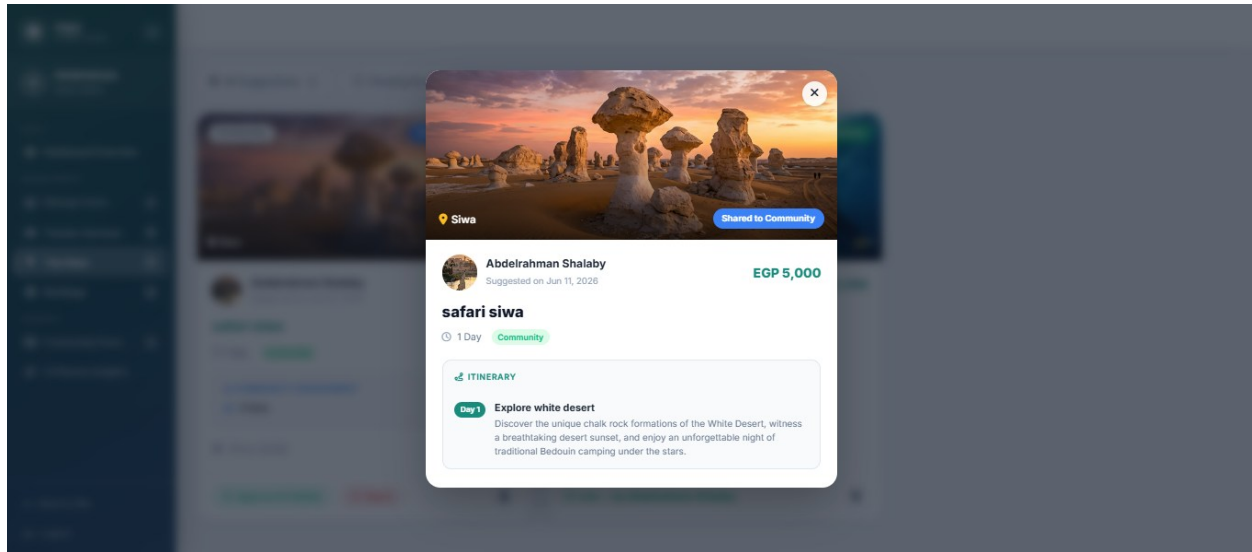


Figure 4.23 manage trips suggest ideas

Users can suggest new trips. The admin reviews them here:

Approve: Make the user's idea a real trip on the site.

Reject: If the idea is not suitable.

Bookings details

MANAGEMENT							
All Bookings							
Q Search bookings...							
All Confirmed Pending Cancelled							
REF #	USER #	LISTING / SERVICE #	DATES #	AMOUNT #	METHOD #	STATUS #	ACTIONS
YOC-EA5D2F1C	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	White Desert Safari	2026-06-12 → 2026-06-13	EGP 2,460	Credit / Debit Card	Confirmed	X
YOC-03AFDB06	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	mena mousa hotel	2026-06-13 → 2026-06-13	EGP 3,960	Credit / Debit Card	Confirmed	X
YOC-44F6B3A8	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	deep diving	2026-05-28 → 2026-05-29	EGP 5,560	Credit / Debit Card	Confirmed	X
YOC-7F39F831	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	The Majesty of Luxor: A 3-Day Royal Escape	2026-05-04 → 2026-05-06	EGP 5,460	Credit / Debit Card	Confirmed	X
YOC-072B030B	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	White Desert Safari	2026-04-28 → 2026-04-29	EGP 4,460	Credit / Debit Card	Confirmed	X
YOC-66F041E1	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	deep diving	2026-04-26 → 2026-04-27	EGP 5,560	InstaPay 0100112804	Confirmed	X
YOC-72B32A1F	akli ehab ehab@gmail.com	deep diving	2026-04-23 → 2026-04-24	EGP 5,560	Credit / Debit Card	Confirmed	X
YOC-9F6140BE	Abdelrahman Shalaby abdelrahmanshalaby2@gmail.com	deep diving	2026-04-23 → 2026-04-24	EGP 3,010	InstaPay 0100112804	Confirmed	X

Figure 4.24 booking details page

A full list of every trip booked on the site. The admin can:

Check if the user paid or not.

See the payment method (Card or InstaPay).

Community Posts

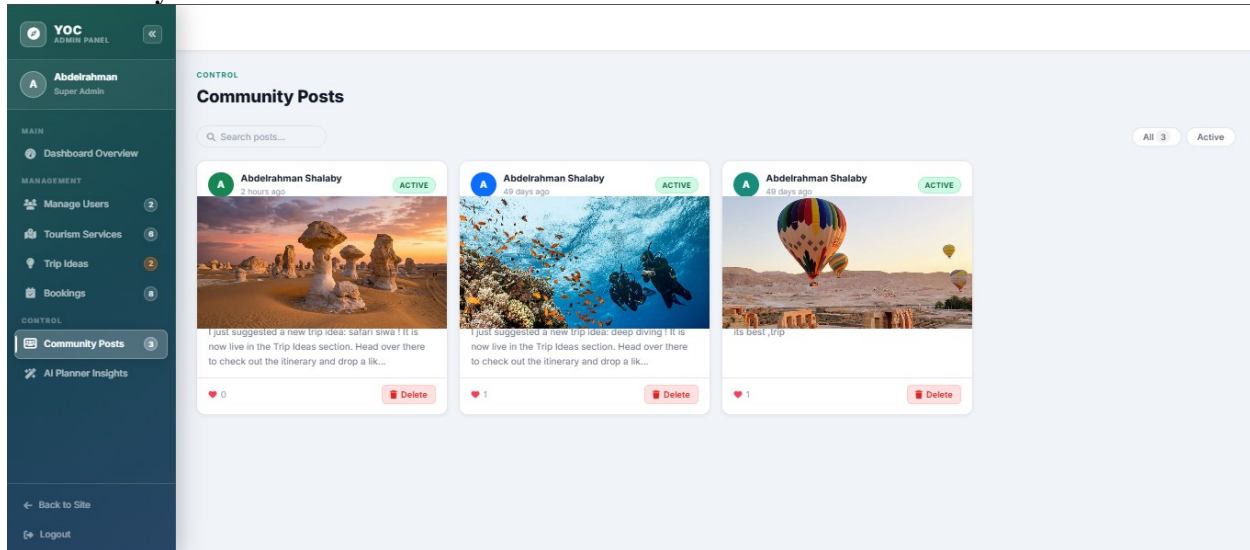


Figure 4.25 community post management page

The admin checks what users post in the community:
Review photos and stories shared by travelers.
Delete: Remove any bad or inappropriate posts.

Ai Planner Insights

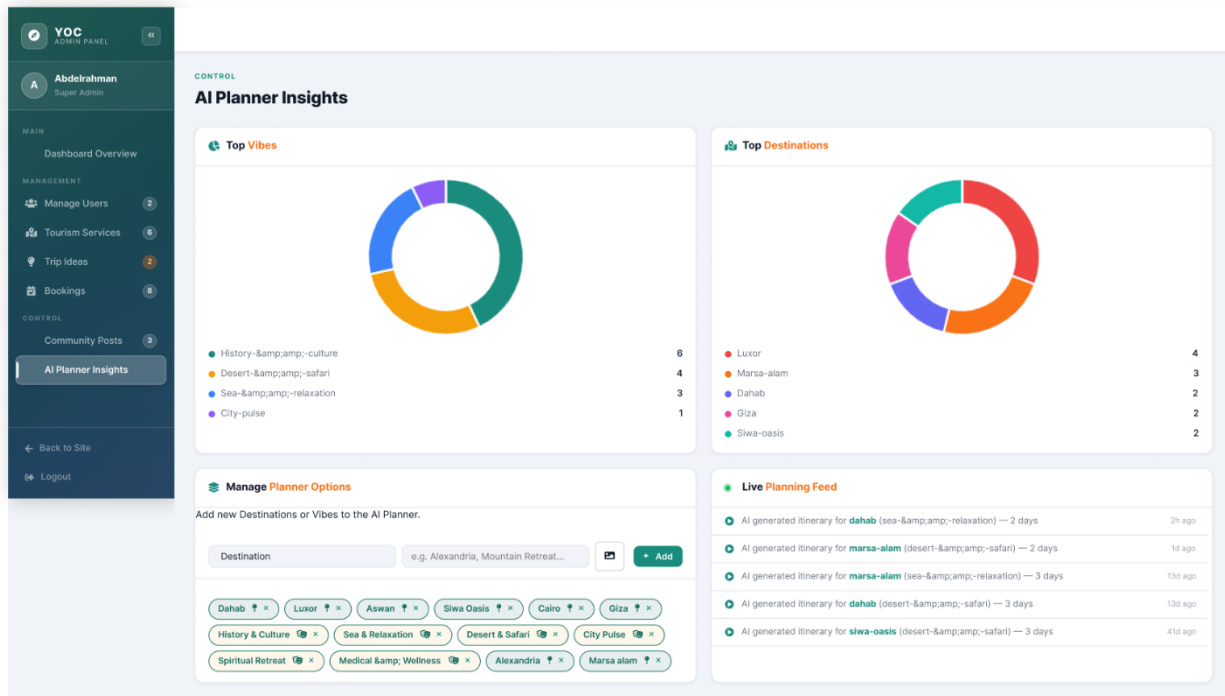


Figure 4.26 Ai planner insight page

- The admin checks what users post in the community:
- This screen shows an AI travel planner system in the YOC Travel Platform.
- It provides an overview of how users interact with the AI trip planner through visual analytics and statistics.
- Top Vibes, Top Destinations, and Manage Planner Options show the most popular travel styles and destinations chosen by users and also allow the admin to add or update new destinations and travel preferences in the AI system.

Chapter Five Conclusion and Future work

5.1 Conclusion

The current system represents the initial implementation of the Your Only Chance (YOC) tourism web platform. It is a web-based system designed to support tourism activities inside Egypt by providing users with a centralized digital environment to explore destinations, discover travel options, and complete bookings online.

The platform targets both local and international travelers and focuses on simplifying the trip planning process by combining exploration, recommendation, and booking services into one unified system.

This project developed a smart and easy-to-use travel platform , the main goal of the system is to help users plan their trips in Egypt in a simple and modern way.

The system combines different features in one platform, including an AI travel planner, an AI chatbot (Nefertari), a booking system, a community section, and a user profile system. These features help users from the first step of planning until completing the booking process.

The AI planner helps users choose destinations, budgets, and trip types, and then generates a full travel plan. The chatbot also helps users by answering questions about tourist places, travel tips, and suggestions.

Overall, the system makes travel planning faster, easier, and more organized compared to traditional methods.

5.2 Achievements

AI-Powered Travel Planning System

The project successfully implemented an intelligent travel planning system that allows users to generate complete personalized itineraries based on destination, budget, and travel preferences, making trip planning faster and more efficient.

Interactive AI Chatbot (Nefertari)

A smart chatbot was developed to act as a virtual travel assistant, providing users with instant answers, travel guidance, and detailed information about tourist attractions and cultural sites in Egypt.

Comprehensive Booking System

The platform provides a complete booking flow that allows users to select travel dates, number of guests, and preferred payment methods (Visa, InstaPay) ensuring a smooth and structured booking experience.

Integrated Travel Ecosystem

The system combines multiple services in one platform, including trip planning, booking, recommendations, and community interaction, creating a unified digital travel experience.

Community Interaction Platform

A community module was implemented to allow users to share travel experiences, explore posts from other travelers, and engage with the platform content, improving user interaction and engagement.

User Profile Management System

Each user has a dedicated profile where they can manage personal information, view their bookings, and track their travel activities in an organized way.

Trip Suggestion and Exploration Feature

Users can actively suggest new trips and explore available destinations, places, and activities, making the system more dynamic and user-driven.

Improved User Experience and Usability

The platform was designed with a simple and intuitive interface that enhances usability, reduces complexity, and provides a smooth experience for different types of users.

5.3 Limitations

- **Local Database AI Constraints (RAG):** The AI Planner and chatbot generate itineraries using inventory stored strictly within our local database (listing table). The system cannot fetch live pricing or live availability from external global airline or hotel APIs.
- **Manual InstaPay Verification:** While international credit card payments are automated via Stripe, domestic InstaPay transfers require manual administrator review and approval, due to the lack of open public banking verification APIs.
- **Local Media Storage Limitation:** All user avatars, trip photos, and suggestion images are stored directly on the local server filesystem rather than a scalable cloud Content Delivery Network (CDN), which creates a storage bottleneck as the platform scales.
- **External Mail Dependency:** Booking notifications and password resets rely completely on external SMTP mail servers (via PHP Mailer). If the external mail provider experiences latency or reaches its API quota, delivery may be delayed.

5.4 Future Work

Enhance AI Understanding

Improve the chatbot and AI planner to generate more natural and intelligent responses.

Improve Personalization System

Upgrade the current Suggest Trip, Places, and Activities features to provide more accurate recommendations based on user behavior and preferences.

Mobile Application Development

Develop Android and iOS apps to make the platform more accessible.

Real-Time Notification System

Add notifications for bookings, trip updates, and community interactions.

Enhance Community Engagement Features

Improve the existing community system by adding more interactive features such as better content discovery, improved post organization, and richer user interactions to increase engagement between travelers.

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Appendix

6.1 Questionnaire Form

YOC

The project is a **tourism website focused on Egypt**, targeting both international tourists and local users. The platform allows users to **book hotels, camps, and trips** directly.

المشروع عبارة عن موقع إلكتروني سياحي متخصص في مصر، يستهدف السياح الدوليين والمستخدمين المحليين على حد سواء. يتيح المنصة للمستخدمين حجز الفنادق والمخيمات والرحلات مباشرة.

atefk4662@gmail.com [Switch account](#)



✉ Not shared

What is your Gender? *

- ☐ Male
- ☐ Female

What is your age group? *

ما هي فئتك العمرية؟

- ☐ Under 18
- ☐ 18–24
- ☐ 25–34
- ☐ 35–44
- ☐ 45+

Have you traveled within Egypt in the past year? *

هل سافرت داخل مصر خلال آخر سنة؟

☐ Yes

☐ No

Figure 6.1 Survey Page 1

Travel Behavior – سلوك السفر

How many times do you travel inside Egypt per year? *

كم مرة تسافر داخل مصر في السنة؟

- ☐ Never
- ☐ Once
- ☐ 2- 3 times
- ☐ More than 3 times

What type of destinations do you usually travel to? *

عادةً بتسافر إلى أي نوع من الوجهات؟

- ☐ Historical & cultural places
- ☐ Adventure trips
- ☐ Relaxation places
- ☐ Medical tourism
- ☐ Other

How do you usually search for travel destinations? *

كيف تبحث عادةً عن أماكن للسفر؟

- ☐ Facebook / Instagram
- ☐ Friends / Acquaintances
- ☐ Booking websites (Booking, TripAdvisor...)
- ☐ WhatsApp / Telegram groups
- ☐ Travel agencies
- ☐ Other

Figure 6.2 Survey Page 2

Pain Points – المشاكل الحالية

Have you faced any problems while searching for trips or tourist places? *

هل واجهت مشاكل أثناء البحث عن رحلات أو أماكن سياحية؟

- ☐ Yes
- ☐ No

Which of these problems have you faced the most? (Select all that apply) *

أي من هذه المشاكل واجهتها أكثر؟ (اختر ما ينطبق)

- ☐ Difficulty comparing prices صعوبة مقارنة الأسعار
- ☐ Not enough information about the place عدم وجود معلومات كافية عن المكان
- ☐ Lack of clear photos/videos مفقوش صور وفيديوهات واضحة
- ☐ Untrustworthy companies / Fake reviews مزيفة Reviews / شركات غير موثوقة
- ☐ No single platform for everything عدم وجود موقع واحد يجمع كل حاجة
- ☐ Manual booking (WhatsApp messages) الحجز يتم بشكل يدوي
- ☐ No personalized recommendations لا يوجد تخصيص حسب اهتماماتي
- ☐ Payment issues مشاكل في الدفع
- ☐ Other

How satisfied are you with current platforms ? *

ما مدى رضاك عن المنصات الحالية؟

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very bad

Figure 6.3 Survey Page 3

YOC Feature Evaluation – تقييم فكرة المنصة

Would you like an Egyptian platform that combines (Hotels + Camps + *
Trips + Guides) in one system?

بنظام (Hotels + Camps + * Trips + Guides) هل ترغب في وجود منصة مصرية تجمع
واحد؟

- ☐ Definitely Yes
- ☐ Maybe
- ☐ I don't know
- ☐ No

Is it important for you that the platform uses AI to recommend trips *
based on your interests?

هل ترى أن وجود AI يرشح لك رحلات مناسبة لاهتماماتك مهم؟

- ☐ Very Important
- ☐ Important
- ☐ Not Important
- ☐ Not Necessary

what is the most critical feature you look for in travel platform? *

- ☐ All-in-One Booking (hotels, camps, and tours)
- ☐ AI Personalization
- ☐ Trust & Safety
- ☐ Local Experiences
- ☐ Community & Rewards
- ☐ Smart Support

Which features would encourage you to try a new Egyptian tourism *
platform?

- ☐ AI Trip planner
- ☐ Verified Local guides
- ☐ Rich Social Feed
- ☐ Gamification & reward
- ☐ Trip Suggestions
- ☐ Exclusive Local Offers
- ☐ User-Friendly Design

Figure 6.4 Survey page 4

Trust Factors – عوامل الثقة

What makes you trust a booking website? *

ما الذي يجعلك تثق في موقع حجز؟

- ☐ Secure payment methods
- ☐ Verified guides
- ☐ Accurate information
- ☐ Clear customer reviews and rating
- ☐ Transparent pricing with no hiding fees
- ☐ Professional website and easy navigation
- ☐ Flexible cancellation and refund policies
- ☐ Positive reputation

Do you prefer that the system suggests a ready-made itinerary based on (mood + budget + city + interests)? *

هل تفضل أن يقترح لك النظام برنامج رحلة جاهز بناءً على (المود + الميزانية + المدينة + الاهتمامات)؟

- ☐ Yes
- ☐ Sometimes
- ☐ No

Are you comfortable paying online at a trusted website? *

هل تشعر بالراحة عند الدفع عبر الإنترنت على موقع موثوق؟

- ☐ Yes
- ☐ Maybe
- ☐ No

Do you prefer to book through: *

هل تفضّل الحجز من:

- ☐ One integrated platform
- ☐ A travel agency
- ☐ Manual search online

Would you like to share your experiences and photos in the community? *

هل ترغب في مشاركة تجاربك وصورك على الـ Community؟

- ☐ Yes
- ☐ No
- ☐ Depends on the trip

Back

Submit

Page 5 of 5

Clear form

Figure 6.5 Survey Page 5

6.2 Screenshots of Website Emails

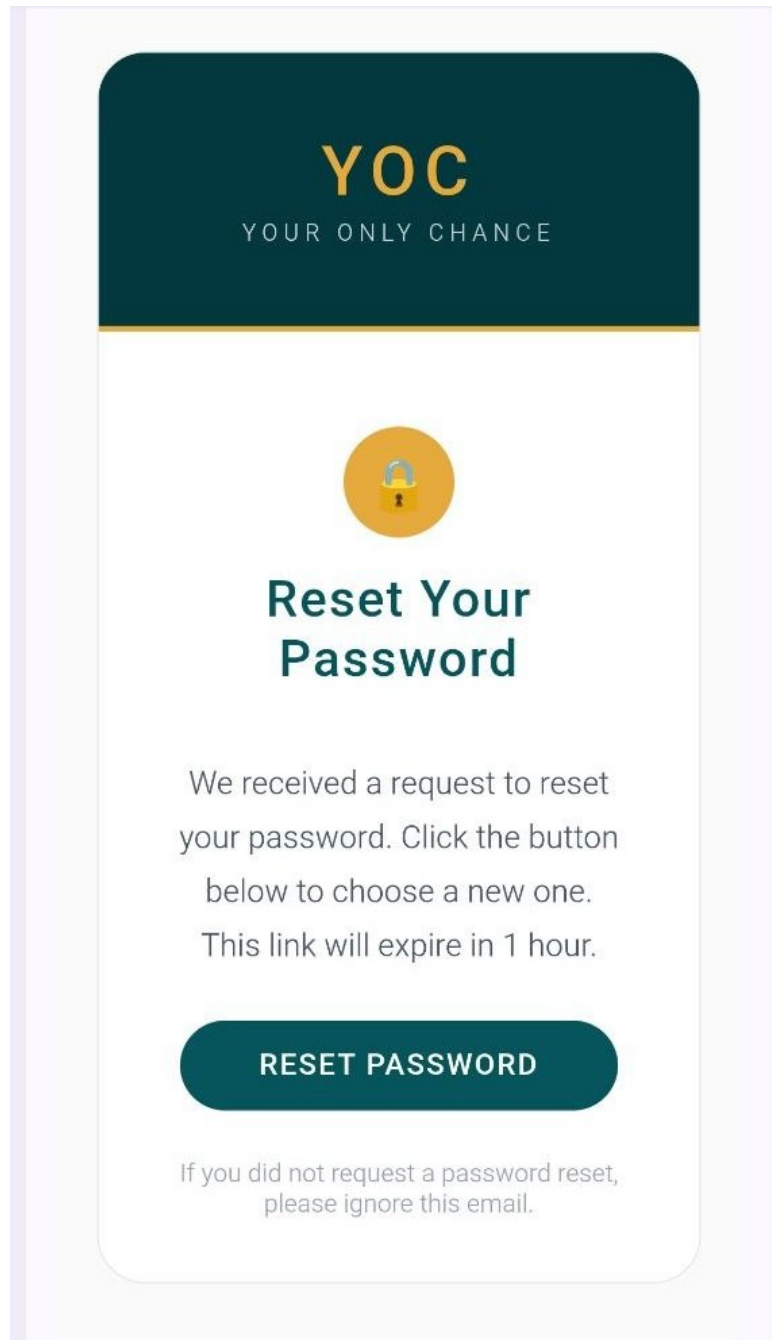


Figure 6.6 Forget Password Email

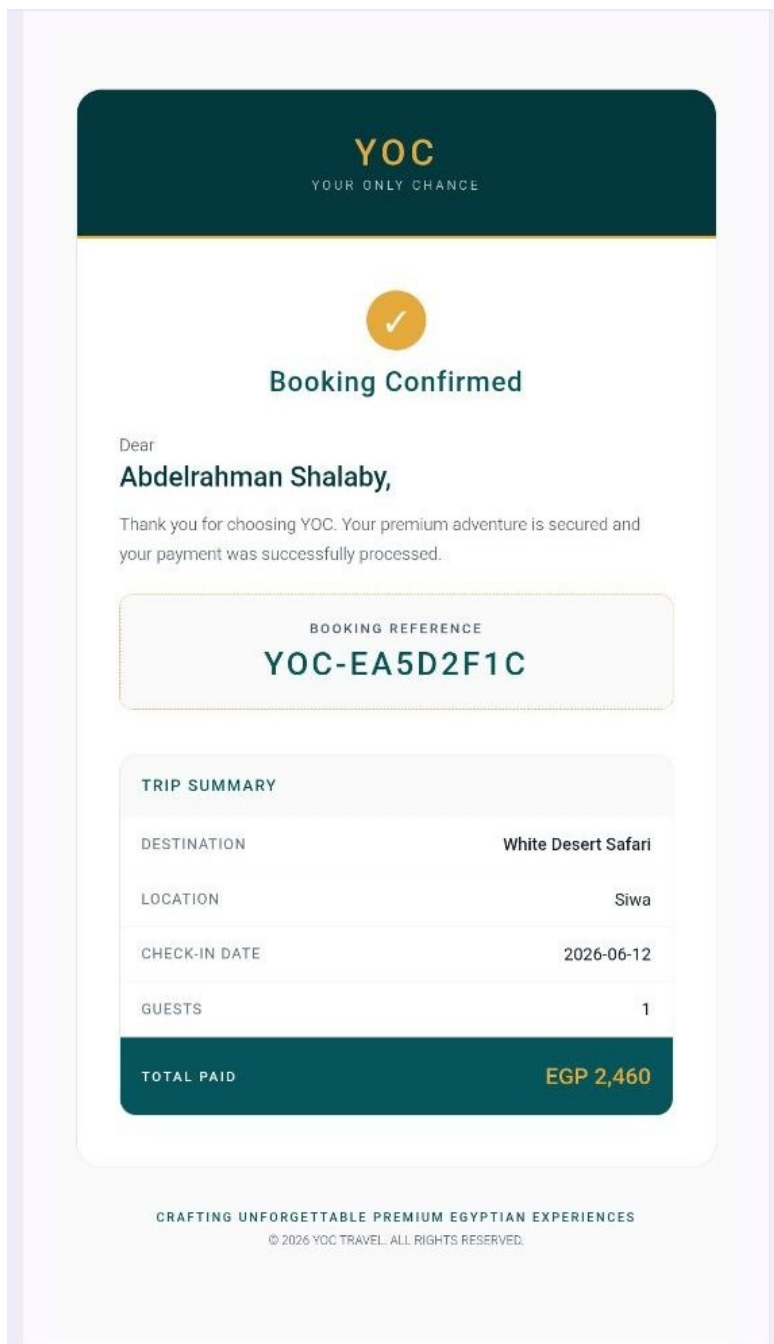


Figure 6.7 booking confirmation

6.3 MySQL Query Sample

```
-- Table structure for table `admin`
CREATE TABLE `admin` (
  `admin_id` int(11) NOT NULL,
  `f_name` varchar(50) NOT NULL,
  `l_name` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
-- Dumping data for table `admin`

INSERT INTO `admin` (`admin_id`, `f_name`, `l_name`, `email`, `password`) VALUES
(1, 'abdelrahman', 'shalaby', 'shalaby@gmail.com', '$2y$10$JBy/tSK.JyFpP/JLEPXzde9wx8vO/jcQ5Y8cBSNRwlmWkqlpJYI9W'),
(2, 'mahmoud', 'assem', 'assem@gmail.com', '$2y$10$BcobXEAOWnGD73PvKNMA2.oh8PAC1xgMTL0MVslrb7QuVJh5EYScC'),
(5, 'mohamed', 'ehab', 'ehab1@gmail.com', '$2y$10$ctpHW43FDxXbAL/6wjBQY./pzitJbbDCGvVuJP/jCzcJhrr4.pBjS');

-- Table structure for table `ai_insights_logs`
CREATE TABLE `ai_insights_logs` (
  `id` int(11) NOT NULL,
  `destination` varchar(100) NOT NULL,
  `vibe` varchar(100) NOT NULL,
  `budget` varchar(50) NOT NULL,
  `days` int(11) NOT NULL,
  `created_at` timestamp NOT NULL DEFAULT current_timestamp()
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

-- Table structure for table `booking`
CREATE TABLE `booking` (
  `booking_id` int(11) NOT NULL,
  `user_id` int(11) NOT NULL,
  `listing_id` int(11) NOT NULL,
  `start_date` date NOT NULL,
  `end_date` date NOT NULL,
  `guests_count` int(11) NOT NULL,
  `status` varchar(50) DEFAULT 'Pending',
  `price` decimal(10,2) NOT NULL,
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),
  `updated_at` timestamp NOT NULL DEFAULT current_timestamp() ON UPDATE current_timestamp(),
  `booking_ref` varchar(20) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

Table structure for table `comment_like`
CREATE TABLE `comment_like` (
  `like_id` int(11) NOT NULL,
  `comment_id` int(11) NOT NULL,
  `user_id` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```